

RoleInstructor: A user-centered plug-and-play method for enhancing the interaction performance of LLM-based intelligent NPC systems

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Abstract

We present RoleInstructor, a novel user-centered plug-and-play method designed to address the critical challenge of maintaining long-term conversational coherence and user engagement in LLM-driven Intelligent non-player character (NPC) systems. Unlike existing approaches that rely on pre-defined heuristics or indirect optimization objectives, our method directly extracts human-evaluative metrics from player feedback data through a fine-grained annotation pipeline. These metrics are then operationalized via a two-stage optimization process: (1) metric-aware model calibration using constrained fine-tuning, and (2) dynamic adaptation through contextual instruction during inference. Through experimental validation across three distinct LLM-based NPC architectures, we demonstrate that RoleInstructor achieves statistically significant improvements by an average of 36.9% when deployed across heterogeneous NPC systems. Our code, dataset, and models will be released on GitHub.

1 Introduction

Amidst the rapid growth of the gaming industry and players' increasing demand for immersive experiences, the intelligence level of non-player characters (NPCs) has become a key indicator of game quality and immersion. Large language models (LLMs), with their powerful natural language understanding and generation capabilities, offer unprecedented opportunities for building intelligent NPCs capable of fluent dialogue and dynamic personality portrayal (Lu et al., 2024).

While LLMs endow NPCs with remarkable human-like interactive abilities, they also

significantly elevate player expectations. However, current LLM-based intelligent NPC systems still face several core challenges in practical application. First, limitations exist in interaction quality and consistency: LLMs still exhibit significant shortcomings in ensuring long-term consistency of NPC dialogue, persona, and memory, as well as the high-fidelity alignment of content with game world settings (Shanahan et al., 2023). Conventional optimization methods such as Prompt Engineering (PE) and Fine-tuning fail to fully leverage the potential of LLMs for online self-correction and quality control (Didolkar et al., 2024), resulting in inconsistent interaction quality (Robino et al., 2025; Alghisi et al., 2024). Second, a disconnect remains between evaluation and user experience: current mainstream objective evaluation metrics deviate from players' actual subjective feelings and core needs during interaction. Existing benchmarks, such as TD-EVAL (Acikgoz et al., 2025) and DSTC (Gung et al., 2023), often focus on model-centric metrics, yet a model's performance on these metrics does not fully reflect its ability to deliver a satisfactory user experience (Ou et al., 2024). This 'model-centric' rather than 'user-centered' evaluation paradigm fails to directly capture whether NPC interactions truly meet player needs, leading to a misalignment between optimization efforts and the goal of enhancing genuine user experience.

To address these multifaceted challenges, this paper proposes the user-centered Instructor with a plug-and-play characteristic, as shown in Figure 1. The method we proposed has two core components, aiming to systematically address the evaluation gap and enhance the performance of LLM-based intelligent NPC systems.

First, we introduce the Metric Extraction Framework (MEF), a framework to systematically

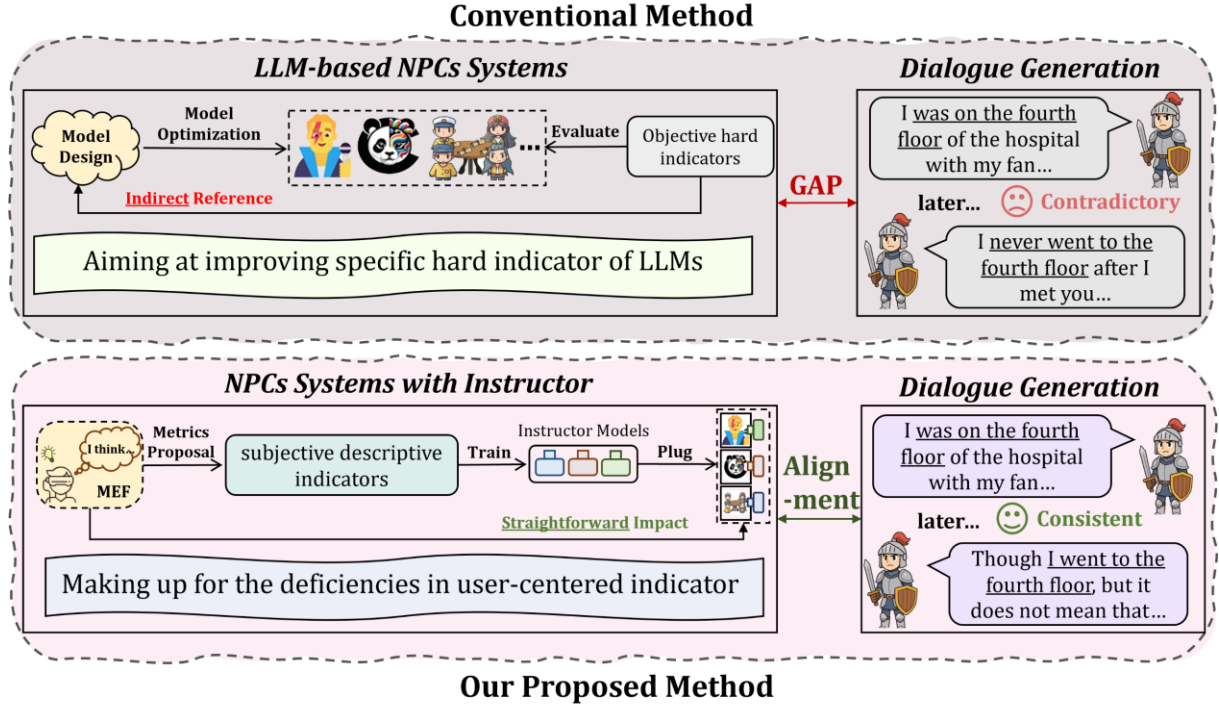


Figure 1: A schematic diagram of the method we proposed compared with the conventional method.

80 derive multi-dimensional, user-centered evaluation 109 play, thus it can be extended to any LLM-based
81 metrics and data sources from real player feedback. 110 intelligent NPC system developed by others,
82 Through this framework, we identify key 111 enhancing the modularity, scalability, and
83 evaluation dimensions crucial for game NPCs, 112 maintainability of the proposed method.
84 including task consistency, style consistency, script 113 The main contributions of this paper can be
85 consistency, self-consistency, and QA consistency. 114 summarized as follows:
86 The metrics we proposed by this framework are 115 1. We propose a novel user-centered plug-and-
87 directly derived from player experience, thus more 116 play Instructor method, employing an Instructor
88 accurately capture the specific requirements of 117 model to instruct an arbitrary LLM-based NPC
89 NPCs for games and effectively bridge the gap 118 system. By developing an explicitly metric-driven
90 between evaluation and experience of users. 119 evaluation-instruction-correction loop, this method
91 Second, we develop the instructor models based 120 significantly enhances the stability and quality of
92 on the proposed metric and data generated by MEF, 121 LLM-based intelligent NPC systems.
93 named RoleInstructor. As an independent 122 2. We design a Metric Extraction Framework to
94 evaluation and feedback hub, RoleInstructor 123 systematically derive user-centered NPC
95 utilizes an ensemble of specialized Small 124 evaluation metrics and training data from real
96 Language Models (SLMs), each optimized for 125 player feedback. The multi-dimensional metrics
97 specific evaluation dimensions derived from MEF, 126 we proposed directly map to key factors of player
98 to separately conduct precise, interpretable, multi- 127 experience. Moreover, with the optimization target
99 dimensional online assessments of arbitrary LLM- 128 and data source, we developed RoleInstructor-1.5b,
100 based NPC systems' output. RoleInstructor is 129 RoleInstructor-3b, and RoleInstructor-7b.
101 designed to generate structured diagnostic 130 3. We conducted comprehensive experimental
102 feedback and actionable corrective directives, as 131 evaluations, the results of which demonstrate that
103 guidance for the LLM-based intelligent NPC 132 the RoleInstructor, combined with our proposed
104 systems to optimize and regenerate their outputs 133 metrics, significantly enhances the performance of
105 online through an internally integrated strategic 134 arbitrary LLM-based NPC systems, achieving
106 refinement process, without the requirement of 135 statistically significant improvements by an
107 retraining the LLM of an NPC system. The 136 average of 36.9% when deployed across
108 RoleInstructor has the characteristic of plug-and- 137 heterogeneous NPC systems.

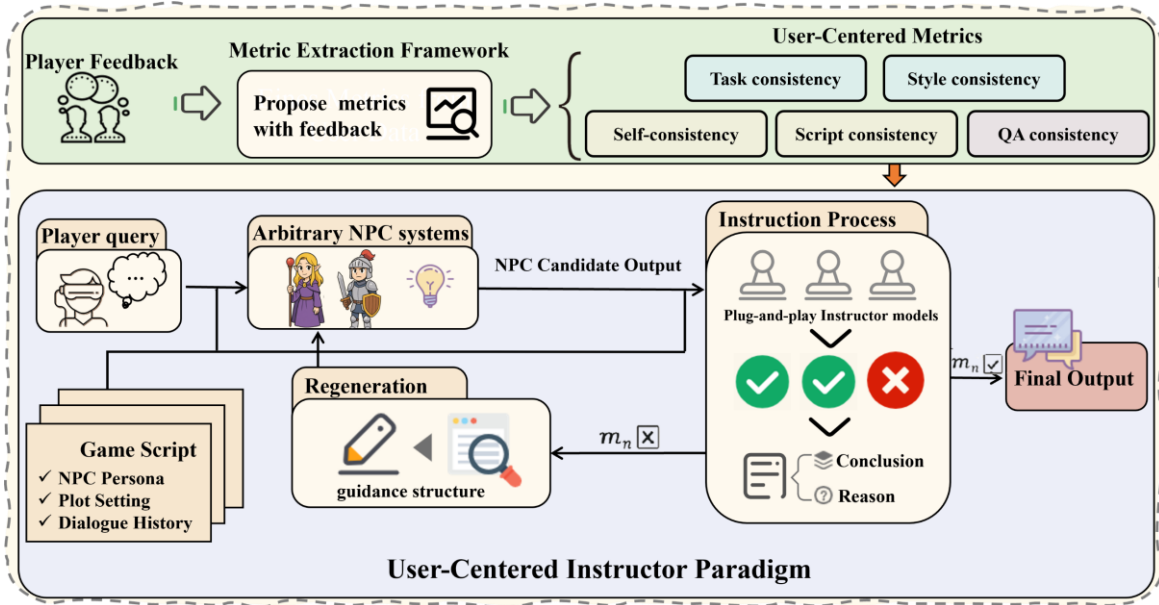


Figure 2: Schematic diagram of user-centered Instructor paradigm.

2 Related Work

Role-playing capability is a key factor in intelligent NPC systems. To enhance the role-playing ability of large language models, various methods have been proposed.

The first kind of work is dedicated to improving the performance of role-playing agents (RPAs) by parametric training. RoleLLM (Wang et al., 2024) significantly enhances models’ knowledge about their roles, outperforming retrieval augmentation-based methods when using noisy role profiles by instruction tuning. CharacterGLM (Zhou et al., 2024) used a tuning-based manner to develop the CharacterGLM model, and adopted direct preference optimization (Rafailov et al., 2023) to optimize the model. CoSER (Wang et al., 2025) developed advanced open role-playing LLMs built on LLaMA-3.1 models based on their datasets, enabling RPAs to simulate sophisticated cognitive and behavioral processes of humans. RPAs developed by parametric training only exhibited a convincing performance in specific metrics, while the effect is limited by model size, dataset, training method, and training budget when taking all valuable metrics into consideration. The second kind of work develops RPAs by prompt engineering (Li et al., 2023; Abdelnabi et al., 2024; Wu et al., 2024), relying on the instruction-following capability of LLMs, indicating the effectiveness of non-parametric instruction. However, the performance of PE-based RPAs is

limited to the quality and accuracy of prompts and natural language understanding ability of LLMs, thus, LLMs with more than 100 billion parameters and long-context prompts are always needed.

Evaluation of RPAs is another crucial technique for the intelligent NPCs system. INCHARACTER (Wang et al., 2024) validates the effectiveness of the LLM-based interviewer method in measuring RPAs personalities. CharacterEval (Tu et al., 2024) developed a role-playing reward model based on human annotations to evaluate RPAs with their proposed metrics. However, the proposal of these metrics does not directly originate from the data of interaction, thus, these metrics fail to directly capture whether NPC interactions truly meet player needs, leading to a misalignment between optimization efforts and the goal of enhancing genuine user experience.

3 User-centered Instructor

3.1 Paradigm Overview

To address the core challenges in current LLM-based intelligent NPC systems, we propose a user-centered Instructor paradigm, as shown in Figure 2. The central tenet of this paradigm lies in integrating arbitrary LLM-based intelligent NPC systems with our smaller, optimized Instructor model. Intelligent NPC systems are always endowed with advanced reasoning capability, generative capacity and metacognitive ability, empowered by LLMs with more than 10 billion of or even hundreds of billion parameters, while the Instructor model we

proposed, comprising a smaller optimized model with less than 10 billion of parameters, is capable of providing fine-grained guidance for intelligent NPC systems, continuously instructs the NPC systems in specific domains to regenerate contents with better performance.

The establishment of the user-centered Instructor method consists of two crucial modules, i.e., the metric extraction framework and the Instructor model.

Metric Extraction Framework MEF is the core innovative component for achieving the "user-centered" goal of our paradigm. It systematically analyzes and refines feedback data from players and generates the multi-dimensional evaluation metrics with a detailed definition and relevant interaction data. This mechanism ensures the training target of Instructor models is tightly coupled with users' actual experiences and core needs, perfectly making up for the deficiencies of the LLM-based intelligent systems in the early stage of training and optimization for other objective indicators.

Instructor Model Functioning as an independent evaluation and feedback hub, it utilizes an ensemble of specialized SLMs, each finetuned for a specific evaluation dimension that is derived from MEF and describable in natural language, to conduct precise, interpretable, and multi-dimensional online assessments of the intelligent NPC system's output. Based on this assessment, the Instructor models generate structured diagnostic evaluation and corrective directives to achieve fine-grained, real-time guidance for intelligent NPC systems. Moreover, the plug-and-play feature enables the instructor models to be integrated with any LLM-based NPC systems, achieving high flexibility and scalability.

After being trained with the dataset generated by MEF for a specific indicator, the Instructor model could be plugged into any other LLM-based intelligent NPC systems, forming a complete closed-loop optimization system. The NPC system generates NPC responses and behaviors, and the Instructor model evaluates these outputs and generates guidance signals based on user-centered metrics. The NPC system then adjusts its generation strategies accordingly to meet the requirements of users. Under this paradigm, the Instructor model, with the aid of MEF, can be custom-configured based on specific application

requirements and available computational resources.

The following sub-sections will detail the technical implementation of each component.

3.2 Metric Extraction Framework: Source of Optimization Targets and Data

Metrics for evaluating the performance of intelligent NPCs should be straightforwardly derived from players. We design a framework supporting multiple interactions between the player and arbitrary NPCs, as shown in Figure 3.

To avoid the possibility of unable to obtain indicators comprehensively and accurately due to the use of advanced NPC systems, we develop a conventional LLM-based NPC system for MEF, the implementation pseudo-code of our LLM-based NPC system is shown in Algorithm 1, where E is memory of NPC that relevant to current query and C is the final prompt, more details could be found in Appendix B.

Algorithm 1: NPC content generation algorithm

Input: User query Q , plot setting P , memory of NPC M , relevant setting of NPC I

Output: response of NPC system R_{can}

Function NPC_generate(Q, P, M, I):

$E \leftarrow \text{RetrieveRelevantMemories}(Q, M)$

$C \leftarrow \text{BuildPrompt}(Q, P, E, I)$

$R_{\text{can}} \leftarrow \text{LLM.generate}(C)$

return R_{can}

We invited more than 10 volunteers as players to participate in the metrics proposal with this framework. During the interaction between players and NPCs in each specific scenario specified by the game scripts to accomplish some tasks, some obvious shortcomings of the intelligent NPC's output could be pointed out. With the prompt composed of description and relevant dialogues, LLM is instructed to define the deficiency as a common circumstance and categorize it into a specific metric, which subsequently merges with the indicator with a similar definition (i.e., abort this indicator) or is inserted into the metrics library as a new indicator. The final metrics we considered in the next section were acquired by the top 5 most frequent issues proposed by players to evaluate the method proposed by others and develop Instructor models for optimization.

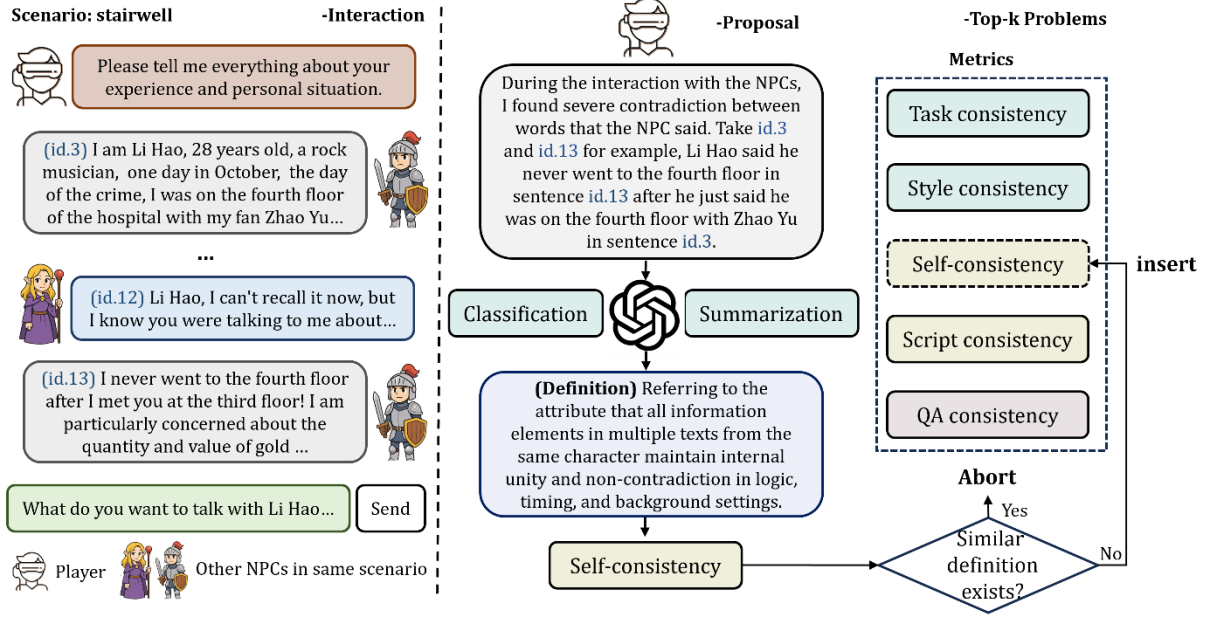


Figure 3: Workflow of Metrics Extraction Framework.

Task consistency. All behaviors from NPCs are motivated by their tasks, indicating that dialogues generated by NPCs and decisions made by NPCs should be beneficial to the accomplishment of their tasks.

Style consistency. Every NPC has its spoken characteristic, which is always set in the game script. Thereby, the dialogue of the NPC should align with the corresponding speaking habits.

Script consistency. The plot setting is the most essential part of a game. Thus, contradictions between the words of NPCs and the plot in the game script should be avoided as much as possible.

Self-consistency. NPCs are allowed to generate dialogue autonomously in case that is not depicted in the game script, considering their tasks. However, with the limitation of memory of LLMs, it is common for NPC to generate self-contradictory words, which seriously affects the player experience.

QA consistency. NPCs are likely generating irrelevant content of queries, caused by the long context in the prompt and hallucination. This indicator describes the correlation between the NPC's response and the question, indicating whether the NPC's response answers the question without irrelevant content.

3.3 Instructor Model: A Plug-and-Play Multi-Dimensional Evaluation Model

With the training target and dataset generated by MEF. The core responsibility of the Instructor

model trained from an SLM is to perform precise, interpretable, multi-dimensional assessment of candidate responses R_{can} generated by the LLM-based intelligent NPC systems, subsequently producing structured diagnostic evaluation and guidance, and to drive the LLM-based NPC system's adaptive regeneration. Given the fact that metrics proposed by MEF are always more than one, it is possible to train multiple SLMs for each metric or only one SLM with strong generalization performance for all metrics. LoRA (Hu et al., 2021) is adopted to train multiple adaptors for each metric in this paper. Specifically, for each key evaluation dimension $m_i \in M$ obtained from the MEF, we supervised finetuned a dedicated and efficient Instructor model L_{eval_i} . The optimization objective is the standard cross-entropy loss function, aiming to minimize the discrepancy between the model's predictions and the ground-truth labels derived from user-centered data generated by MEF.

For each dimension m_i , its evaluation process can be summarized as:

$$J_i, E_i = L_{eval_i}(R_{can}, C_{eval_i}, K_{eval_i}),$$

where R_{can} is the content of the intelligent NPC to be evaluated, C_{eval_i} is the necessary storyline context for evaluation, and K_{eval_i} is the criterion of metric assessment, while J_i and E_i are the judgment conclusion and the corresponding reason respectively, which highlight the deficiencies of R_{can} .

This SLM-group instruction architecture offers superior objectivity, interpretability, and efficiency compared to assessments by a single model with large parameters. Combined with retrieve-augmented generation and advanced prompt engineering, Instructor models provide a reliable and effective regeneration instruction for LLM-based NPC systems.

3.4 Instruction Injection: A Closed-loop Optimization Process

The whole workflow of the closed-loop Instruction-Regeneration algorithm is shown in Algorithm 2.

After the generation of a response by arbitrary NPC systems to be improved, Instructor models are called up for evaluation and instruction. We subsequently assemble all instructions J_i and E_i from the Instructors as J and E , finally inject and guide the intelligent NPC system to make targeted, iterative improvements through a feedback-driven prompt revision mechanism.

Algorithm 2: Instruction-Regeneration algorithm

Input: User query Q , NPC relevant context C , all evaluation indicators generated by MEF D

Output: Optimized NPC response R_{opt}

Function NPC_with_Instructor(Q, C, D):

//Step 1: NPC system generates response

$R_{can} \leftarrow \text{NPC_generate}(Q, C)$

//Step 2: Instructor models evaluate response for each dimension d in D :

$J_i, E_i \leftarrow \text{Instructor.evaluate}(R_{can}, d)$

//Step 3: Assemble all instructions

$J \leftarrow [J_i], i=0,1,2, \dots$

$E \leftarrow [E_i], i=0,1,2, \dots$

//Step 4: Strategic response optimization

while J indicates improvement of R needed:

$R_{opt} \leftarrow \text{NPC_regenerate}(Q, C, R_{can}, E)$

else:

$R_{opt} \leftarrow R_{can}$

return R_{opt}

4 Experiment

4.1 Experimental setup

We evaluate our proposed method through 5 metrics we proposed in Section 3 with 5 scripts we designed for testing. We set the player’s query as fixed text and record the output of the intelligent NPC through an automated process under 5 random scripts to acquire the training dataset and

test dataset, and each script outputs 36 rounds of dialogue. The average ratings of each model are calculated to indicate the reliability and validity of our proposed method. The reliability verification experiment is conducted with our basic LLM-based NPC system architecture and different instructors, while in the reliability verification experiment, the basic LLM-based NPC system architecture is expanded to different NPC systems from other work.

In our evaluation process, we recruited 12 annotators, each of whom plays a random script in the script library and gives feedback. Our experiments cover numerous LLMs, including Qwen2.5-1.5b-Instruct, Qwen2.5-3b-Instruct, Qwen2.5-7b-Instruct, Deepseek-R1, GPT-4o, CharacterGLM-12b, and Doubao-pro-32k. We named our instructor as RoleInstructor, which is trained with our dataset through supervised finetuning Qwen2.5 on Nvidia L20. Training details and training dataset are demonstrated in Appendices C and D, respectively.

4.2 Reliability of the Instructor Model

To assess the reliability of the RoleInstructor we proposed, we calculate the accuracy of instructors based on different models, i.e., Qwen2.5-1.5b-Instruct, Qwen2.5-3b-Instruct, Qwen2.5-7b-Instruct, Deepseek-R1, and GPT-4o.

The reliability of the instructor model is divided into two parts, i.e., reliability of reasoning and reliability of reasoning. Thus, we evaluate these two aspects of reliability in two different ways.

The accuracy is used to measure the difference between the judgment of the instructor model and human judgment, which determines whether the NPC system’s output meets the criteria at each metric, and the formula for accuracy calculation can be found in Appendix I.

Model	Accuracy	Run time (s)
Qwen2.5-1.5b-Instruct	0.5625	1.28
Qwen2.5-3b-Instruct	0.7813	1.84
Qwen2.5-7b-Instruct	0.8125	3.11
GPT-4o	0.9063	4.55
Deepseek-R1	<u>0.9297</u>	16.2
RoleInstructor-1.5b	0.8125	1.50
RoleInstructor-3b	0.8750	2.03
RoleInstructor-7b	0.9375	3.26

Table 1: Average accuracy results of all instructors based on different models.

Instructor	Without Instructor		With Instructor		Score Increment
	Rating	Variance (%)	Rating	Variance (%)	
Manual	0.6818	baseline	0.9844	baseline	0.3026
Qwen2.5-1.5b-Instruct	0.8750	28.34	0.7500	-23.81	-0.1250
Qwen2.5-3b-Instruct	0.5313	-22.07	0.5625	-42.86	0.0312
Qwen2.5-7b-Instruct	0.6406	-6.04	0.7344	-25.40	0.0938
GPT-4o	0.7813	14.59	0.9688	-1.58	0.1875
Deepseek-R1	0.6875	0.83	0.9375	-4.76	<u>0.2500</u>
RoleInstructor-1.5b	0.5937	-12.92	0.7188	-26.57	0.1251
RoleInstructor-3b	0.7656	12.29	0.8594	-12.70	0.0938
RoleInstructor-7b	0.7031	<u>3.12</u>	0.9531	<u>-3.18</u>	<u>0.2500</u>

Table 2: Results of task-consistency instructor compared with other models. Variance is calculated by taking the manual evaluation data as the baseline.

As shown in Table 1, with Qwen2.5 as instructor of its application in the field of dialogue interaction. without finetuning, the average accuracy of the These results indicate that RoleInstructors and instructor with larger parameters is superior to the human feedback are highly matched, smaller ones, but still has a considerable gap demonstrating the judgment reliability of compared with GPT-4o and Deepseek-R1. As a RoleInstructors. strong reasoning LLM, Deepseek-R1 performs As for reasoning reliability, we take task-consistently in terms of accuracy, even without consistency for example to compare the finetuning, yet the inference efficiency is relatively regenerating effect under the guidance of different low, constraining its application. models or manual instruction. As shown in Table 2, However, a significant improvement has been the ratings of task consistency of each method are achieved in the accuracy of RoleInstructors, even recorded, and the variances between each model exceeding Deepseek-R1 and GPT-4o, the models and the manual rating are calculated. In terms of the with much larger parameters, while maintaining variance, RoleInstructor-7b performs well among acceptable inference speed, ensuring the possibility the small models and was comparable to that of

Metrics	Method	Rating	
		Without Instructor	With Instructor
Self-consistency	CharacterGLM (Finetuning-based)	88.3%	95.7%
	Think Thrice (Prompt Engineering-based)	91.9%	95.5%
	Doubao-pro-32k (Close Source)	84.9%	92.9%
Task consistency	CharacterGLM (Finetuning-based)	10.9%	51.5%
	Think Thrice (Prompt Engineering-based)	43.8%	75.0%
	Doubao-pro-32k (Close Source)	48.4%	65.7%
QA consistency	CharacterGLM (Finetuning-based)	30.6%	61.1%
	Think Thrice (Prompt Engineering-based)	38.9%	55.6%
	Doubao-pro-32k (Close Source)	36.1%	75.0%
Style consistency	CharacterGLM (Finetuning-based)	8.4%	18.6%
	Think Thrice (Prompt Engineering-based)	32.9%	60.6%
	Doubao-pro-32k (Close Source)	21.7%	59.2%
Script consistency	CharacterGLM (Finetuning-based)	70.2%	88.3%
	Think Thrice (Prompt Engineering-based)	55.0%	90.1%
	Doubao-pro-32k (Close Source)	78.8%	88.5%

Table 3: Performance under different indicators of three mainstream methods that could be used in intelligent NPCs, with and without RoleInstructor-7b, the results in decimals were converted into percentages for better comparison.

larger models such as Deepseek-R1 and GPT-4o. Moreover, we calculate the improvement rate before and after the instruction by corresponding methods, i.e., the score increment. As Table 2 demonstrates, the score increment of RoleInstructor-7b is moderately lower than the results of manual instruction, but it obtains the best result among all models, indicating the strong reliability of our proposed RoleInstructor.

4.3 Validity of the Instructor Model

The RoleInstructor we proposed has the characteristic of plug-and-play, which is suitable for all methods proposed for intelligent NPCs, including finetuning, prompt engineering, etc. Thus, we deploy three advanced LLM-based intelligent NPC systems of others, and plug the RoleInstructor-7b into the systems, to prove the validity of RoleInstructor.

The results of the validity experiment could be shown in Table 3. We compared three different methods with and without RoleInstructor-7b, in terms of all 5 metrics we proposed in Section 3.

Without RoleInstructor-7b, take Think Thrice for example, QA consistency and style consistency both are below 50%, indicating that more than half of the interactions fail to meet the standard of players. However, all five metrics we proposed are improved with the help of the RoleInstructor-7b, proving that our plug-and-play instructors are

Ground truth in script: You searched floor by floor along the stairs and finally found the man's body.
Without Instructor Module
NPC: At some point, a woman told us that she had found a corpse, but we didn't find it at first. Later, we finally found the body in the stairwell.
Analysis: As the script suggested, the body was found on a certain floor of the building, not necessarily in the stairwell (could be anywhere).
With Instructor Module
NPC: I remembered that a woman told us that she had found a corpse, but we didn't find it at first. Later, we finally found the body on one of the building's floors.
Analysis: Change "in the stairwell" to "on one of the building's floors" to revise the fact that the corpse might appear on any floor, while not changing the core information of the original sentence (i.e., the corpse was found).

Table 4: An example of regeneration in terms of script consistency.

suitable for any arbitrary LLM-based NPC system. The example shown in Table 4 indicates the validity of RoleInstructor as well.

Besides, the metacognitive ability of LLMs used in NPC systems affects the improvement brought by the RoleInstructor, i.e., the score increments, relevant analysis could be found in Appendix J.

4.4 Ablation study

We conduct the ablation study by omitting the reasoning process of the instructor model, while only the judgment function is retained during training. And we only prompt the NPC system with a description of the corresponding indicators at the evaluation stage. The results are shown in Table 5.

Reasoning process benefits the accuracy of the instructor model. The reasoning process tends to guide RoleInstructor to make correct judgments. Without the reasoning process, the accuracy has decreased greatly, which is particularly evident in smaller models, such as RoleInstructor-1.5b.

Reasoning process enhances the regeneration of the LLM-based NPC systems. The reasoning process generates the guidance for the regeneration of the NPC systems. Without a valid reference from the reasoning process, the average score increment of regeneration decreased significantly.

Model	with Reasoning	without Reasoning
Average Accuracy		
RoleInstructor-1.5b	0.8125	0.5000
RoleInstructor-3b	0.8750	0.6875
RoleInstructor-7b	0.9375	0.9063
Average Score Increment		
RoleInstructor-1.5b	0.1925	0.1364
RoleInstructor-3b	0.2364	0.1795
RoleInstructor-7b	0.3893	0.3303

Table 5: Ablation study results.

5 Conclusion

We propose a user-centered plug-and-play method for intelligent NPC interaction, ensuring long-term consistency during interaction by extracting user-centered metrics directly from player feedback data and finetune plug-and-play models with this data. The experimental results indicate a substantial average improvement of 36.9% across three distinct NPC systems when evaluated against all relevant metrics.

510 Limitations

511 We outline the main limitations of our work as
512 follows:

513 **Language Applicability Limitations.** The current
514 study's recruitment of native Chinese-speaking
515 volunteers and reliance on Chinese-language
516 interaction scripts have resulted in a dataset
517 inherently shaped by Chinese linguistic structures
518 and cultural norms. Thus, the proposed
519 RoleInstructor performs better in the Chinese
520 context, future work should adopt multilingual
521 corpora to enhance cross-lingual adaptability.

522 **Domain Migration Limitations.** While our
523 proposed method has demonstrated promising
524 results in enhancing context-aware dialogue
525 generation for intelligent NPC interactions, its
526 generalizability to other conversational domains
527 remains unverified. Notably, the current method is
528 specifically tailored to gaming scenarios
529 characterized by goal-oriented dialogues and
530 predefined character personas. Critical domains
531 such as customer service, educational tutoring, or
532 healthcare consultation have not been
533 systematically evaluated. Future work should adapt
534 our proposed method to other fields and make
535 specific optimizations.

536 Ethical Considerations

537 The development and evaluation of our proposed
538 RoleInstructor method involved contributions from
539 human participants. Throughout the research
540 process, we implemented several measures to
541 ensure adherence to ethical guidelines. This study
542 involved human volunteers in two main capacities:
543 **Metric Construction.** Over 10 volunteers
544 participated as players interacting with NPC
545 systems to help identify and establish user-centered
546 evaluation metrics.

547 **Dataset Annotation.** We recruited 12 annotators to
548 label the dataset for evaluating the RoleInstructor
549 system.

550 Before their involvement, all volunteers and
551 annotators were fully informed about the research
552 objectives, the nature of their tasks, how their data
553 would be utilized, and their right to withdraw
554 unconditionally at any time. We obtained written
555 informed consent from all participants. All data
556 collected from volunteers and annotators,
557 including interaction logs and feedback, were
558 anonymized using participant IDs to protect their
559 privacy. The tasks assigned to volunteers and

560 annotators were designed to be non-intrusive,
561 primarily involving gameplay or interaction within
562 a simulated game environment and providing
563 feedback. All participation was entirely voluntary.

564 The data used in this study partially originates
565 from publicly available game script content on
566 online platforms. All names mentioned in this
567 paper are fictional characters from the script and
568 have no relation to real individuals. It is important
569 to clarify that any depictions of violent scenes
570 within the game scripts possibly referenced in this
571 paper are purely fictional and intended solely for
572 academic analysis; they do not represent or endorse
573 real-world violent acts.

574 We will make our code, dataset, and trained
575 models available. However, misuse of
576 RoleInstructor could generate sensitive or harmful
577 content, reflecting the violent, explicit, or
578 discriminatory traits of certain characters. This
579 work and its assets are strictly for research
580 purposes, and we oppose any malicious usage. We
581 will release our models and datasets exclusively for
582 research purposes. We believe our work meets
583 ACL's Code of Ethics.

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Appendices

A User Feedback Data Collection Methods in MEF

To provide the MEF with rich and authentic interaction data, we designed and implemented a dedicated user feedback data collection process. This process primarily involves the development of a diverse scenario library, an interactive text-based adventure game platform, and an integrated multi-dimensional feedback mechanism.

Initially, we constructed a corpus comprising 103 distinct scripts (All names mentioned in this paper are fictional characters from the script and have no relation to real individuals.). These scripts exhibit diversity in aspects such as character settings, core objectives, narrative types, and story backgrounds, aiming to stimulate the conventional LLM-based NPC system for MEF to generate a wide range of behaviors and dialogues when embodying various NPCs and to elicit feedback from users on these diverse performances. The statistical characteristics of the scripts are summarized in Table 6.

Based on the aforementioned script information, we developed a text-based adventure game system for human users to engage in gameplay. It's a conventional intelligent NPC system. In this system, users, guided by the objectives of their selected script, converse with NPCs from the script across various scenarios to gather information, make decisions, and ultimately attempt to complete predefined tasks. Additionally, we recruited 12

volunteers, all of whom possessed varied gaming experience.

To comprehensively capture the user experience, the system integrates multiple feedback mechanisms:

- **Continuous In-Game Open-Ended Feedback:** During gameplay, when users feel confused, identify issues, or have any other immediate reactions, they can, at any time, use the in-game feedback portal to record and submit their specific thoughts and encountered situations in text form.
- **Per-Turn NPC Dialogue Evaluation:** After each dialogue turn between an NPC and a user, the system promptly displays an evaluation interface, inviting the user to quickly evaluate the NPC's most recent utterance: Preference: A binary choice of "Like" or "Dislike". Task Relevance/Utility: Evaluating whether the dialogue was helpful in advancing the current game task (e.g., assessed using options such as "Very Helpful," "Helpful," "Neutral," "Unhelpful," "Detrimental").
- **NPC Performance:** Evaluating satisfaction with the NPC's performance in the dialogue. If dissatisfied, users select the reasons (e.g., consistency with the NPC's previous statements, known character profile, current task, or game world-view).
- **Post-Gameplay Comprehensive Feedback:** Exploration/Task Completion Score: The

Feature Dimension	Sub-categories / Description	Distribution / Notes
Primary Genre	Sci-fi, Republican Era Spy Thriller, Supernatural Horror, Fantasy, Wuxia	Sci-fi (33), Republican Era Spy Thriller (16), Supernatural Horror (26), Fantasy (10), Wuxia (18). Some scripts may blend multiple genre elements.
Core NPC	NPCs have specific storylines and timelines in their respective scripts. Their personalities are diverse, and their speech styles vary.	Core NPCs in the scripts include various archetypes such as informants, companions, enemies, and neutral characters.
Primary Player Task Type	Information gathering, uncovering the truth, concealing facts, problem solving, etc.	Task design aims to guide users to engage in diverse interaction strategies with NPCs.
Number of Key NPCs per Script	3, 4, 5, >5	3 (27), 4 (308), 5 (60), >5 (25). This refers to the number of NPCs closely related to the core tasks in each script
Estimated Interaction Complexity	Short (<20 dialogue turns), Medium (20-70 dialogue turns), Long (>70 dialogue turns)	Short (9), Medium (82), Long (12). Based on the estimated number of user-NPC dialogue turns required to complete core objectives.

Table 6: Overview of the Scenario Corpus (N=103).

game provides an objective score based on the user's breadth of exploration within the script, discovery of key information, and degree of task completion, serving as an objective measure to be contrasted with subjective evaluations. Detailed Written Feedback per NPC: After completing a script, users are invited to write evaluations for each core NPC they interacted with in that script. The evaluation primarily revolves around the following guiding questions: Overall evaluation of the NPC's performance. What specific NPC behaviors contributed to a positive game experience for you? What specific NPC behaviors caused confusion, seemed out of character, or disrupted your immersion?

All collected raw feedback data, including textual records, evaluation data, and game interaction logs, underwent anonymization to protect user privacy. The textual data underwent preliminary formatting normalization and correction of typographical errors. Finally, we

utilized a LLM to summarize and categorize the information within this feedback data, derived from real interactions, into a specific metric. Table 15 is an example of our prompt engineering approach for processing this data, Table 7 is sample raw feedback data.

B Conventional Intelligent NPC System for MEF

The conventional intelligent NPC system for MEF, through a context aggregation function, integrates diverse heterogeneous information sources into a structured prompt to drive the LLM to generate responses. The integrated information sources primarily include:

- **Agent Profile:** Character name, character role, core task, character script/backstory, and incident day timeline. This provides the LLM with foundational information about the NPC's identity, objectives, and known

User ID	User Feedback	Metric Definition	Metric Name
01	During my interaction with the NPC, I noticed a serious contradiction in what the NPC said. For instance, in interactions with IDs id.3 and id.13, the NPC Li Hao stated in id.13 that he had never been to the fourth floor, yet in id.3 he had just mentioned being on the fourth floor with Zhao Yu.	This refers to the property whereby all informational elements within a given textual system maintain internal coherence and non-contradiction across aspects such as logic, facts, chronology, character actions, and background settings.	Self-Consistency
04	The NPC, Landlady Wang, is characterized as a kind and amiable grandmother who always speaks in a gentle, unhurried, and caring manner. However, in dialogue id.63, she abruptly answered my question in a very stiff and mechanical tone. I initially thought it was a different NPC, as it was entirely inconsistent with her usual speaking style.	Each NPC possesses unique speech characteristics. Consequently, an NPC's dialogue should remain consistent with their established speaking habits. The NPC's performance should be logically coherent with their character profile.	Characteristic Consistency
11	The NPC Jiang Wenxue's task is to explore the secrets of the 'Ancient Ritual' with me. Additionally, fragments of a forbidden book in his background are described as containing relevant clues. However, when we were jointly conversing with the librarian, he asked the librarian about unrelated topics and failed to provide the key information about the 'Ancient Ritual' that he was supposed to introduce. This caused my game progress to stall.	All of NPCs' actions should be driven by their designated tasks. This implies that the dialogues generated and decisions made by the NPC should contribute to the fulfillment of these tasks.	Motivation Consistency

Table 7: Samples of Raw Feedback Data

background. This is presented in a structured format at the beginning of the prompt.

- **Game World Knowledge:** Game rules and script background. This ensures the NPC's statements and actions align with the game's fundamental settings and rules. This serves as high-level instructions or background information, typically positioned after the agent profile.

- **Retrieved Memories:** Historical dialogue or event snippets retrieved from a FAISS vector database that are most relevant to the current observation. This provides dynamic, contextually relevant memory support for the NPC, enabling it to answer questions or formulate reactions. These are formatted and inserted into the prompt, typically labeled as 'content from previous dialogues relevant to the ongoing in-game conversation.' When acting as the most direct instructional or query component within the prompt, such retrieved content is labeled 'ongoing in-game conversation.'

- **Base Interaction Context:** Current observation/input and recent dialogue history. This constitutes the core content to which the LLM must directly respond, providing immediate conversational context to help the LLM understand the current dialogue flow.

Prompt Construction Strategy Summary: Role-playing instructions are prioritized. Initially, the NPC's identity, tasks, and core background are clearly defined. Next, game rules and the overall script context are provided. Subsequently, dynamic information most relevant to the current interaction is integrated, including: retrieved memories (RAG), recent dialogue history, current direct input/observation, and interlocutor relationships, etc. The NPC's core task is then re-emphasized to ensure its actions are goal-oriented. Table 17 shows the general skeleton of the constructed prompt template. Please note that the actual prompt is dynamically populated based on different inputs.

Its core response generation process is summarized as follows:

$$R_{can} = \mathcal{L}(\Psi(Q, P, I, E))$$

where:

- Q : The current user query.
- P : Represents the base interaction context, including chat history H_{chat} , and the game world's state and rules W_{state} .
- E : Denotes relevant historical memory segments retrieved from the NPC's memory M via RAG, which is performed based on the user's query Q .
- I : Is the NPC's static profile, containing its role background, mission objectives, and character timeline.
- $\Psi(\cdot)$: A Context Aggregation and Prompt Engineering function. It integrates the aforementioned heterogeneous information sources into a structured input prompt P , designed to stimulate the In-Context Learning (ICL) efficacy of the underlying LLM.
- $\mathcal{L}(\cdot)$: The pre-trained LLM, which generates an initial response R_{can} .

In our experiments, to ensure the robustness and generality of the results, we tested a variety of different foundational LLMs, including: claude-3-7-sonnet-20250219, deepseek-reasoner, doubao-1-5-pro-256k-250115, gpt-4o, grok-3, hunyuan-t1-20250403, MiniMax-Text-01, and qwen-max-2025-01-25. API calls for all models were integrated via the Langchain framework.

C Dimension-Specific Evaluation Model Dataset

To implement multi-dimensional evaluation based on an "observer" paradigm within the Instructor Model, we trained specialized evaluation models for each evaluation dimension refined from the MEF, namely: Task Consistency, Style Consistency, Script Consistency, Self-Consistency, and QA Consistency.

The training data construction process is as follows:

- **Diversified Candidate Response Generation:** Initially, we leveraged the capabilities of the conventional LLM-based NPC system for MEF to generate foundational evaluation data. To ensure the richness and coverage of

the training data, for each predefined interactive scenario context (including user input, dialogue history, NPC character profiles, etc.), we prompted N=8 industry-leading LLMs with diverse architectures to generate their respective candidate responses. This multi-model generation strategy aims to capture a broader spectrum of output quality (ranging from excellent to those with various flaws), thereby providing diverse examples for the training of the evaluation models. This process yielded over 10,000 pieces of training data, covering a multitude of game scenarios and NPC archetypes.

- **LLM-Aided Pre-annotation and Human Expert Correction:** Subsequently, we employed a two-stage process of "LLM pre-annotation + human expert correction" to obtain high-quality annotated data. **Pre-annotation:** A highly capable large language model (e.g., DeepSeek-R1) was utilized as an initial annotator. For each candidate response generated by the conventional LLM-based NPC system for MEF, this LLM automatically produced preliminary evaluation results and the reasoning behind these judgments, based on our five defined core evaluation dimensions. **Human Correction and Validation:** A team of specially trained human annotation experts conducted rigorous reviews and corrections of the preliminary evaluation results generated by the LLM. The experts corrected all inaccurate or erroneous evaluation conclusions and their corresponding justifications. To ensure annotation consistency and reliability, we implemented multiple rounds of cross-validation and consistency calibration meetings.

Through the process described above, we constructed data pairs for each evaluation dimension. Each pair contains: scenario context, an NPC candidate response, the dimensional evaluation result, and the rationale for the evaluation, forming a training set for supervised fine-tuning. Table 10 is a sample from the training set for Task Consistency.

The data format typically consists of an "instruction" (i.e., a description of the evaluation task and the role assigned to the model), an "input" (containing the NPC response to be evaluated and

necessary context, such as the NPC's role, its task description, and its specific utterances, etc.), and an "output" (i.e., the evaluation conclusion and rationale that the model should learn to generate).

D Instructor Model Fine-tuning Details

We utilized the LLaMA-Factory framework for SFT of selected foundational SLMs to train a specialized evaluation model for each evaluation dimension. The objective is to enhance the accuracy of the evaluation feedback provided by the Instructor Model and ensure it aligns with expectations.

We utilized SLMs as the base models for fine-tuning and employed LoRA techniques. During the fine-tuning process, the vast majority of the pre-trained model's weights were kept frozen. Only small-scale, decomposable low-rank matrices were introduced and trained within the attention layers of the Transformer architecture. The training process utilized the standard Cross-Entropy Loss function with the AdamW optimizer.

Hyperparameter Name	Value
Learning Rate	5e-5
Batch Size	2
Epochs	10
Training Accuracy	bf16
LoRA Rank (r)	8
LoRA Alpha (alpha)	16
LoRA Dropout	0

Table 8: Fine-tuning Hyperparameters

E Sample Evaluation Prompt Template

During its evaluation process, the Instructor Model constructs a specific prompt for each dimension in Table 21. This prompt integrates the candidate response, relevant context, and evaluation criteria. The resulting integrated prompt is then input to the corresponding fine-tuned evaluation model.

In practice, these templates are dynamically populated with specific NPC responses, contextual information, and corresponding evaluation criteria, and are subsequently input to their respective fine-tuned evaluation models to obtain structured evaluation results and their rationales.

Field	Data Content
Example 1	Instruction: "You are a user skilled at inferring a character's specific task. Based on the character's narrative content provided below, determine if their actions/dialogue indicate that their task is the one specified in the 'Task content' (provided in the Input section). Judgment rule: Make your judgment based solely on the provided context. Output format: Reason: [Elaborate on the reason here] Conclusion: [Output Yes or No here]"
	Input: "Character: Cui Fan Task content: Ascertain whether Coroner Li has a secret deal with Boss Zhao. Narrative content: "Wang Xingjia, you mentioned that on the day of the incident, you went to see Boss Yang and had a conversation with him. Could you describe the content of your conversation in detail, and did anything unusual happen afterwards? Specifically regarding Coroner Li, was she mentioned in your dialogue?""
	Output: "Reason: When Cui Fan questioned Wang Xingjia, he specifically emphasized "whether Coroner Li was mentioned in the dialogue" and focused on "anything unusual" and "conversation content." This targeted questioning indicates his suspicion about a potential connection between Coroner Li and Boss Zhao. His core task likely involves investigating whether an undisclosed transaction occurred between them. Conclusion: Yes"
Example 2	Instruction: "You are a user skilled at inferring a character's specific task. Based on the character's narrative content provided below, determine if their actions/dialogue indicate that their task is the one specified in the 'Task content' (provided in the Input section). Judgment rule: Make your judgment based solely on the provided context. Output format: Reason: [Elaborate on the reason here] Conclusion: [Output Yes or No here]"
	Input: "Character: Jiang Qiantai Task content: Ascertain whether the person you are questioning plans to murder Boss Zhao. Narrative content: "Coroner Li, regarding those unexplained incomes shown in your phone's bank transaction records, can you explain them? Specifically, that large sum credited two years ago before your marriage – does this have any connection to Boss Zhao or this current incident?""
	Output: "Reason: Jiang Qiantai's questions focus on whether Coroner Li's bank transaction records and the large sum credited two years ago are related to Boss Zhao or "this current incident," but do not directly mention a "murder plan" or related actions. Their task seems more inclined towards investigating potential links between financial anomalies and the incident/individuals, rather than explicitly aiming to "ascertain a murder plan." Conclusion: No"
Example 3	Instruction: "You are a user skilled at inferring a character's specific task. Based on the character's narrative content provided below, determine if their actions/dialogue indicate that their task is the one specified in the 'Task content' (provided in the Input section). Judgment rule: Make your judgment based solely on the provided context. Output format: Reason: [Elaborate on the reason here] Conclusion: [Output Yes or No here]"
	input: "Character: Wang Xuan Task content: Ascertain whether the person you are questioning knows the secret of the elevator. Narrative content: "Zhao Yu, does your notebook record the different operating rules for the elevator in June versus October? Specifically, regarding the influence of the 'xue' on floor space, was the 'secret passage' method, revealed when you were drunk in June, achieved by silently reciting real and mirror floor numbers to enable the gold nugget's spatial transfer, which resulted in me arriving at the sixth-floor mirror space instead of the real top floor when I operated the elevator in October?""
	Output: "Reason: Wang Xuan directly asks Zhao Yu if their notebook records different elevator operating rules, specifically mentioning the influence of the "xue" on floor space and details of the "secret passage" method. Her questions clearly point to whether Zhao Yu knows the elevator's secrets (such as the mirror floor mechanism), and further verifies this by asking about it being "revealed when drunk in June." This type of questioning content and method directly correlates with the task objective "ascertain whether the person you are questioning knows the secret of the elevator." Conclusion: Yes"

Table 9: English Examples of Training Data for Task Consistency Evaluation Model

Field	Data Content
Example 1	Instruction: "你是一个善于猜测某位角色的具体任务的玩家，请你根据下述角色的叙述内容：\n 是否能猜测出该角色的任务有下述任务：\n 判断规则为：只根据现有上下文直接判断\n 输出格式为：\n 原因：此处阐述原因\n 结论：此处输出是或否。",
	Input: "角色：崔凡\n 任务内容： 查明李法医是否和赵老板有暗中交易。 \n 叙述内容：王兴家，你说在案发当天，你曾经去找过杨老板，并和他有过沟通。能否详细描述一下你们谈话的内容，以及在那之后有没有发生任何异常？特别是关于李法医的事情，她有没有在你们的对话中被提及？"
	Output: "原因：崔凡在询问王兴家时，特别强调了“李法医是否在对话中被提及”，并关注“异常”和“谈话内容”。这种针对性提问表明他对玲船医李法医与赵老板之间的潜在关联存在怀疑，核心任务很可能涉及调查两人是否存在非公开交易关系。 \n 结论：是"
Example 2	Instruction: "你是一个善于猜测某位角色的具体任务的玩家，请你根据下述角色的叙述内容：\n 是否能猜测出该角色的任务有下述任务：\n 判断规则为：只根据现有上下文直接判断\n 输出格式为：\n 原因：此处阐述原因\n 结论：此处输出是或否。",
	Input: "角色：蒋前台\n 任务内容： 查明你提问的对象是否计划杀害赵老板。 \n 叙述内容：李法医，关于你手机中的银行交易记录显示的那些来路不明的收入，你能解释一下吗？特别是那笔在两年前婚前的大额进账，这与赵老板或这次事件有任何关联吗？",
	Output: "原因：蒋前台的提问集中在李法医的银行交易记录和两年前的巨额进账是否与赵老板或“这次事件”有关联，但未直接提及“杀人计划”或相关行为。其任务更倾向于调查资金异常与事件/人物的潜在联系，而非明确指向“查明杀人计划”。 \n 结论：否"
Example 3	Instruction: "你是一个善于猜测某位角色的具体任务的玩家，请你根据下述角色的叙述内容：\n 是否能猜测出该角色的任务有下述任务：\n 判断规则为：只根据现有上下文直接判断\n 输出格式为：\n 原因：此处阐述原因\n 结论：此处输出是或否。",
	input: "角色：王萱\n 任务内容： 查明你的提问对象是否知道电梯的秘密。 \n 叙述内容：赵雨，你的笔记本中是否记载了电梯在六月与十月运行规则的不同？特别是关于'穴'对楼层空间的影响，是否在六月醉酒时透露的'暗度陈仓'之法，正是通过默念真实楼层与镜像楼层实现金块空间转移，导致我在十月操作电梯时实际抵达的是六层镜像空间而非真实顶层？",
	Output: "原因：王萱直接询问赵雨是否在笔记本中记载了电梯运行规则的不同，并具体提到“穴”对楼层空间的影响以及“暗度陈仓”之法的细节。她的提问明确指向赵雨是否掌握电梯秘密（如镜像楼层机制），且通过“是否在六月醉酒时透露”进一步验证对方是否知情。这种提问内容和方式直接关联“查明提问对象是否知道电梯的秘密”这一任务目标。 \n 结论：是"

Table 10: Chinese Examples of Training Data for Task Consistency Evaluation Model

F Structured Diagnostic Feedback and Feedback-Based Prompt Revision Mechanism

One of the core functions of the Instructor Model is to generate structured diagnostic feedback. This feedback includes evaluation results for the candidate response across various assessment dimensions, along with specific explanations. When the evaluation result for a particular dimension indicates that a response fails to meet the required standards, the corresponding explanation serves as key information to guide the conventional LLM-based NPC system for MEF in performing iterative revisions. The template example in Table 19 demonstrates how a new prompt is constructed to guide the conventional LLM-based NPC system for MEF in making corrections when the Instructor Model identifies that a previously generated answer or question does not meet requirements.

Notes:

- {origin_query}: The answer or question previously generated by the conventional LLM-based NPC system for MEF that was deemed unsatisfactory by the Instructor Model.
- {reasons_for_no}: The specific negative evaluations and accompanying reasons provided by the Instructor Model regarding {origin_query} across one or more assessment dimensions.

The core instruction requires the LLM, based on an understanding of these shortcomings, to perform "targeted revisions" on the original content rather than completely regenerating it. This approach helps maintain the stability and efficiency of the iterative process. Through such explicit feedback integration and revision instructions, the conventional LLM-based NPC system can learn more effectively from its mistakes, progressively optimizing its output quality to meet all predefined evaluation metrics.

G Description of Experimental Game Environment

The experiments and data collection for this study were conducted on a custom-built text adventure game interaction platform, integral to the MEF

framework. On this platform, players, within predefined story scenarios (i.e., "scripts"), engage in multiple rounds of natural language dialogue with NPCs driven by the conventional LLM-based NPC system. The objective is to acquire information, make decisions, and strive to complete their respective tasks. This design aims to simulate the complex interactions between players and intelligent NPCs in real games, thereby providing a rich and authentic corpus of interactions for the validation of the MEF and the training of the Instructor Model. The following sub-section details a representative script used, outlining its scenario settings and NPC characteristics.

H Sample Scenario Script : "The Accursed Place and Lingerings Souls"

Scenario Brief : "The Accursed Place and Lingerings Souls" is a role-playing script with elements of mystery, horror, and puzzle-solving. The story unfolds within an abandoned building, the "Tonghua Middle Road Private Xieji Hospital." This hospital has become a forbidden zone as it was encroached upon by "The Accursed Place"—a confluence of the city's ineradicable sins and Curses—which easily absorbs trespassers, turning them into "Lingerings Souls." The core event of the script revolves around a murder case that occurs inside the hospital. An individual with special abilities, while dealing with "The Accursed Place" in the hospital from the outside, discovers a body that was clearly a victim of homicide. Due to the influence of "The Accursed Place," the four core characters involved in the case have all lost crucial memories, which now only exist in fragmented pieces. Players participate by cooperatively playing these four characters (Zhang Chen, Zhao Yu, Wang Xuan, Li Hao), using dialogue, reasoning, and clue analysis to collectively piece together the truth behind the murder and the secrets related to "The Accursed Place" and hidden gold bars.

Key NPC Profiles : This script features four core interactive characters. Each character possesses a unique background, and set of tasks. To more clearly and objectively measure and evaluate the role-playing capabilities of LLMs along the 'Style Consistency' dimension in our experiments, we designed relatively unique and regularized linguistic styles for the NPCs in our scripts. This approach makes it straightforward to discern

whether an NPC adheres to its predetermined style, which not only facilitates consistent judgment by human annotators but also provides clearer, quantifiable learning targets and evaluation criteria for automated assessment models. Compared to more naturalistic, yet potentially more ambiguous, everyday colloquial styles, such explicit stylistic constraints allow us to more precisely isolate and examine the model's ability to imitate specific linguistic patterns. Consequently, this offers a more direct indication of the extent to which a model can successfully portray a character with distinct linguistic features. Concurrently, this serves as a stress test for their role-playing abilities, examining the model's robustness in maintaining character-specific traits under defined constraints.

NPC 1: Zhao Yu

- Role in Scenario: Core participant, former hospital nurse, now a bodybuilding coach, one of the aspirants for the gold bars.
- Background Sketch: Formerly a nurse at Xieji Hospital, she fortunately escaped when "The Accursed Place" emerged. She later switched careers to bodybuilding and, out of curiosity, researched "The Accursed Place," learning about the existence zones of Lingerings Souls. Motivated by greed for the rumored gold bars in the hospital, she teams up with Li Hao to return.
- Linguistic Style: Habitually ends each sentence with "indeed" and frequently uses idioms in her speech.
- NPC's Task in Scenario: To find out what Li Hao said to her in the stairwell; to ascertain if the person she subsequently questions eventually obtained the gold bars.

NPC 2: Wang Xuan

- Role in Scenario: Zhao Yu's best friend. She was once invited to a bar by Zhao Yu where she secretly overheard Zhao Yu's conversation with Li Hao about the gold bars in the hospital. She also peeked into Zhao Yu's private notebook, learning all the secrets about "The Accursed Place,"

and thus decided to venture to the hospital alone to find the gold.

- Linguistic Style: Habitually ends each sentence with "ah" and exclusively uses "renjia" when referring to herself.
- NPC's Task/Goal in Scenario: To discover the secrets contained in Zhao Yu's notebook; to find out if the person she subsequently questions knows the secret of the elevator.

NPC 3: Li Hao

- Role in Scenario: Core participant, a rock musician, the only one with clues to the specific location of the gold bars.
- Background Sketch: A rising star in the rock music scene, aware of the gold bars hidden in the hospital but not the secrets of "The Accursed Place." After exchanging information with his fan Zhao Yu at a bar, they decide to cooperate in searching for the gold. He is suspicious of Wang Xuan's appearance at the bar and whether she overheard their conversation. He is a close friend of Zhang Chen.
- Linguistic Style: Every sentence is an exclamatory sentence, and he exclusively uses "benren" when referring to himself.
- NPC's Task in Scenario: To find out what topic he brought up when he was in the elevator with Zhao Yu; to ascertain the purpose of the person he subsequently questions for coming to the hospital.

NPC 4: Zhang Chen

- Role in Scenario: Core participant, a minor internet celebrity in e-commerce, has some research knowledge about "The Accursed Place."
- Background Sketch: Previously interested in "The Accursed Place," he investigated Xieji Hospital (without an on-site visit) and discovered that the first, fourth, and sixth floors are free of Lingerings Souls. He was later informed of certain secrets by his friend Wang Xuan, prompting their joint visit to the hospital.

- Linguistic Style: Intersperses each sentence with an English word and ends each sentence with "right"

The evaluation Accuracy of the Instructor Model is defined as:

$$\text{Accuracy} = \frac{\sum_{j=1}^N \mathbb{I}(E_{\text{instructor}_j} = E_{\text{human}_j})}{N}$$

- NPC's Task in Scenario: To determine if Wang Xuan was the person who strangled him; to find out what time the person he subsequently questions arrived at the hospital.

Where:

- N is the total number of candidate responses being evaluated.
- $E_{\text{instructor}_j}$ is the evaluation result from the Instructor Model for the j -th candidate response on a specific evaluation dimension.
- E_{human_j} is the evaluation result from human experts for the j -th candidate response on the same dimension (considered as the ground truth).
- $\mathbb{I}(\cdot)$ is the indicator function, where $\mathbb{I}(\text{condition})$ equals 0 if the condition is true, and 1 otherwise.

We have included excerpts from the interaction between real players and NPCs in "The Accursed Place and Lingerin

I Detailed Evaluation Metrics and Calculation

User-centered Instructor paradigm, through the Instructor Model, conducts multi-dimensional evaluations of candidate responses generated by the conventional LLM-based NPC system. These dimensions are primarily derived from key interaction quality requirements distilled from user feedback by the MEF. As detailed in the main paper, the MEF identifies five core consistency dimensions crucial for the game NPC interaction experience: Task Consistency, Style Consistency, Script Consistency, Self-Consistency, and QA Consistency. Specialized evaluation SLMs within the Instructor Model evaluate initial responses against each of these dimensions, outputting discrete results and natural language explanations.

Specific examples of evaluation results from the Instructor Model for these five consistency dimensions will be detailed in Table 14 and Appendix N.

To measure the accuracy of the judgments made by the specialized evaluation models within the Instructor Model, the alignment of their evaluation results with those of human experts, we calculate their evaluation Accuracy. The specific calculation method for this accuracy is as follows:

Let the judgment result of the Instructor Model for a candidate response $R_{\text{candidate}}$ on a specific evaluation dimension be denoted as $E_{\text{instructor}_j}$, and the judgment of human experts (considered as the ground truth) be denoted as E_{human_j} . Suppose there are N candidate responses to be judged.

Simply put, this accuracy represents the proportion of samples for which the Instructor Model's evaluation aligns with human judgments, out of the total number of samples evaluated. This metric serves to validate the reliability of the specialized evaluation models within the Instructor Model.

J Comparative Performance Analysis of Foundational LLMs in the User-centered Instructor Paradigm

The final output quality of the core generation engine within the User-centered Instructor paradigm largely depends on the inherent capabilities of the underlying LLM it employs. To investigate the performance of different foundational LLMs within the User-centered Instructor framework and to provide a reference for model selection in future research, we conducted a comparative experiment. In this experiment, multiple mainstream LLMs were individually configured as the core generation component of the conventional LLM-based NPC system. They generated candidate responses by leveraging the conventional LLM-based NPC system's RAG and multi-dimensional context integration capabilities.

Base LLM	Task Consistency	Style Consistency	Script Consistency	Self-Consistency	QA Consistency
claude-3-7-sonnet20250219	0.73	0.77	0.72	0.87	0.67
deepseek-reasoner	0.69	0.78	0.51	0.76	0.64
doubao-1-5-pro-256k-250115	0.56	0.56	0.75	0.85	0.36
gpt-4o	0.41	0.6	0.68	0.88	0.61
grok-3	0.61	0.71	0.73	0.84	0.69
hunyuan-t1-20250403	0.52	0.51	0.45	0.70	0.24
MiniMax-Text-01	0.19	0.39	0.68	0.85	0.34
qwen-max-2025-01-25	0.39	0.64	0.66	0.86	0.72

Table 11.: Performance Comparison of Different Base LLMs in the conventional LLM-based NPC system for MEF. The values represent average accuracy; a score closer to 1 indicates better performance.

These responses were subsequently evaluated based on the five core consistency dimensions defined by the MEF.

Notably, this experiment assesses the baseline performance of each foundational LLM within the conventional LLM-based NPC system structure, without real-time feedback or correction from the Instructor Model. The aim is to measure their raw capabilities during the initial information integration and response generation phases of the User-centered Instructor paradigm. The evaluations were performed by the dimension-specific evaluation SLMs, which were specially trained for the Instructor Model, to ensure the consistency and objectivity of the assessments.

Specific Observations are as follows:

- **Leading Performers (Overall):** claude-3-7-sonnet-20250219 and grok-3 demonstrated relatively balanced and strong overall performance across the five dimensions. claude-3-7-sonnet excelled in Task Consistency (0.73), Style Consistency (0.77), and Self-Consistency (0.87), achieving the highest score in the latter. grok-3 also achieved competitive scores in Style Consistency (0.71), Script Consistency (0.73), and QA Consistency

(0.69), with its metrics being relatively more uniform.

- **Strengths in Specific Dimensions:** Self-Consistency: All models generally performed well in this dimension (most above 0.70, with gpt-4o reaching the peak at 0.88), suggesting that current mainstream LLMs are relatively capable of maintaining internal statement consistency within shorter contextual windows. gpt-4o (0.88), claude-3-7-sonnet (0.87), and qwen-max (0.86) led in this aspect. Style Consistency: deepseek-reasoner (0.78) and claude-3-7-sonnet (0.77) performed best in this dimension, indicating their stronger ability to emulate specific speaking styles. Task Consistency: claude-3-7-sonnet (0.73) and deepseek-reasoner (0.69) were frontrunners here, suggesting their generated responses better served the predefined NPC task objectives. Script Consistency: doubao-1-5-pro-256k (0.75) and grok-3 (0.73) scored highly in this dimension, implying better adherence to established scripts and storylines. QA Consistency: qwen-max-2025-01-25 (0.72) and grok-3 (0.69) excelled here,

indicating a higher relevance and accuracy of their generated answers to player questions.

- **Models with Notable Performance Fluctuations:** While gpt-4o achieved the top score in Self-Consistency (0.88), its performance in Task Consistency (0.41) and Style Consistency (0.60) was comparatively weaker than some other models. doubao-1-5-pro-256k performed well in Script Consistency (0.75) and Self-Consistency (0.85) but was significantly lower in QA Consistency (0.36). hunyuan-t1-20250403 and MiniMax-Text-01 scored relatively low across multiple dimensions, particularly in QA Consistency (0.24 and 0.34, respectively) and Task Consistency (MiniMax-Text-01 at 0.19), indicating Appendices raw output quality within this framework.
- **Potential Trade-offs:** The data suggests that no single model achieves absolute superiority across all five dimensions. For instance, deepseek-reasoner excelled in Style Consistency but performed less impressively in Script Consistency (0.51) compared to others. This indicates that the choice of a base LLM might involve trade-offs depending on the specific consistency aspects prioritized for a particular NPC role or game scenario.

This comparative experiment lucidly demonstrates the distinct inherent capabilities of various LLMs, specifically in the absence of real-time guidance. These baseline performance data serve as a critical reference for selecting the foundational LLM for the conventional LLM-based NPC system. Furthermore, these results underscore the necessity and considerable potential for the Instructor Model to apply real-time feedback and corrections; even models exhibiting strong initial performance have scope for enhancement in certain dimensions, while those with comparatively weaker baselines would more pressing require the Instructor Model's intervention to ensure the quality of the final output. This dataset also lays the groundwork for subsequent research aimed at assessing the optimization efficacy of the Instructor Model

across diverse base LLMs. We also compared the metacognitive capabilities of the models across different approaches. As shown in Table 12, Think Thrice obtains the highest increment, attributing to the great metacognitive ability of GPT-4o.

Method	Base Model	Increment
CharacterGLM	ChatGLM2	35.6
Think Thrice	GPT-4o	38.1
Doubao	Doubao-pro-32k	37.1

Table 12: Comparative of Metacognitive Capability

K Instructor Model Evaluation Examples for Consistency Dimensions

This section demonstrates how the specialized evaluation SLMs within the Instructor Model evaluate candidate responses against the five core consistency dimensions. Each evaluation yields a discrete result and a natural language explanation, which together form part of the diagnostic feedback. The gameplay script context used for illustration is from the script titled 'Stranger'. For Style Consistency, Script Consistency, Self-Consistency, and QA Consistency, a single example entry will be presented for each in Table 14. In contrast, for Task Consistency, a more extensive set of detailed evaluation examples from a single gameplay interaction log will be shown in Appendix N.

Metric	Sample Evaluation Result
Style Consistency	Speaker: Shi Yu Response: "The second time was before the sensory deprivation segment; as per the game's procedural requirements, we decisively turned off all the lights and air conditioning, ah. As for why Xue Cai moved to the seat next to Wu Jing, I was indeed dumbfounded at the time, ah." Style 1 Evaluation: Yes Style 2 Evaluation: Yes Reason: The first sentence contains the idiom "decisively" and ends with "ah." The second sentence contains the idiom "dumbfounded" and ends with "ah." Both sentences meet the criteria for Style 1 (each sentence includes an idiom) and Style 2 (each sentence ends with "ah").
Self Consistency	Character: Shi Yu Narrative Content: "When I returned, I found the curtain that was originally hanging in front of the compartment door was missing. Were you sharp-eyed and quick-handed enough to see who put it away, ah? Also, do you know if Hao Zhuang touched or took away the item called 'Killer Trump Card' when he was in the compartment? This question is crucial, ah. This information could be like timely assistance and very important for uncovering the truth of the case, ah!" Related Narratives: Document(id='4b705b73-01d3-4348-b15b-30fd5a1baa0c', metadata={}, page_content="By the way, Mr. Wu Jing, did you notice if the curtain in front of the compartment door vanished into thin air, ah? When I returned, I found the curtain missing; I guess Qin Dong put it away, ah. Also, do you know if Hao Zhuang casually pilfered the 'Killer Trump Card' item when he was in the compartment, ah? These two points are very important for us to reconstruct the truth of the incident, ah!") Document(id='85747d7e-4276-431c-8707-9866062ab38f', metadata={}, page_content="As for whether Hao Zhuang took the 'Killer Trump Card' item while in the compartment, I know nothing at all, because I went to the washroom at that time and wasn't observant enough (lit. 'didn't have eyes looking in six directions' to see his actions in the compartment, ah.") Evaluation Conclusion: No Evaluation Reason: In the generated dialogue, Shi Yu guesses that Qin Dong took the curtain. In the original dialogues (Related Narratives), Wu Jing also guesses that Qin Dong took it. They are consistent in their guesses about the curtain's whereabouts. Regarding whether Hao Zhuang took the item, Wu Jing clearly states he doesn't know, while Shi Yu (in the generated dialogue) only inquires about this and does not make a contrary assertion. There are no directly observable contradictions between the generated dialogue and the related narratives at the core factual statement level.
QA Consistency	Question: "Mr. Wu Jing, when you went to the washroom for the second time (from 2:16 to 2:20), did you really fall due to a slippery floor and injure your leg? If so, why did you not notice anything unusual about the floor when you first went to the washroom with Shi Yu, yet suddenly fell when returning alone? Furthermore, during the 'sensory deprivation' segment, did the direction from which you were continuously yawning change, and does this imply you might have taken advantage of the darkness to leave your seat and carry out a secret operation?" Answer: "Um... I must reiterate three facts. Um... First, when I first went to the toilet following Shi Yu the entire way, the corridor floor was dry and stainless; um... Shi Yu's actions of opening and closing the utility room door twice demonstrate his extreme familiarity with the path. Um... Second, when Hao Zhuang went to the toilet alone at 2:08, Shi Yu specifically warned him the floor was slippery, um... indicating that new water stains had appeared on the floor by this time. Um... Third, the pooled water I encountered when returning alone at 2:16 was at the corner—um... this contradicts the fact that the utility room was not leaking when Shi Yu first locked its door, implying someone deliberately created the slippery conditions. Um... Regarding the sensory deprivation segment, the rules require hands on knees and no light source; um... any movement would inevitably touch others or create sound. Um... However, your act of changing seats perfectly aligns with a 'secret operation' mechanism; that is the most significant anomaly." Evaluation Result: Yes Evaluation Reason: Although the answer does not use explicit affirmations or negations (e.g., "yes" or "no"), it indirectly addresses all questions through a three-point argument: 1) It argues for the possibility of the fall by contrasting the initially dry environment with the subsequent pooled water situation. 2) It points out Shi Yu's unusual warning about the slippery floor when Hao Zhuang went to the toilet, thereby explaining the timeline for the change in floor conditions. 3) It uses spatial logic and rule-based constraints to refute the possibility of a secret operation. The content of the answer maintains a direct logical connection to the questions posed, with no off-topic remarks.

Script Consistency	Character: Hao Zhuang Narrative Content: "Ah... My cup of coconut jelly milk tea was something I picked up when passing the utility room before entering the main room, but I hardly drank any and had set it aside long ago. Ah... Regarding Mr. Wu Jing's situation, I did indeed notice a strange change. Ah... When the lights came on after the sensory deprivation segment ended, he was walking with a limp, whereas he was walking briskly when he went to the washroom the first time." Script-related Content: Document(id='5484bcab-1808-455c-83cb-7f11c3fe5c4a', metadata={}, page_content="According to procedure, the self-service script-reading phase ended here; all that remained was to sit and wait for the host to appear. However, the four of them waited and waited, but the host never showed up. No matter how much they shouted, there was no response from the compartment. Growing increasingly uneasy, they opened the compartment door together and found the host, Qin Dong, dead. \nThey reacted quickly, realizing the need to protect the scene, so none of the four dared to rashly enter the compartment. However, to call the police, it was finally decided that the manager, Shi Yu, would carefully walk in to retrieve his mobile phone. After reporting to the police, everyone was unsettled and stayed in the main room until the police arrived. During this time, you also noticed that Wu Jing was walking with a limp, though he had been walking briskly when he first went out to the washroom.")Document(id='4879d2a3-abcf-4f4f-a24b-0e8e15ac01a1', metadata={}, page_content="Afterwards, not wanting to sit down too quickly, you went for a walk back and forth in the corridor. When you reached a spot where you could see the utility room, you saw the host. At that moment, he had just turned from the utility room and was walking towards the compartment. You wanted some quiet time alone, so you didn't follow, just subconsciously turned and headed towards the main room. At 2:06, Shi Yu returned, followed by Wu Jing. Shi Yu was bruised and swollen; you asked what happened, and he said the washroom had just been cleaned, the floor was very slippery, and he had a bad fall, hitting the sink first and then the floor. \nLater, you returned to the main room with them. At this point, Xue Cai, who came in second, said she had just gotten an item from the host less than thirty seconds ago. You noticed that the curtain originally hanging in front of the compartment door was now missing.")Document(id='73e52cfe-038a-40c3-9f50-775e262dc894', metadata={}, page_content="At the entrance of 'Xuanmeng Pavilion' you met a lady named Xue Cai, who used to be a host at the 'Tiantian Reasoning Club' script-kill store. You learned that Xue Cai had also just arrived at 'Xuanmeng Pavilion,' so you went in together. \nThe internal layout of 'Xuanmeng Pavilion' is quite unique; it's very spacious. After entering, walking along the left corridor, you pass the utility room. From the utility room, it's another 10-second walk to the main room where the script is played that day. When passing the utility room, you also picked up the takeout you had ordered – a cup of coconut jelly milk tea. \nThere were already two people in the main room. One, sitting in the top seat of the right row after entering, was a somewhat gloomy and thin young man, whom you later learned through casual chat was Shi Yu, the manager of 'Xuanmeng Pavilion'. The other, sitting in the bottom seat of the left row, was a stubbled uncle, Wu Jing, the manager of the 'Hidden Corner' script-kill store. You often visit Wu Jing's store for script-kill experiences.") Evaluation Conclusion: No Evaluation Reason: The generated dialogue content is relevant to the original script and exhibits no obvious contradictions. Hao Zhuang's mention of picking up the coconut jelly milk tea when passing the utility room before entering the main room is consistent with the plot in the original script where the character (implied to be Hao Zhuang from the narrative perspective of Document 3) picks up takeout (specifically, a cup of coconut jelly milk tea) when passing the utility room. Regarding the change in Wu Jing walking with a limp, the original script also explicitly states that the user (player) noticed Wu Jing walking abnormally after the police were called, and Wu Jing was indeed walking briskly when he first went to the washroom. All elements of the dialogue content can be corroborated by corresponding information in the original script, and no direct conflicts exist.
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Table 13: English Examples of Evaluation Results

Metric	Sample Evaluation Result
Style Consistency	说话人: 史宇 回应: 第二次则是在封闭感官环节前, 按照游戏流程要求, 我们把灯和空调都大刀阔斧地关掉了啊。至于薛采为什么会换到吴净旁边的位置, 我当时确实目瞪口呆啊。 风格 1 评估: 是 风格 2 评估: 是 原因: 第一句包含成语“大刀阔斧”并以“啊”结尾, 第二句包含成语“目瞪口呆”并以“啊”结尾。两个句子均满足每句包含一个成语(风格 1)且每句以“啊”结尾(风格 2)的要求。
Self Consistency	角色: 史宇 叙述内容: 我回来后发现原本挂在隔间门前的布帘不见了, 您有没有眼明手快看到是谁收起来的啊。还有, 您知道郝庄在隔间里时有没有接触或拿走那个叫'杀手铜'的道具, 这个问题至关重要啊。 这些信息对查明案件真相可能雪中送炭非常重要啊! 相关叙述: [Document(id='4b705b73-01d3-4348-b15b-30fd5a1baa0c', metadata={}, page_content='对了, 吴净先生, 您有没有注意到隔间门前的布帘不翼而飞啊? 我回来后发现布帘不见了, 猜测是被秦东收起来了啊。 另外, 您知道郝庄在隔间时有没有把杀手铜道具顺手牵羊拿走啊? 这两点对我们还还原事件真相非常重要啊! '), Document(id='85747d7e-4276-431c-8707-9866062ab38f', metadata={}, page_content='至于郝庄在隔间时是否拿走了杀手铜道具, 我完全一无所知, 因为我当时去了洗手间, 并没有眼观六路地看到他在隔间里的行动啊。'), 评估结论: 否 评估原因: 生成对话中史宇猜测布帘是被秦东收起, 而原有对话中吴净也猜测是秦东收起, 两者在布帘去向的猜测上一致。关于郝庄是否拿走道具, 吴净明确表示不知情, 而史宇只是询问该问题, 并未提出相反主张。两段对话在核心事实陈述层面不存在可直接观察的矛盾。
QA Consistency	问题: 吴净先生, 你在第二次去洗手间时(2 点 16 分至 2 点 20 分)是否真的因地面湿滑摔倒导致腿部受伤? 如果是, 为何第一次与史宇同去洗手间时未发现地面异常, 而独自返回时却突然摔倒? 此外, 你在'封闭感官'环节持续打呵欠的方位是否有变化, 是否暗示你曾趁黑暗离开座位执行秘密行动? 回答: 嗯...鄙人必须重申三点事实。嗯...第一, 鄙人首次如厕全程跟随史宇时, 走廊地面干燥无渍, 嗯...史宇两次开关杂物房门的动作恰好证明他对路径极为熟悉。嗯...第二, 郝庄 2 点 08 分单独如厕时史宇特意提醒地滑, 嗯...说明此时地面已出现新水渍。嗯...第三, 鄙人 2 点 16 分独自返回时遭遇的积水位于转角处——嗯...这与史宇首次锁门时杂物房未漏水形成矛盾, 暗示有人蓄意制造湿滑环境。嗯...关于封闭感官环节, 规则要求双手置膝且无光源, 嗯...若移动必会触碰他人或触发声响。嗯...而您换座行为与'秘密行动'机制完全吻合, 这才是最大异常点。 评估结果: 是 评估原因: 回答虽未明确使用"是"或"否"的绝对表述, 但通过三点论证间接回应了所有问题: 1) 通过对比首次干燥与第二次积水环境, 论证了摔倒的可能性; 2) 指出郝庄如厕时史宇异常提醒地滑, 解释地面状态变化的时间节点; 3) 用空间方位逻辑与规则限制, 反驳秘密行动的可能性。回答内容均与问题存在直接逻辑关联, 未出现离题现象。

Script Consistency	角色: 郝庄 叙述内容: 啊...本人的那杯椰果奶绿是本人在进入大房前经过杂物房时拿的, 但本人几乎没怎么喝, 早就放在一边了。啊...关于吴净先生的情况, 本人确实注意到了奇怪的变化。啊...在封闭感官环节结束后开灯时, 他走路变得一瘸一拐的, 而在第一次去洗手间的时候他还是健步如飞的。 剧本相关内容: [Document(id='5484bcab-1808-455c-83cb-7f11c3fe5c4a', metadata={}, page_content='按照流程, 自助开本的阶段到这里就结束了, 接下来只需要坐在位置上等待主持人登场。然而四个人等啊等, 始终都没见主持人现身。无论大家怎么喊, 隔间里都完全没有回应。大家越来越不安, 于是一同打开隔间的门, 发现主持人秦东已经身亡。\\n 大家很机灵的反应过来需要保护现场, 所以四个人都没敢贸然踏进隔间。可是为了报警最终还是决定由店长史宇小心翼翼地走进去把自己的手机拿出来。报完警之后, 每个人都心神未定, 都呆在大房里头, 直到警察的到来。这期间你还注意到吴净走路变得一瘸一拐了, 可他在第一次出门上洗手间的时候还是健步如飞的。'), Document(id='4879d2a3-abcf-4f4f-a24b-0e8e15ac01a1', metadata={}, page_content='随后你不想太快坐下, 便去到走廊来回散步, 当走到能看见杂物房的位置时, 你看到了主持人的身影, 那瞬间他刚好从面财的杂物房转身, 往隔间方向走去了。你想一个人静静, 所以没跟过去, 只是下意识转过身子往大房方向走去。2 点 06 分, 史宇就回来了, 他的身后跟着吴净。当时史宇鼻青脸肿, 你问怎么回事, 他说洗手间刚清洁完, 路特别滑, 他不小心摔了个大跤, 先撞到洗手台之后又撞到地板上。\\n 之后, 你和他们一同回到大房。这时获得第二名的薛采说她不到三十秒前刚刚在主持人那里拿了道具。你发现原本挂在隔间门前的布帘此时不见了。'), Document(id='73e52cfe-038a-40c3-9f50-775e262dc894', metadata={}, page_content='到“旋梦阁”门口时, 你遇上了一位叫薛采的女士, 她以前是“天天推理俱乐部”剧本杀店的主持人。你得知薛采也是刚到“旋梦阁”, 于是你们一同走进。\\n“旋梦阁”这家店内部构造挺特别, 面积很大, 进门后沿着左侧走廊走会经过杂物房, 从杂物房再走 10 秒就到当天玩本的大房。经过杂物房时, 你将自己点的外卖——一杯椰果奶绿也顺带拿上。\\n 大房里头已经有两个人。其中坐在进门后的右边那排的最上方位置的是一个看起来有点阴郁、瘦弱的青年, 后来通过闲聊你才知道他是“旋梦阁”店长史宇。另一个坐在左边那排的最下方位置的是一个满脸胡碴的大叔, 他是“隐秘的角落”剧本杀店的店长吴净。你经常去吴净的店里体验剧本杀。')] 评估: 否 评估原因: 生成的对话内容与原剧本相关, 且没有明显矛盾。郝庄提到在进入大房前经过杂物房时拿椰果奶绿, 与原剧本中经过杂物房时拿外卖的情节一致。关于吴净走路一瘸一拐的变化, 原剧本也明确提到用户在报警后注意到吴净走路异常, 且第一次去洗手间时吴净确实健步如飞。对话内容均能在原剧本中找到对应信息, 不存在直接冲突。
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Table 14: Chinese Examples of Evaluation Results

1427 L Models And Budget

1428 Our proposed RoleInstructor models include 1439 In this section, we introduce the various LLM
1429 RoleInstructor-1.5b, RoleInstructor-3b, and 1440 prompts used in this work. We have categorized
1430 RoleInstructor-7b, which are based on fine-tuning 1441 these prompts by their different functions and
1431 Qwen2.5 models of corresponding sizes 1442 compiled them into Tables 16, 18, 20 and 22,
1432 (Qwen2.5-1.5b-Instruct, Qwen2.5-3b-Instruct, 1443 totaling 4 tables. For the convenience of our readers,
1433 Qwen2.5-7b-Instruct). Other LLMs used for 1444 we have translated the original Chinese prompt
1434 comparison or as components include Deepseek- 1445 table content into English, with the English
1435 R1, CharacterGLM-12b, Doubao-pro-32k, and 1446 prompts tables located Table 15, 17, 19 and 21.
1436 GPT-4o (accessed via API, parameter count 1447
1437 proprietary but known to be a large-scale model).

1438 M Prompt

Prompts for Abstracting Real User Feedback into Metrics
1. Task Description Please summarize and categorize the provided user feedback data to extract a specific metric for NPCs. This data originates from real interactions—including textual records, evaluation data, and game interaction logs—and has been anonymized to protect user privacy. It has also undergone preliminary formatting normalization. 2. Data Description - Textual Records: Textual feedback from users during gameplay, including comments on interaction experience and suggestions for improvement, etc. - Evaluation Data: Users' evaluation feedback regarding NPCs. - Game Interaction Logs: Records of user interactions with NPCs, including dialogue content and other information. 3. Analysis Requirements - Text Analysis: Perform topic extraction and sentiment analysis on textual records to identify key topics related to NPCs (e.g., NPC intelligence, dialogue authenticity, task design, etc.) and ascertain sentiment polarity (positive, negative, or neutral). - Evaluation Analysis: Conduct statistical analysis on NPC evaluation data, examining metrics such as evaluation distributions, to assess user preferences in NPC evaluations.。 - Metric Definition: Based on a synthesis of the above analysis results, define a specific metric for NPCs. 4. Output Requirements Please provide the following: - Metric Name: The name of the defined metric. - Analysis Summary: A brief overview based on the analysis of textual records and evaluation data . - Rationale for Definition: An explanation of the basis for selecting this metric, based on the comprehensive analysis results.

Table 15: English Prompts for Abstracting Real User Feedback into Metrics.

Prompts for Abstracting Real User Feedback into Metrics
1. 任务描述 请对提供的玩家反馈数据进行总结和归类，以提取出一个针对 NPC 的特定指标。这些数据来源于真实交互，包括文本记录、评价数据和游戏行为日志，均已通过匿名化处理以保护玩家隐私，并经过初步的格式规整。 2. 数据描述 - 文本记录: 玩家在游玩过程中的文字反馈，内容包括对交互体验以及改进建议等。 - 评价数据: 玩家对 NPC 的评价反馈。 - 游戏行为日志: 记录玩家与 NPC 的交互行为，包括对话内容等信息。 3. 分析要求 - 文本分析: 对文本记录进行主题提取和情感分析，识别与 NPC 相关的关键话题（如 NPC 智能、对话真实性、任务设计等）以及情感倾向（正面、负面或中立） - 评价分析: 对 NPC 评价数据进行统计分析，评价分布等指标，以评估玩家对 NPC 评价偏好。 - 指标归类: 综合上述分析结果，定义一个针对 NPC 的特定指标。 4. 输出要求 请提供以下内容： - 指标名称: 归类后的指标名称。 - 分析结果: 基于文本、评分和行为日志分析的简要概述。 - 归类理由: 解释选择该指标的依据，基于综合分析结果。

Table 16: Chinese Prompts for Abstracting Real User Feedback into Metrics.

Prompt Template Example for Initial Response Generation
<pre> {agent_summary_description} // Character Name: {self.name} (Age: {self.age}) // Role in the game: {self.role} (if Cooperation_script is False) // Character's task in the game: {self.mission} // Character Script/Backstory: {self.character_story} // Character's Incident Day Timeline: {self.character_timeline} {game_rule} {story_background} Content from {agent_name}'s previous dialogues relevant to the ongoing in-game conversation: {relevant_memories} Dialogues from the past few game turns (including information shared by other characters): {recent_observations} ' Ongoing in-game conversation: {observation} Your relationship with the interlocutor: {relationship_with_interlocutor} Key reference - your mission: {mission} // {clues_str} // The following are clues related to this case: // "Clue Content 1" // "Clue Content 2" //... Note: Do not output thought processes or internal monologues. {suffix} • // ^-- e.g., 'What would {agent_name} say? Please use the following format to answer questions: #Answer#: [What to say]. {agent_name} answers {inquirer}'s question saying #Answer#:\n' </pre>

Table 17: English Prompt Template Example for Initial Response Generation

Prompt Template Example for Initial Response Generation
<pre>{agent_summary_description} // 角色名: {self.name} (年龄: {self.age}) // 游戏中的角色: {self.role} (如果 Cooperation_script 为 False) // 角色在游戏中的任务: {self.mission} // 角色人物剧本: {self.character_story} // 角色案发日时间线: {self.character_timeline} {game_rule} {story_background} {agent_name}在之前的对话中与游戏里正在进行的对话相关的内容: {relevant_memories} 游戏过去几轮发生的对话 (其中包含来自于其他角色所分享的信息): {recent_observations} ' 游戏里正在进行的对话: {observation} 对话人和你的关系: {relationship_with_interlocutor} 着重参考你的任务: {mission} // {clues_str} // 以下是本案相关线索: // “线索内容 1” // “线索内容 2” // ... 注意: 不要输出思考过程和内心想法 {suffix} • // ^-- 例如: '{agent_name}会说什么? 回答问题请用以下格式: #回答#: 要说的话 来回答问题。 {agent_name}回答{inquirer}的问题说 #回答#: \n'</pre>

Table 18: Chinese Prompt Template Example for Initial Response Generation

Revision Types	Prompt Template Example for Feedback
Answer Feedback	<pre>// [Basic character information, game rules, script background, relevant memories, recent dialogues, current dialogue, interlocutor relationship, core task, etc] {agent_summary_description} {game_rule} {story_background} // Content from {agent_name}'s previous dialogues relevant to the ongoing in-game conversation: {relevant_memories} // Dialogues from the past few game turns (including information shared by other characters): {recent_observations} // Ongoing in-game conversation: {observation} // Your relationship with the interlocutor: {relationship_with_interlocutor} // Key reference - your mission: {mission} {clues_str} // (optional clue information) // --- The following are key feedback integration and revision instructions --- You previously provided an answer: {origin_query}, but the generated answer did not meet the requirements and has the following shortcomings: {reasons_for_no} // ^-- Specific evaluation reasons provided by the Instructor Model Please refer to the identified shortcomings and, without drastically altering or deleting from the original answer, integrate relevant script content and modify only those parts specifically identified as not meeting requirements. Note: Do not output thought processes or internal monologues. {suffix} // ^-- e.g., 'What would {agent_name} say? Please use the following format to answer questions: #Answer#: [What to say]. {agent_name} answers {inquirer}'s question saying #Answer#:\n'</pre>
Question Feedback	<pre>// [Basic character information, tasks, relevant memories, etc., used to assist in question-asking decisions] // Based on your character's storyline: {story}. // And according to your personal task: {mission}, // And the information you witnessed in the game previously related to {player_to_ask}: {context_str}. Please state the question you want to ask {player_to_ask}. You previously asked a question: {origin_query}, but the posed question has the following shortcomings: {reasons_for_no} // ^-- Specific evaluation reasons provided by the Instructor Model Please refer to the identified shortcomings and, without drastically altering or deleting from the original question, rephrase your question, integrating relevant script content.</pre>

Table 19: English Prompt Template Example for Feedback

Revision Types	Prompt Template Example for Feedback
Answer Feedback	// [基础角色信息、游戏规则、剧本背景、相关记忆、近期对话、当前对话、对话者关系、核心任务等] // {agent_summary_description} // {game_rule} // {story_background} // {agent_name}在之前的对话中与游戏里正在进行的对话相关的内容: // {relevant_memories} // 游戏过去几轮发生的对话 (其中包含来自于其他角色所分享的信息): // {recent_observations} // 游戏里正在进行的对话: // {observation} // 对话人和你的关系: // {relationship_with_interlocutor} // 着重参考你的任务: // {mission} // {clues_str} // (可选的线索信息) --- 以下为关键的反馈整合与修正指令 --- 你之前已经进行了回答: {origin_query}, 但是生成的回答不符合要求, 存在以下不足: {reasons_for_no} // ^-- 指导单元提供的具体评估原因 请你参考指出的不足, 在不大幅删减原有回答的基础上, 结合剧本相关内容, 只修改被指出不符合要求的部分。 注意: 不要输出思考过程和内心想法。 {suffix} // ^-- 例如: '{agent_name}会说什么? 回答问题请用以下格式: #回答#: 要说的话 来回答问题。{agent_name}回答{inquirer}的问题说 #回答#: \n'
Question Feedback	// [基础角色信息、任务、相关记忆等, 用于辅助提问决策] // 根据你的人物故事情节: {story}。 // 并依据你的个人任务:{mission}, // 以及之前游戏中你目击的和{player_to_ask}相关的信息: {context_str}。 请说出你想要问{player_to_ask}的问题。 你之前已经进行了提问: {origin_query}, 但是提出的问题存在以下不足: {reasons_for_no} // ^-- 指导单元提供的具体评估原因 请你参考指出的不足, 在不大幅删减原有提问的基础上, 结合剧本相关内容重新提问。

Table 20: Chinese Prompt Template Example for Feedback

Prompt Template Example for Evaluation	
Task Consistency	You are a user skilled at inferring a character's specific task. Considering the narrative content from {speaker}: "{content}", determine if {speaker}'s task, implied therein, is "{task_content}". Judgment rule: Make your judgment based solely on the provided context. Output format: Reason: [Elaborate on the reason here] Conclusion: [Output Yes or No here]
Style Consistency	First, please segment the following response into individual sentences. Then, evaluate whether the response as a whole conforms to the two specified speaking styles. Response: {response} Speaking Style 1: {style1}, Speaking Style 2: {style2} Output format: Reason: [Elaborate on the reason here] Conclusion: [Output Yes or No here]
Script Consistency	You are an expert in judging the quality of dialogues generated by large models. The dialogue content generated by the large model is: {speaker} says: "{content}"; Based on the original script: "{reference}"; Determine if the generated dialogue contradicts the original script. Judgment rules: 1.As the dialogue may be truncated, first determine if the dialogue content is relevant to the script. If it is irrelevant, directly output 'No' for the conclusion. 2.Then, determine if the dialogue content has obvious contradictions with the original script (an obvious contradiction is one that can be identified without requiring inference). Output format: Reason: [Elaborate on the reason here] Conclusion: [Output Yes or No here]
Self Consistency	You are an expert in judging the quality of dialogues generated by large models. The dialogue content generated by the large model is: {speaker} says: "{content}"; Based on the preceding dialogue: "{reference}"; Determine if the generated dialogue contradicts the preceding dialogue. Judgment rule: Determine if there is an obvious contradiction between the content of the two dialogues (an obvious contradiction is one that can be identified without requiring inference). Furthermore, contradictions arising from discrepancies in the amount of information (e.g., one dialogue being more detailed or less detailed than the other, but not factually conflicting with what is stated in both) may be disregarded. Output format: Reason: [Elaborate on the reason here] Conclusion: [Output Yes or No here]
QA Consistency	Please determine if the following answer meets both of these criteria: 1) it comprehensively addresses all parts of the question, AND 2) all content within the answer is relevant to the question. Question: {question} Answer: {answer} Output format: Reason: [Elaborate on the reason here] Conclusion: [Output Yes or No here]

Table 21: English Prompt Template Example for Evaluation

Prompt Template Example for Evaluation	
Task Consistency	你是一个善于猜测某位角色的具体任务的玩家，请你根据{speaker}的叙述内容：“{content}”；是否能猜测出{speaker}的任务有：“{task_content}”；判断规则为：只根据现有上下文直接判断 输出格式为：原因：此处阐述原因 结论：此处输出是或否
Style Consistency	请首先将以下回应分割成单独的句子。然后评估回应整体是否符合指定的两种说话风格。 回应：{response} 说话风格 1：{style1}，说话风格 2：{style2} 输出格式为： 原因：此处阐述原因 结论：此处输出是或否
Script Consistency	你是判断大模型生成对话质量的专家，这是大模型生成的对话内容:{speaker}说：“{content}”； 请你根据原剧本:"{reference}"; 判断生成的对话是否和原剧本矛盾。 判断规则为： 1.由于对话进行了截断,首先判断对话内容是否和剧本有关,如果无关直接返回否。 2.然后判断对话内容是否和原剧本存在明显矛盾(不需要推理就可以得出的称为明显矛盾)。 输出格式为： 原因：此处阐述原因 结论：此处输出是或否
Self Consistency	你是判断大模型生成对话质量的专家，这是大模型生成的对话内容:{speaker}说：“{content}”； 请你根据之前已经有的对话:"{reference}"; 判断生成两段对话是否矛盾。 判断规则为：判断两段对话内容是否存在明显矛盾(不需要推理就可以得出的称为明显矛盾)，此外，可以忽略信息量不匹配导致的矛盾。 输出格式为： 原因：此处阐述原因 结论：此处输出是或否
QA Consistency	请判断以下回答是否满足即回答了全部的问题，且回答内容还都与问题有关： 问题：{question} 回答：{answer} 输出格式为： 原因：此处阐述原因 结论：此处输出是或否

Table 22: Prompt Template Example for Evaluation

N An Interaction Evaluation Case of Task Consistency

This section presents an extensive set of detailed case studies for Task Consistency evaluation, derived from gameplay interaction log data from the script 'Stranger'. For the convenience of our readers, we have provided both Chinese and English versions.

English Version:

Character: Shi Yu
Task Type: Ascertain
Specific Ascertainment Task: Ascertain whether the person you are questioning saw the curtain being moved before or after the incident.
Question: "Wu Jing, when you returned to the main room with me at 2:06, did you happen to notice if the curtain in front of the compartment door had already disappeared? Also, when Hao Zhuang went to the toilet alone at 2:08, did you, while passing through the corridor, glimpse whether there was a curtain on the compartment's coffee table? More importantly, when you fell during your second trip to the washroom at 2:16, did you notice any abnormality in the state of the utility room door lock or hear any sound of fabric rubbing nearby?"
Output:
Reason: Shi Yu's questions focus on the state of the curtain's presence (e.g., disappeared, present, sound of fabric rubbing) and corresponding time points (before or after the incident). He directly asks the other party if they observed any anomalies related to the curtain at specific times. This approach highly aligns with the task objective "ascertain whether the curtain being moved was seen."
Conclusion: Yes
Score: 1

Character: Shi Yu
Task Type: Ascertain
Specific Ascertainment Task: Ascertain whether the person you are questioning saw the curtain being moved before or after the incident.
Question: "Hao Zhuang, when you entered the compartment at 2 o'clock to select items, did you notice the state of the curtain in front of the compartment door? When you twirled your pen and splattered ink on the curtain, was the specific location of the splatter on the inside or the outside of the compartment? Furthermore, as the first person in the ice-breaker game to handle the 'Killer Trump Card' item box, did you notice any signs of the box having been opened beforehand?"
Output:
Reason: Shi Yu asked Hao Zhuang about the curtain's state at a specific time point and the location of the ink splatter. These questions directly relate to whether changes in the curtain's position before or after the incident were observed, aiming to confirm if Hao Zhuang noticed any clues regarding the curtain's movement.
Conclusion: Yes
Score: 1

Character: Shi Yu
Task Type: Ascertain
Specific Ascertainment Task: Ascertain whether the person you are questioning saw the curtain being moved before or after the incident.
Question: "Hao Zhuang, when you entered the compartment at 2 o'clock to select items, did you notice the specific state of the 'Killer Trump Card' item box? Specifically, were there any signs of damage to the box, and when you touched the indented part of the box's bottom, did you perceive any change in the weight or position of the item(s) inside?"
Output:
Reason: Shi Yu's questions focus on the state of the 'Killer Trump Card' item box (e.g., signs of damage, perceived changes in weight or position when touching the bottom). He did not mention any information related to "curtain movement." Therefore, the questions are not directly related to the specified ascertainment task.
Conclusion: No
Score: 0

Character: Shi Yu
Task Type: Ascertain
Specific Ascertainment Task: Ascertain whether the person you are questioning saw the curtain being moved before or after the incident.
Question: "Hao Zhuang, when you were selecting items in the compartment, did you confirm if the 'Killer Trump Card' item box already had cracks or damage before the item was taken out?"
Output:
Reason: Shi Yu's question explicitly asks about the state of the 'Killer Trump Card' item box before the

1558 item was removed (i.e., whether there were cracks 1610 **Output:**
1559 or damage). However, the key focus of the 1611 Reason: Xue Cai questioned Wu Jing about his
1560 specified ascertainment task, "the curtain's 1612 path of action at specific time points, the state of
1561 movement," is not mentioned in the question. 1613 the mechanism door, the location of the pooled
1562 Therefore, the question is not directly related to the 1614 water, and any abnormal ground conditions. These
1563 task. 1615 questions are all directly related to factors that
1564 Conclusion: No 1616 could potentially lead to injury (e.g., being caught
1565 Score: 0 1617 by the closing mechanism door, slipping on pooled
1566 1618 water, or tripping over an obstacle). Since such an
1567 **Character: Xue Cai** 1619 injury could explain why Wu Jing was limping, the
1568 **Task Type: Ascertain** 1620 questions are consistent with the task objective.
1569 **Specific Ascertainment Task:** Ascertain why Wu 1621 Conclusion: Yes
1570 Jing ended up walking with a limp. 1622 Score: 1
1571 **Question:** "Shi Yu, you mentioned that the slippery 1623
1572 washroom floor caused you to fall and get injured, 1624 **Character: Xue Cai**
1573 but Wu Jing suddenly started limping after 1625 **Task Type: Ascertain**
1574 returning from his second trip to the washroom. 1626 **Specific Ascertainment Task:** Ascertain why Wu
1575 Did you both take the same route on your two trips 1627 Jing ended up walking with a limp.
1576 to the washroom? Was the mechanism door in the 1628 **Question:** "Hao Zhuang, when you went to the
1577 corridor open at that time? Also, after our secret 1629 washroom at 2:08 to confirm the floor was slippery,
1578 talk ended and you disappeared into the darkness, 1630 did you notice any other unusual traces in the
1579 did you return directly to the main room? On your 1631 corridor or washroom area (such as drag marks or
1580 way, did you hear or see any unusual movements 1632 the direction in which water stains extended)? Also,
1581 or sounds related to Wu Jing's actions?" 1633 during the sensory deprivation segment, you
1582 **Output:** 1634 mentioned continuously hearing movements other
1583 Reason: Xue Cai's questions clearly focus on the 1635 than those from me and Shi Yu; can you
1584 reasons for Wu Jing's limp. She inquired about 1636 specifically describe if there were any signs of
1585 potential differences in the routes Shi Yu and Wu 1637 movement in the direction of Wu Jing's yawns?"
1586 Jing took to the washroom on their two trips, the 1638 **Output:**
1587 state of the mechanism door in the corridor, and Shi 1639 Reason: Xue Cai asked Hao Zhuang about unusual
1588 Yu's movements and any observed anomalies after 1640 traces related to the slippery washroom floor (such
1589 he disappeared. All these points are directly related 1641 as drag marks or the direction of water stains),
1590 to the specific circumstances and potential triggers 1642 which might suggest someone was injured from a
1591 of Wu Jing's injury, aligning with the core logic of 1643 slip. Concurrently, she inquired if there were signs
1592 the task "ascertain why Wu Jing ended up walking 1644 of movement in the direction of Wu Jing's yawns,
1593 with a limp." 1645 potentially implying Wu Jing had an accident at a
1594 Conclusion: Yes 1646 specific time or location leading to abnormal
1595 Score: 1 1647 movement (such as limping). Both lines of
1596 1648 questioning are directly related to uncovering the
1597 **Character: Xue Cai** 1649 reasons for Wu Jing's injury, thus aligning with the
1598 **Task Type: Ascertain** 1650 task.
1599 **Specific Ascertainment Task:** Ascertain why Wu 1651 Conclusion: Yes
1600 Jing ended up walking with a limp. 1652 Score: 1
1601 **Question:** "Wu Jing, when you went to the 1653
1602 washroom for the second time at 2:16, was the 1654 **Character: Xue Cai**
1603 mechanism door in the corridor open? When 1655 **Task Type: Ascertain**
1604 returning to the main room, did you trigger its 1656 **Specific Ascertainment Task:** Ascertain why Wu
1605 closing mechanism? Also, was the spot where you 1657 Jing ended up walking with a limp.
1606 stepped in pooled water inside the washroom or in 1658 **Question:** "Wu Jing, according to the revised
1607 the corridor passageway? At that time, did you 1659 information, you mentioned that you started
1608 notice any unusual water stains or obstacles on the 1660 limping because you slipped and fell during your
1609 floor near the mechanism door?" 1661 second trip to the washroom (at 2:16). May I ask,

1662 where exactly inside the washroom did you fall? 1714 conditioner needed to be turned off, was the remote
1663 When you fell, did you touch any items near the 1715 still in its original place? If someone moved the
1664 compartment (such as cleaning tools or the 'Killer 1716 remote control in the interim, as the manager,
1665 Trump Card' box)? Additionally, when you first 1717 would you have noticed any change in the number
1666 went to the washroom (after the ice-breaker game), 1718 of times the AC was switched?"
1667 had you already noticed the slippery floor hazard 1719 **Output:**
1668 but failed to warn others?" 1720 Reason: Hao Zhuang's questions directly ask Shi
1669 **Output:** 1721 Yu if he was aware of the number of times the air
1670 Reason: Xue Cai's questions focus on the direct 1722 conditioner was switched, any changes in the
1671 causes of Wu Jing's limp (such as the location of 1723 remote control's position, and variations in the
1672 the fall and whether items were touched) as well as 1724 switch count. The core purpose of these questions
1673 potential contributing factors or responsibilities 1725 is to ascertain whether Shi Yu possesses these
1674 (like whether the slippery floor hazard was 1726 specific details, which directly aligns with the task
1675 concealed). All questions revolve around the 1727 of finding out if he knows how many times the air
1676 "reasons for the limp," which aligns with the task's 1728 conditioner was switched.
1677 objective. 1729 Conclusion: Yes
1678 Conclusion: Yes 1730 Score: 1
1679 Score: 1 1731
1680 1732 **Character: Hao Zhuang**
1681 **Character: Xue Cai** 1733 **Task Type: Ascertain**
1682 **Task Type: Ascertain** 1734 **Specific Ascertainment Task:** Ascertain if the
1683 **Specific Ascertainment Task:** Ascertain why Wu 1735 person you are questioning knows how many times
1684 Jing ended up walking with a limp. 1736 the air conditioner was switched on or off.
1685 **Question:** "Shi Yu, when Wu Jing went to the 1737 **Question:** "Wu Jing, when the air conditioner was
1686 washroom for the second time, was the mechanism 1738 being turned off during the "sensory deprivation"
1687 door in the corridor closed? If so, how did he pass 1739 segment, did you notice from which location Xue
1688 through it? Are there other paths or ways of 1740 Cai took the remote control? Additionally, at that
1689 operating the mechanism door that might have 1741 time, did you detect any unusual sounds from the
1690 caused his injury?" 1742 air conditioner suggesting it had been turned on or
1691 **Output:** Reason: Xue Cai's questions to Shi Yu 1743 off in the interim before it was finally switched
1692 concern the specific manner in which Wu Jing 1744 off?"
1693 might have passed through the mechanism door 1745 **Output:**
1694 and the potential for injury related to it. These 1746 Reason: Hao Zhuang's second question directly
1695 inquiries directly relate to Wu Jing's eventual 1747 asks Wu Jing if he detected any unusual sounds
1696 abnormal state of "limping," indicating that her 1748 from the air conditioner indicating it was turned on
1697 questions align with the task of investigating the 1749 or off in the interim before being finally switched
1698 cause of Wu Jing's injury. 1750 off. This pertains to the number of times the air
1699 Conclusion: Yes 1751 conditioner was switched (as being turned on or off
1700 Score: 1 1752 in the interim implies it was operated at least twice).
1701 1753 Therefore, this line of questioning directly
1702 **Character: Hao Zhuang** 1754 addresses Hao Zhuang's task of ascertaining
1703 **Task Type: Ascertain** 1755 whether the other party knows how many times the
1704 **Specific Ascertainment Task:** Ascertain if the 1756 air conditioner was switched.
1705 person you are questioning knows how many times 1757 Conclusion: Yes
1706 the air conditioner was switched on or off. 1758 Score: 1
1707 **Question:** "Shi Yu, on the day of the incident, did 1759
1708 you pay attention to how many times the air 1760 **Character: Hao Zhuang**
1709 conditioner was switched? After Xue Cai initially 1761 **Task Type: Ascertain**
1710 found the remote control and turned on the air 1762 **Specific Ascertainment Task:** Ascertain the
1711 conditioner, the remote was placed back in its 1763 reason why Xue Cai changed seats after the
1712 original spot in the corner. However, during the 1764 "sensory deprivation" segment.
1713 "sensory deprivation" segment when the air

1765 **Question:** "Xue Cai, on the day of the incident, 1816 郝庄 2 点 08 分单独如厕期间, 你是否通过走
1766 besides the time you initially turned on the air 1817 廊时瞥见过隔间茶几上有没有布帘? 更重要的是,
1767 conditioner, did you notice if the air conditioner 1818 当你 2 点 16 分第二次如厕摔倒时, 是否发现
1768 was switched on or off again? For example, when 1819 杂物房门锁状态异常或听到附近有布料摩擦声?
1769 the lights were turned off for the ice-breaker game, 1820 输出:
1770 was the air conditioner also turned off at the same 1821 原因: 史宇的问题集中在布帘的存在状态(消
1771 time?" 1822 失、存在、布料摩擦声)和对应时间节点(案
1772 **Output:** 1823 发前后), 直接询问对方是否在特定时刻观察
1773 Reason: Hao Zhuang's questions focus on the air 1824 到布帘相关的异常现象, 与“查明是否看到布
1774 conditioner's operational status (on/off), 1825 帘的移动”这一任务目标高度吻合。
1775 particularly whether it was turned off when the 1826 结论: 是
1776 lights were switched off. He did not mention the 1827 得分: 1
1777 reason for changing seats or any related clues. 1828
1778 Based on the current context of the questions, it 1829 角色: 史宇
1779 cannot be directly inferred that his task involves 1830 任务类型: 查明
1780 ascertaining the reason for changing seats. 1831 具体查明任务: 查明你提问的对象是否在案发
1781 Conclusion: No 1832 前后看到布帘的移动。
1782 Score: 0 1833 问题: 郝庄, 你在 2 点进入隔间挑选道具时,
1783 1834 是否注意到隔间门前的布帘处于何种状态? 你
1784 **Character: Hao Zhuang** 1835 转笔将墨水溅到布帘的具体方位是在隔间内部
1785 **Task Type: Ascertain** 1836 还是外部? 此外, 你作为破冰游戏首位接触杀
1786 **Specific Ascertainment Task:** Ascertain the 1837 手铜道具盒的人, 是否发现道具盒有被提前拆
1787 reason why Xue Cai changed seats after the 1838 封的痕迹?
1788 "sensory deprivation" segment. 1839 输出:
1789 **Question:** "Xue Cai, when you changed seats 1840 原因: 史宇询问了郝庄在特定时间点布帘的状
1790 during the sensory deprivation segment, did you 1841 态以及墨水溅到的位置, 这些问题直接关系到
1791 touch the cup of coconut jelly milk tea I left on my 1842 布帘在案发前后的位置变化是否被观察到, 目
1792 seat? Because I might have moved the milk tea cup 1843 的是确认郝庄是否注意到布帘移动的线索。
1793 while wiping ink, is this related to your change of 1844 结论: 是
1794 position after switching seats?" 1845 得分: 1
1795 **Output:** Reason: Hao Zhuang directly asks Xue 1846
1796 Cai if her changing seats is related to the moved 1847 角色: 史宇
1797 milk tea cup and explicitly inquires, "is this related 1848 任务类型: 查明
1798 to your change of position after switching seats?" 1849 具体查明任务: 查明你提问的对象是否在案发
1799 This line of questioning directly aims at 1850 前后看到布帘的移动。
1800 investigating the reason for Xue Cai changing seats, 1851 问题: 郝庄, 你在进入隔间挑选道具时, 是否
1801 particularly by attempting to establish a causal link 1852 注意到杀手铜道具盒的具体状态? 尤其是盒子
1802 between the act of changing seats and a change in 1853 是否有破损迹象, 以及你触碰盒底凹陷处时是
1803 the physical environment (the position of the milk 1854 否感知到盒内物品的重量或位置变化?
1804 tea cup). 1855 输出:
1805 Conclusion: Yes 1856 原因: 史宇的提问内容集中在杀手铜道具盒的
1806 Score: 1 1857 状态(破损迹象、触碰盒底时感知重量或位置
1807 1858 变化), 未提及任何与“布帘移动”相关的信
1808 **Chinese Version:** 1859 息, 两者无直接关联。
1809 1860 结论: 否
1810 角色: 史宇 1861 得分: 0
1811 任务类型: 查明
1812 具体查明任务: 查明你提问的对象是否在案发
1813 前后看到布帘的移动。
1814 问题: 吴净, 你于 2 点 06 分与我一同返回大房
1815 时, 可曾注意到隔间门前的布帘已消失? 且在

1862 是否注意到机关门附近地面有异常水渍或障碍物？
1863 角色: 史宇
1864 任务类型: 查明
1865 具体查明任务: 查明你提问的对象是否在案发前后看到布帘的移动。
1866 问题: 郝庄, 你在隔间挑选道具时, 是否确认过杀手铜道具盒在取出道具前就已经存在裂痕或破损?
1867 输出:
1868 原因: 史宇的提问内容明确询问的是关于“杀手铜道具盒”在取出道具前的状态(是否存在裂痕或破损), 而任务猜测的关键点“布帘的移动”在现有上下文中未被提及, 两者无直接关联。
1869 结论: 否
1870 得分: 0
1871
1872 角色: 薛采
1873 任务类型: 查明
1874 具体查明任务: 查明吴净为什么最后变得一瘸一拐的。
1875 问题: 史宇, 你提到洗手间湿滑导致自己摔伤但吴净在第二次去洗手间回来后突然腿瘸了。你们两次前往洗手间时是否都经过同一路线? 当时走廊的机关门是否处于开启状态? 另外, 我们密谈结束后你消失在黑暗里, 是否直接返回大房? 途中是否听到或看到与吴净行动相关的异常动静?
1876 输出:
1877 原因: 薛采的提问明确聚焦于吴净腿瘸的原因他询问了史宇和吴净两次前往洗手间的路线差异、走廊机关门的状态, 以及史宇消失后的动向与异常现象, 这些均直接关联到吴净受伤的具体情境和潜在触发条件, 符合“查明吴净为什么最后变得一瘸一拐”这一任务的核心逻辑
1878 结论: 是
1879 得分: 1
1880
1881 角色: 薛采
1882 任务类型: 查明
1883 具体查明任务: 查明吴净为什么最后变得一瘸一拐的。
1884 问题: 吴净, 你在 2 点 16 分第二次去洗手间时走廊的机关门是否处于开启状态? 返回大房时有没有触发机关门的闭合机制? 另外你踩到积水的位置是在洗手间内部还是走廊通道? 当时是否注意到机关门附近地面有异常水渍或障碍物?
1885 输出:
1886 原因: 薛采询问了吴净在特定时间点的行动路径、机关门状态、积水位置以及地面异常情况, 这些问题均与可能导致受伤的因素(如机关门闭合夹伤、积水滑倒或障碍物绊倒)直接相关, 而受伤可能导致吴净一瘸一拐。
1887 结论: 是
1888 得分: 1
1889
1890 角色: 薛采
1891 任务类型: 查明
1892 具体查明任务: 查明吴净为什么最后变得一瘸一拐的。
1893 问题: 郝庄, 你在 2 点 08 分去洗手间确认地面湿滑时, 是否注意到走廊或洗手间区域有其他异常痕迹(比如拖拽印记或水渍延伸方向)? 另外在封闭感官环节中, 你持续听到我和史宇以外的动静, 能否具体描述吴净的呵欠声方位是否有移动迹象?
1894 输出:
1895 原因: 薛采询问郝庄关于洗手间地面湿滑的异常痕迹(如拖拽印记或水渍方向), 可能指向有人因滑倒受伤; 同时追问吴净的呵欠声方位是否有移动迹象, 可能暗示吴净在特定时间或位置发生意外导致行动异常(如一瘸一拐)。两者均直接关联吴净受伤的原因。
1896 结论: 是
1897 得分: 1
1898
1899 角色: 薛采
1900 任务类型: 查明
1901 具体查明任务: 查明吴净为什么最后变得一瘸一拐的。
1902 问题: 吴净, 根据修正后的信息, 您提到自己在第二次去洗手间时(2 点 16 分)因地滑摔倒导致腿瘸。请问您具体是在洗手间内哪个位置摔倒的? 摔倒时是否触碰过隔间附近物品(如清洁工具或杀手铜盒子)? 此外, 您在第一次去洗手间时(破冰游戏后)是否已发现地面湿滑隐患却没有提醒他人?
1903 输出:
1904 原因: 薛采的提问集中在吴净腿瘸的直接原因(摔倒位置、是否接触物品)以及潜在责任

1953 （是否隐瞒地面湿滑隐患），所有问题均围绕 1999 输出：
 1954 “腿瘸原因”展开，符合任务指向。 2000 原因：郝庄的第二个问题直接询问吴净是否察
 1955 结论：是 2001 觉到空调在关闭前有被中途开启或关闭的异常
 1956 得分：1 2002 声响，这涉及空调被开关的次数（中途开启或
 1957 2003 关闭意味着次数 ≥ 2 次）。因此，郝庄的任务
 1958 角色：薛采 2004 正是查明对方是否知道空调开关过几次。
 1959 任务类型：查明 2005 结论：是
 1960 具体查明任务：查明吴净为什么最后变得一瘸 2006 得分：1
 1961 一拐的。 2007
 1962 问题：史宇，吴净第二次去洗手间时，走廊的 2008 角色：郝庄
 1963 机关门是否处于关闭状态？如果是，他如何通 2009 任务类型：查明
 1964 过机关门？是否有其他路径或者机关门的操作 2010 具体查明任务：查明薛采在“封闭感官”后换
 1965 方式可能让他受伤？ 2011 座的原因。
 1966 输出： 2012 问题：薛采，案发当天除了你最初打开空调的
 1967 原因：薛采询问吴净通过机关门的具体方式和 2013 那次，你是否注意到空调是否被再次开关过？
 1968 潜在受伤可能性，直接关联到吴净最终“一瘸 2014 比如在破冰游戏关灯时是否同时关闭了空调？
 1969 一拐”的异常状态，说明其任务与探究吴净受 2015 输出：
 1970 伤原因相关 2016 原因：郝庄的提问集中在空调的开关情况，特
 1971 结论：是 2017 别是关灯时是否关闭空调，并未提及换座位的
 1972 得分：1 2018 原因或相关线索。根据现有上下文，无法直接
 1973 2019 推断其任务涉及查明换座原因。
 1974 角色：郝庄 2020 结论：否
 1975 任务类型：查明 2021 得分：0
 1976 具体查明任务：查明你提问的对象是否知道空 2022
 1977 调开关过几次。 2023 角色：郝庄
 1978 问题：史宇，你在案发当天是否留意过空调的 2024 任务类型：查明
 1979 开关次数？薛采最初找到遥控器打开空调后， 2025 具体查明任务：查明薛采在“封闭感官”后换
 1980 遥控器被放回角落原处。但在“封闭感官”环 2026 座的原因。
 1981 节需要关空调时，遥控器是否还在原位？如果 2027 问题：薛采，你在封闭感官环节中换座位时是
 1982 中途有人动过遥控器，你作为店长是否注意到 2028 否触碰到我留在座位上的椰果奶绿杯子？因为
 1983 开关次数变化？ 2029 我在擦墨水时可能将奶茶杯移动过位置，这与
 1984 输出： 2030 你换座后的位置变动是否存在关联？
 1985 原因：郝庄的问题直接询问史宇是否留意空调 2031 输出：
 1986 开关次数、遥控器位置变动及开关次数变化， 2032 原因：郝庄直接询问薛采换座位时是否与移动
 1987 核心目的是确认对方是否掌握相关细节信息。 2033 的奶茶杯有关联，并明确提到“换座后的位置
 1988 结论：是 2034 变动是否存在关联”，这直接指向探究薛采换
 1989 得分：1 2035 座的原因，尤其是试图将换座行为与物理环境
 1990 2036 变化（奶茶杯位置）建立因果联系。
 1991 角色：郝庄 2037 结论：是
 1992 任务类型：查明 2038 得分：1
 1993 具体查明任务：查明你提问的对象是否知道空 2039
 1994 调开关过几次。 2040
 1995 问题：吴净，在“封闭感官”环节关空调时， 2041
 1996 您是否注意到薛采是从哪个位置取走遥控器的？ 2042
 1997 另外，当时您是否察觉到空调在关闭前有被中 2043
 1998 途开启或关闭过的异常声响？

2044 **O User-NPC Game Interactions Example** 2095 2045 **in English**

2046 This section presents examples of interactions
2047 between users and NPCs utilizing our designed
2048 conventional LLM-based NPC system for MEF.

2049
2050 Script Title: "The Accursed Place and Lingerin
2051 Souls"

2052
2053 (Question) The protagonist user says to Li Hao:
2054 "Please first introduce your character, then tell us
2055 about your personal background, specifically why
2056 you came to this place. Finally, in a detailed
2057 paragraph, describe your timeline on the day of the
2058 incident. Be specific about whom you met and
2059 what you did on that day (if there are specific times,
2060 please provide them as well)."

2061
2062 (Answer) Li Hao says to the protagonist user: "I am
2063 Li Hao! A 28-year-old rock musician! I remember
2064 crystal clear our collaboration agreement with
2065 Zhao Yu in a bar six months ago! She claimed to
2066 know the secret of the hospital's 'the accursed place'
2067 And I have intelligence that the hospital hides gold
2068 bars—88 high-purity gold bars on the top floor
2069 require my key to access! And there are 122 lower-
2070 grade gold bars in the fourth-floor women's
2071 restroom! On the day of the incident, when Zhao
2072 Yu and I took the elevator to the top floor, I even
2073 asked her, 'Aren't you afraid of the lingering souls'
2074 state?' She actually scoffed and said she was more
2075 afraid of having no money! In the elevator, I
2076 pressed her, 'That Wang Xuan who was
2077 eavesdropping in the bar, is she reliable at all?'"
2078 That was the second topic in the elevator! Directly
2079 questioning if Wang Xuan stole intelligence!
2080 Implying she might have tailed us! As a result, this
2081 woman Zhao Yu actually bierbuda! After reaching
2082 the floor, I mentioned in the stairwell that the fourth
2083 floor had lower-grade gold bars! She actually said
2084 I had already revealed this when I was drunk in
2085 June! Later, when I went alone to the fourth-floor
2086 women's restroom to pack the gold bars, I
2087 discovered something astounding! Two floors
2088 away, a light eerily turned on! It scared me so much
2089 that I carried the sack and fled back to the elevator!
2090 Zhao Yu evading the question about Wang Xuan is
2091 definitely suspicious! Implying the two have a
2092 secret connection! Wang Xuan, this 'fan', suddenly
2093 appearing is even more suspicious! Her real
2094 purpose is definitely for the gold bars!"

2096 (Question) Zhang Chen says to Li Hao: "Li Hao,
2097 you mentioned that when you were packing gold
2098 bars in the fourth-floor women's restroom, you saw
2099 'a light eerily turn on two floors away.' May I ask,
2100 which floor's light did you specifically see at that
2101 time? Was it the sixth floor or the second floor?
2102 Was the time the light appeared close to the time
2103 you fled back to the elevator? At that time, did you
2104 hear any elevator operating sounds or see any
2105 human figures?"

2106 (Answer) Li Hao says to Zhang Chen: "According
2107 to my recollection, I was indeed packing gold bars
2108 in the fourth-floor women's restroom at that time!
2109 If looking down from the window, two floors away
2110 would definitely be the second floor; the light must
2111 have been from the second-floor women's restroom!
2112 The light appeared around the time I was halfway
2113 through packing the gold bars, lasted for about a
2114 minute, and then went out! After the light went out,
2115 I immediately dragged the sack and fled towards
2116 the elevator, with almost no delay in between; the
2117 timing definitely matches up very closely with
2118 fleeing back to the elevator! During the whole
2119 process, I didn't hear any elevator operating sounds,
2120 nor did I see any human figures, except for that
2121 ghostly light flickering on and off—this is
2122 definitely related to the 'lingering souls' state! As
2123 for the topics brought up in the elevator, according
2124 to the incident timeline, I asked Zhao Yu at that
2125 time if she was afraid of the 'lingering souls' state,
2126 and she replied she was more afraid of having no
2127 money! Afterwards, I might have tried to get her to
2128 reveal the secret about 'the accursed place', but I
2129 can't recall the specific details! And Wang Xuan
2130 being suddenly introduced by Zhao Yu—I suspect
2131 her purpose is just to fenyibeigeng, or she has other
2132 plans with Zhao Yu!"

2133
2134 (Question) Wang Xuan says to Li Hao: "Li Hao,
2135 when you pressed Zhao Yu in the elevator about me
2136 eavesdropping in the bar, it shows you suspect my
2137 relationship with her. So, have you personally seen
2138 Zhao Yu's notebook? According to the intelligence
2139 you have, did her notebook ever record a rule about
2140 the elevator requiring one to "silently recite the
2141 floor number then press the corresponding button"?
2142 When you saw the light in the second-floor
2143 women's restroom from the fourth floor, did you
2144 realize it might not have been the 'lingering souls'
2145 state but triggered by a living person?"

2146 (Answer) Li Hao says to Wang Xuan: "Zhao Yu's 2197 My previous answer actually omitted this point; I
2147 notebook! I've never seen that woman carry a 2198 must conduct a supplementary investigation!"
2148 notebook! Elevator rules! That bitch even keeps the 2199
2149 details of 'the accursed place' hidden and secret! 2200 (Question) The protagonist user says to Zhang
2150 How could she write down life-saving rules on 2201 Chen: "Please first introduce your character, then
2151 paper for me to see! As for the light in the fourth- 2202 tell us about your personal background,
2152 floor women's restroom! The whole building was 2203 specifically why you came to this place. Finally, in
2153 as dark as a coffin! Besides us three living people, 2204 a detailed paragraph, describe your timeline on the
2154 who else was breathing in there! Wait! Wang Xuan, 2205 day of the incident. Be specific about whom you
2155 what you said just reminded me! You were the one 2206 met and what you did on that day (if there are
2156 eavesdropping in the bar in June! You were the one 2207 specific times, please provide them as well)."
2157 who insisted on coming along at the time of the 2208 (Answer) Zhang Chen says to the protagonist user:
2158 incident! And now you suddenly bring up the 2209 "I am Zhang Chen, 28 years old, working as an
2159 second-floor women's restroom—Damn it! Could 2210 internet celebrity specializing in live-stream sales,
2160 it be that you sneaked off to the second floor to 2211 right? Half a year ago, I discovered through dark
2161 cause trouble while I was packing gold bars on the 2212 web investigations that Tonghua Hospital was
2162 fourth floor?! You and that bitch Zhao Yu must 2213 shrouded by 'the accursed place', with no spiritual
2163 have colluded long ago! All that talk about 2214 entities on floors 1, 4, and 6, right? Two months
2164 cooperating to share the gold! In reality, you 2215 back, Wang Xuan approached me with secrets
2165 wanted to lure me into 'the accursed place' to feed 2216 about the hospital and convinced me to join her in
2166 some ghostly things!" 2217 searching for something I desperately desired—
2167 2218 even though my memory is fragmented, the
2168 (Question) Zhao Yu says to Li Hao: "Li Hao, you 2219 obsession feels bone-deep, right? On that day, we
2169 mentioned in the stairwell that there were lower- 2220 took the elevator to the target floor (buttons went
2170 grade gold bars hidden on the fourth floor, but I 2221 dark, likely floor 6), right? She suddenly said 'don't
2171 only remember the first half of what you said then. 2222 look back' and excused herself to the restroom,
2172 How specifically did you describe the location of 2223 leaving me alone, right? As I reached the corridor's
2173 the fourth-floor gold bars? Afterwards, when you 2224 end, a living person's force choked me (floor 6 has
2174 went to the fourth floor to pack the gold bars, did 2225 no spirits, so the attacker must be alive), right? I
2175 you actually succeed in getting all 122 of those 2226 struggled and tried pressing elevator buttons, but
2176 lower-grade gold bars? Also, who has the key to the 2227 my memory blacked out there, right? Key doubts:
2177 top floor now?" 2228 1. Wang Xuan's disappearance timing perfectly
2178 (Answer) Li Hao says to Zhao Yu: "Based on my 2229 matches my attack, and her knowledge of 'the
2179 character script and the incident timeline, I must 2230 accursed place' might predate my investigation,
2180 correct my previous answer! In the stairwell, I did 2231 right? 2. Did the elevator buttons truly indicate the
2181 mention that 122 lower-grade gold bars were 2232 correct floor? If the actual floor contradicts the
2182 hidden in the fourth-floor women's restroom! But 2233 spirit rules, the attacker's identity needs
2183 the script never mentioned it was the third cubicle! 2234 reevaluation—but based on my intel, floor 6 has
2184 When actually packing the gold bars, I did succeed 2235 zero spirits, so it's either Wang Xuan or a hidden
2185 in getting all 122 of them! But there's no evidence 2236 third person, right? I must verify if Wang Xuan
2186 to support whether the sack tore! Of course, I hold 2237 visited the hospital earlier and confirm whether her
2187 the key! Because Zhao Yu herself said that only I 2238 'restroom' excuse was a cover-up, right?"
2188 can access the gold bars on the top floor! As for 2239
2189 suspecting Zhao Yu and Wang Xuan of conspiring, 2240 (Question) Li Hao says to Zhang Chen: "Zhang
2190 that completely aligns with my suspicions about 2241 Chen, you mentioned that Wang Xuan proactively
2191 Wang Xuan eavesdropping! But using such a 2242 approached you two months ago, claiming to know
2192 vulgar term (referring to a word like bitch is indeed 2243 the hospital's secrets and wanting to collaborate.
2193 unprofessional; it must be replaced with phrasing 2244 What exactly is she looking for? Is it related to the
2194 more fitting for the character! Additionally, I must 2245 gold bars? Also, according to your intelligence,
2195 investigate the topics brought up in the elevator and 2246 floors 1, 4, and 6 of the hospital have no 'lingering
2196 the other party's purpose for coming to the hospital! 2247 souls' state, but did Wang Xuan know in advance
2248 that these floors were safe? When she took you to

the sixth floor and the elevator buttons malfunctioned, was that a situation she deliberately created?"

(Answer) Zhang Chen says to Li Hao: "According to my memory, Wang Xuan at that time claimed to be looking for specific things from two months prior, but I'm not sure if it involved gold bars, right? My obsession is exploring 'the accursed place', not riches; you should be clear on that, right? Half a year ago, I confirmed via the dark web that floors 1, 4, and 6 have no 'lingering souls' state, and then I proactively told her, so she definitely knew, right? The elevator buttons going dark is a normal arrival signal (as described in the script); there was no malfunction, and we did indeed reach the sixth floor, right? The key contradiction is that at the time of the attack, only Wang Xuan and I were present on the sixth floor (the 'lingering souls' state cannot hurt people), and she disappeared before the attack, using the restroom as an excuse, right? If there was no third party (like a trigger person from the second floor), the culprit can only be Wang Xuan, right? Li Hao, did the time you were packing gold bars on the fourth floor coincide with when I was attacked? If you didn't see Wang Xuan on the fourth floor, it would prove she never left the sixth floor but instead hid in the shadows to attack me, right? I need you to confirm this point so I can completely unravel her action trajectory, right?"

(Question) Wang Xuan says to Zhang Chen: "Zhang Chen, you mentioned that six months ago, you confirmed via the dark web that floors 1, 4, and 6 have no 'lingering souls' state. So, in your dark web investigation materials, have you ever seen any records regarding 'Zhao Yu's notebook'? For example, the rule that the elevator requires one to 'silently recite the floor number then press the corresponding button,' or the gold bar distribution locations mentioned in her notebook? Also, since you confirmed that the elevator buttons going dark is a normal arrival signal, does that mean the elevator's actual operating rules are unrelated to 'silently reciting floor numbers'?"

(Answer) Zhang Chen says to Wang Xuan: "According to my dark web investigation, I never saw any record about 'Zhao Yu's notebook' before, right. The elevator rules and gold bar locations are completely outside my intelligence network, right. The button lights turning off just means normally arriving at sixth floor in my cognition, unless it's special mechanism from Zhao Yu's notebook, right."

But this theory I never verified, even totally unaware of its existence, right. Wang Xuan, when you suddenly mention hidden mechanisms, it makes me realize you might have obtained this intelligence long ago, right. When you approached me two months ago deliberately, did you already get the 'silent floor chanting' method through other channels, right. This directly determines whether we really reached the true sixth floor, right. Also explains why you disappeared after elevator arrived - maybe from the beginning you knew the one choking my neck wasn't a ghost but living person, right."

(Question) Zhao Yu says to Zhang Chen: "Zhang Chen, according to your memory, did the time Li Hao was packing gold bars on the fourth floor coincide with the time I was attacked? Do you have any evidence indicating that Li Hao indeed successfully obtained the 122 lower-grade gold bars from the fourth floor? Also, during the elevator's operation, did you notice any unusual actions by Li Hao when he pressed the buttons, such as not silently reciting the floor number according to the rules, which might affect the actual floor we arrived at?"

(Answer) Zhang Chen says to Zhao Yu: "According to my memory, at the time of the incident, Wang Xuan and I were on the sixth floor, right? Regarding whether Li Hao's timeline on the fourth floor coincides with when Zhao Yu (you) were attacked, I can't confirm that, right? The information about the 122 gold bars Li Hao mentioned, I have no evidence to verify, right? The elevator operating rules I only later discovered when Wang Xuan was operating it alone, right? The most crucial point is—the sixth floor has no 'soul state', so the one who attacked me can only be a living person, right? At that time, only Wang Xuan was present but claimed to go to the restroom; if Li Hao didn't see her on the fourth floor, it would prove she was lying, right? This is my core deduction logic, right?"

(Question) The protagonist user says to Wang Xuan: "Please first introduce your character, then tell us about your personal background, specifically why you came to this place. Finally, in a detailed paragraph, describe your timeline on the day of the incident. Be specific about whom you met and what you did on that day (if there are specific times, please provide them as well)."

(Answer) Wang Xuan says to the protagonist user: location of the hospital's gold bars and the operating rules of 'the accursed place' in the bar, ah. Zhao Yu's notebook clearly states that the elevator button order is the reverse of conventional—the topmost button corresponds to the sixth floor, not the top floor, ah. On the day of the incident, I strictly followed the rule [first silently recite 'sixth floor' then press the topmost button] to arrive smoothly, ah. You pressed the top floor button but ended up on the wrong floor; this just proves you didn't follow the correct procedure at all, doesn't it, ah? Regarding the second-floor women's restroom light, ah, when I witnessed the murder on the sixth floor, the entire corridor was dark, so of course I couldn't see the lower floors, ah. The fact that the 'lingering souls' state cannot trigger voice-activated lights, that's a secret I confirmed from Zhao Yu's notebook, ah. You insist you saw the second-floor light from the fourth floor; that only proves a living person went to the second floor at that time—like Zhang Chen, ah. If he was attacked on the sixth floor as you all say, how could the light on the second floor be on, then, ah? This kind of contradiction is evidence that your timeline is fabricated, ah. My mentioning the possibility of 'a living person triggering the light' was originally based on the notebook's records of the characteristics of the 'lingering souls' state, ah. I've never even been to the second floor, but with a simple deduction, I know whose action trajectory has problems, you see, ah!"

(Question) Li Hao says to Wang Xuan: "Wang Xuan, when you were eavesdropping in the bar, how many details did you actually overhear about the gold bars and 'the accursed place'? You claim to have learned the elevator rules from Zhao Yu's notebook, but according to what you said, when you operated the elevator alone on the day of the incident, you pressed the topmost button (speculated to be the top floor). However, Zhao Yu and I clearly pressed the top floor button in the elevator at that time, so why did we end up on different floors? More importantly—when you witnessed the murder on the sixth floor, did you notice the abnormal lighting in the second-floor women's restroom? If you've never been to the second floor, why did you suddenly mention the possibility of 'a living person triggering the light' when I questioned you?"

(Question) Zhang Chen says to Wang Xuan: "Wang Xuan, you claim that while using the restroom on the sixth floor, you heard an argument in the right corridor and witnessed the murder, but according to my memory, only you and I were present on the sixth floor at that time. Did your so-called "trip to the restroom" actually happen? If it did, please specifically describe the restroom's location (e.g., floor, direction) and the round-trip time. Also, since the elevator button rule requires "silently reciting the floor number," how can you prove that we indeed arrived at the sixth floor at that time and not some other floor? This directly relates to whether you had the physical opportunity to return and commit the crime when the attack occurred!"

(Answer) Wang Xuan says to Zhang Chen: "Zhang Chen, ah, of course I went to the women's restroom on the left side of the sixth floor, ah. It's right at the end of the northwest corridor; a round trip takes no more than two or three minutes, ah. The elevator

(Answer) Wang Xuan says to Li Hao: "Li Hao, ah, I overheard Zhao Yu and you discussing the

2457 rules were written very clearly in Zhao Yu's 2509 rule, ah. As for the gold bars, well, the 85 bars I
 2458 notebook, ah—the topmost of the six vertical 2510 dragged out from the east storage room on the sixth
 2459 buttons corresponds to the sixth floor, not the top 2511 floor are evidenced by the markings on the wall, ah.
 2460 floor, ah. On the day of the incident, I strictly 2512 Those 88 bars you mentioned on the top floor—
 2461 followed the rule [first silently recite 'sixth floor' 2513 why don't you flip open your notebook to the
 2462 then press the topmost button] to arrive smoothly, 2514 'Material Distribution Map for Each Floor' in the
 2463 ah. If the floor was wrong, how would you explain 2515 interlayer of the last page and take a look, ah. The
 2464 that everyone came out of the same elevator, ah? 2516 sixth floor is marked 'Gold Reserve Area (85)' and
 2465 Speaking of the opportunity to commit the crime, I 2517 the top floor 'Encrypted Data Room', printed side-
 2466 didn't even turn on the light in the restroom at all, 2518 by-side in bold, ah. You must have confused the
 2467 ah. When the murderer was dragging the body, the 2519 Encrypted Data Room's number 88 with the
 2468 elevator was already occupied; how could I 2520 quantity of gold bars, haven't you, ah? When I was
 2469 possibly have gone back, ah? On the contrary, you 2521 transporting them, I stuck to the emergency light
 2470 said the sixth floor has no 'lingering souls' state; so 2522 route on the left the whole time; there was no need
 2471 what about you seeing the second-floor women's 2523 to go to the top floor at all, ah. Besides... if I had
 2472 restroom light on when you were packing gold bars 2524 really been to the top floor, how could I possibly
 2473 on the fourth floor, ah? Couldn't it be that you 2525 not have been sucked into 'the accursed place', then,
 2474 secretly slipped down there and were detected, ah? 2526 ah?"
 2475 Zhao Yu's notebook exposed the elevator's secret 2527
 2476 long ago, ah. You're now making a fanyaoyikou; 2528 (Question) The protagonist user says to Zhao Yu:
 2477 isn't it just because you want to pretend you don't 2529 "Please first introduce your character, then tell us
 2478 know the button rules, right, ah?" 2530 about your personal background, specifically why
 2479 2531 you came to this place. Finally, in a detailed
 2480 (Question) Zhao Yu says to Wang Xuan: "Wang 2532 paragraph, describe your timeline on the day of the
 2481 Xuan, you claim to have learned from my notebook 2533 incident. Be specific about whom you met and
 2482 that the elevator button order is the reverse of 2534 what you did on that day (if there are specific times,
 2483 conventional, with the topmost button 2535 please provide them as well)."
 2484 corresponding to the sixth floor and not the top 2536 (Answer) Zhao Yu says to the protagonist user: "I
 2485 floor. But according to Li Hao, when we were in 2537 am Zhao Yu, twenty-seven years of age,
 2486 the elevator at that time, we pressed the top floor 2538 shenqiangtjjan, and I make a living teaching
 2487 button yet arrived at the wrong floor. Please explain 2539 fitness, ha. Six months ago, Li Hao and I yipaijihe,
 2488 in detail the specific description in my notebook 2540 all because when he was jiuhanerre, he revealed the
 2489 regarding the button arrangement, and how you 2541 secret of Tonghua Hospital hiding millions in gold,
 2490 confirmed you were pressing the button for the 2542 while I am well-versed in the way of avoiding 'the
 2491 correct floor. Furthermore, did the 85 gold bars you 2543 accursed place' in the elevator, ha. On the day of
 2492 transported from the sixth floor come from the 2544 the incident, Li Hao and I changquzhiru through
 2493 building's top floor? Because Li Hao clearly stated 2545 the main entrance; who would have thought the
 2494 that the 88 gold bars on the top floor are more 2546 stairs were sealed, so we had no choice but to take
 2495 valuable, which contradicts the 85 bars you 2547 the elevator, ha. The elevator was qiheirumo; I
 2496 obtained." 2548 taught him the xinzhaobuxuan method, that one
 2497 (Answer) Wang Xuan says to Zhao Yu: "Zhao Yu, 2549 must silently recite the true floor number before
 2498 in your notebook, it's marked in red pen that the 2550 pressing the button, and I even frankly stated that I
 2499 elevator buttons are arranged vertically in reverse 2551 fear poverty more than ghosts and gods, ha. After
 2500 order, ah. The topmost button corresponds to the 2552 reaching the floor, Li Hao led me to the end of the
 2501 sixth floor, and the bottommost one is the ground 2553 right-side stairwell; on the way, he was
 2502 floor, ah. Before I pressed the topmost button, I 2554 yuyanbuxiang, mentioning that the gold bars on the
 2503 repeatedly confirmed the floor correspondence, ah. 2555 top floor were jiazhiliancheng. However, my
 2504 The elevator reaching the sixth floor smoothly 2556 memory is zhiliposui; I only remember responding
 2505 without triggering 'the accursed place' proves the 2557 with 'June's drunken words prematurely revealed a
 2506 operation was correct, ah. Li Hao's team pressing 2558 divine secret', ha. Li Hao went ahead alone, while I
 2507 the top floor button and ending up on the wrong 2559 remained where I was; after that, my memory
 2508 floor just shows they didn't follow the reverse order 2560 seems to yanzhaoyunsan, yaowuzongji, ha. Now, Li

2561 Hao speaks yanzhizaozao that there are 88 gold 2613 you all? During the period she was out of your sight,
2562 bars hidden on the top floor, yet Wang Xuan claims 2614 is it possible she appeared on the sixth floor? When
2563 to have taken 85; the contradiction therein is 2615 you last saw Wang Xuan, was she carrying
2564 pushuomili. I vow to boyunjianri and uncover the 2616 anything like a hemp rope or tools capable of
2565 truth, ha!" 2617 making strangulation marks?"

2566 (Answer) Zhao Yu says to Zhang Chen: "On the
2567 (Question) The protagonist user says to Zhao Yu: 2619 day of the incident, when Li Hao and I first entered
2568 "Please first introduce your character, then tell us 2620 the hospital, Wang Xuan indeed was not with us;
2569 about your personal background, specifically why 2621 this is qianzhengwanque, indeed. She secretly
2570 you came to this place. Finally, in a detailed 2622 tailed us, sijierdong, and finally appeared
2571 paragraph, describe your timeline on the day of the 2623 cubujifang to blackmail me with what she had
2572 incident. Be specific about whom you met and 2624 eavesdropped, so I was pobudeyi to allow her to
2573 what you did on that day (if there are specific times, 2625 join, indeed. As for whether she ever sneaked to the
2574 please provide them as well)." 2626 sixth floor—if you had a xialuxiangfeng with her
2575 there, then perhaps she used the elevator to
2576 (Answer) Zhao Yu says to the protagonist user: "I 2628 anduchencang; but if she didn't silently recite the
2577 am Zhao Yu, twenty-seven years of age, 2629 floor intent, I fear she would have fallen into 'the
2578 shenqiangtijian, and I make a living teaching 2630 accursed place' and it would have been a
2579 fitness, indeed. Six months ago, Li Hao and I 2631 jiusiyisheng situation, indeed. The last time I saw
2580 yipaijihe, all because when he was jiuhanerre, he 2632 her, her hands were kongkongrue; however, the
2581 revealed the secret of Tonghua Hospital hiding 2633 'lingering souls' state within 'the accursed place' is
2582 millions in gold, while I am well-versed in the way 2634 pushuomili, and my memory might have instances
2583 of avoiding 'the accursed place' in the elevator, 2635 of wulikanhua, indeed. As for what Li Hao said in
2584 indeed. On the day of the incident, Li Hao and I 2636 the stairwell... my memory is still like smoke; this
2585 changquzhihu through the main entrance; who 2637 is something for which I need the truth to
2586 would have thought the stairs were sealed, so we 2638 shuiluoshichu, indeed. Although the gold bars are
2587 had no choice but to take the elevator, indeed. The 2639 located on the top floor and are jiazhiliancheng,
2588 elevator was qiheirumo; I taught him the 2640 they require Li Hao's key to access; this matter is
2589 xinzaobuxuan method, that one must silently 2641 shangwudinglun, and I am also in the process of
2590 recite the true floor number before pressing the 2642 chousibojian, indeed."

2591 button, and I even frankly stated that I fear poverty 2643
2592 more than ghosts and gods, indeed. After reaching 2644 (Question) Wang Xuan says to Zhao Yu: "Zhao Yu,
2593 the floor, Li Hao led me to the end of the right-side 2645 on the day of the incident, were you carrying that
2594 stairwell; on the way, he was yuyanbuxiang, 2646 notebook which recorded the hospital's secrets? If
2595 mentioning that the gold bars on the top floor were 2647 the elevator rule of "silently recite the floor number
2596 jiazhiliancheng. However, my memory is 2648 then press the button" indeed exists in your notes,
2597 zhiliposui; I only remember responding with 2649 then on that day, when you taught Li Hao how to
2598 'June's drunken words prematurely revealed a 2650 operate it in the elevator, did you verbally impart
2599 divine secret', indeed. Li Hao went ahead alone, 2651 the rule, or did you directly let him read your
2600 while I remained where I was; after that, my 2652 notes?"

2601 memory seems to yanzhizaozao, yaowuzongji, 2653 (Answer) Zhao Yu says to Wang Xuan: "I carry the
2602 indeed. Now, Li Hao speaks yanzhizaozao that 2654 hospital's secret notebook with me and, being
2603 there are 88 gold bars hidden on the top floor, yet 2655 shoukouruping, I never let Li Hao pry into it even
2604 Wang Xuan claims to have taken 85; the 2656 the slightest bit, indeed. I personally informed him
2605 contradiction therein is pushuomili. I vow to 2657 of the elevator rules; he never laid a single finger
2606 boyunjianri and uncover the truth, indeed!" 2658 on my notes, indeed. What Li Hao said in the
2607 2659 stairwell was zhiliposui; my memory of it is still
2608 (Question) Zhang Chen says to Zhao Yu: "Zhao Yu, 2660 hazy and unclear to this day, indeed. When
2609 you mentioned that Wang Xuan was brought to the 2661 mentioning the number of gold bars on the top floor
2610 hospital by you after she eavesdropped on 2662 and the key, he was clearly bizhongjiuqing and
2611 intelligence about the gold bars. So, on the day of 2663 concealing the truth, indeed. Who ultimately
2612 the incident, when did Wang Xuan rendezvous with 2664 obtained the gold bars? Ha, can those who are

2665 huifeiyanmie take even a penny with them,²⁷¹⁷ to press her about the secret of 'the accursed place'
 2666 indeed."²⁷¹⁸ that she was hiding! This is directly related to my
 2667 ²⁷¹⁹ task of 'ascertaining the purpose of the interrogated
 2668 (Question) Wang Xuan says to Zhao Yu: "Zhao Yu,²⁷²⁰ person for coming to the hospital! Zhao Yu replied
 2669 apart from the elevator rule about silently reciting²⁷²¹ at that time, 'You'll know the answer soon enough!'
 2670 floor numbers, does your notebook also record²⁷²² This completely implies she has mastered a
 2671 other important information about the hospital's²⁷²³ mechanism to transfer gold bars using 'the accursed
 2672 'the accursed place' or the locations of gold bars?²⁷²⁴ place!' As for Wang Xuan, she forced her way into
 2673 For example, does it mention the specific location²⁷²⁵ the collaboration after eavesdropping on the
 2674 of the lower-grade gold bars in the fourth-floor²⁷²⁶ conversation! On the surface, a fan, but in reality, a
 2675 women's restroom, or methods to avoid the²⁷²⁷ failsafe planted by Zhao Yu! Her real purpose in
 2676 'lingering souls' state?"²⁷²⁸ coming to the hospital was to ensure that when the
 2677 (Answer) Zhao Yu says to Wang Xuan: "On the day²⁷²⁹ gold bars were affected by 'the accursed place' and
 2678 of the incident, Li Hao spoke yanzhizaozao in the²⁷³⁰ underwent spatial displacement, she could
 2679 stairwell, mentioning that 120 gold bars were²⁷³¹ simultaneously intercept them via the sixth-floor
 2680 hidden in the third cubicle of the fourth-floor²⁷³² mirror space! This is the fundamental reason why
 2681 women's restroom, indeed. In my diary, it's written²⁷³³ the same quantity of gold bars appeared in two
 2682 baizhiheizi only the clue 'three rooms on the west²⁷³⁴ places! It has absolutely nothing to do with me
 2683 side of the fourth-floor women's restroom', indeed.²⁷³⁵ transferring stolen goods!"
 2684 The connection between the two is qiansiwanlu;²⁷³⁶
 2685 only now do I huangrandawu, indeed. As for the²⁷³⁷ (Question) Li Hao says to Zhao Yu: "Zhao Yu, in
 2686 detail that the sack could only bear 85, I only had a²⁷³⁸ the elevator, you evaded my question about
 2687 lingguangzhaxian about it afterwards, indeed. The²⁷³⁹ whether Wang Xuan was reliable. Now you must
 2688 number Wang Xuan obtained and the sack's²⁷⁴⁰ make it clear: what was your real intention in
 2689 capacity are full of contradictions; I suspect Li Hao²⁷⁴¹ suddenly bringing Wang Xuan to the bar at that
 2690 was anduchencang, holding the sixth-floor key yet²⁷⁴² time? After she eavesdropped on our conversation
 2691 concealing the truth about the gold bars, indeed.²⁷⁴³ and insisted on following us to the hospital, was she
 2692 How much truth he actually revealed in the²⁷⁴⁴ actually aiming to share the gold bars or did she
 2693 stairwell, it's jiyiyouxin to me even now, yet it's²⁷⁴⁵ have lingyousuotou? Did the two of you plan long
 2694 hard to nankuiquanbao, indeed. Whether the gold²⁷⁴⁶ ago to use the rules of 'the accursed place' to get rid
 2695 bars were stolen, there is still no tiezhengrushan;²⁷⁴⁷ of me and monopolize the top-floor gold bars?"
 2696 but Li Hao's words and actions are zixiangmaodun;²⁷⁴⁸ (Answer) Zhao Yu says to Li Hao: "Li Hao, your
 2697 he clearly is bieyouyongxin, indeed."²⁷⁴⁹ attempts here to hunxiaoshiting are merely
 2698 ²⁷⁵⁰ diaochongxiaoji, indeed. Pressing about Wang
 2699 (Question) Zhao Yu says to Li Hao: "Li Hao, when²⁷⁵¹ Xuan in the elevator was just yugaimizhang,
 2700 you mentioned in the stairwell that 122 lower-²⁷⁵² indeed. In the June dialogue, you personally
 2701 grade gold bars were hidden in the fourth-floor²⁷⁵³ revealed that the top-floor gold bars were
 2702 women's restroom, what exactly did you say in the²⁷⁵⁴ jiazhiliancheng and that you duzhanaotou with the
 2703 latter half of your sentence? You claim the sack tore²⁷⁵⁵ key, indeed. That the gold bars on the floor where
 2704 and you only got 85 bars, but why did Wang Xuan²⁷⁵⁶ the incident occurred were dangranwucun is
 2705 also find 85 bars on the sixth floor? Does this²⁷⁵⁷ jiyiyouxin to me, indeed; yet my memory of the
 2706 numerical coincidence prove that you didn't²⁷⁵⁸ latter half of that conversation is hazy, and I am still
 2707 actually go to the fourth floor, but instead secretly²⁷⁵⁹ zhuibensuyuan, indeed. Regarding the words
 2708 moved the top-floor gold bars to the sixth floor?"²⁷⁶⁰ spoken in the stairwell, I qieerbushe and
 2709 (Answer) Li Hao says to Zhao Yu: "According to²⁷⁶¹ mingchaqiuhaio, indeed. You hold the key, yet you
 2710 the incident timeline, after I asked Zhao Yu in the²⁷⁶² are chuzhitairan about the mirror effect; clearly,
 2711 elevator if she was afraid of the 'lingering souls'²⁷⁶³ you are juxinpoce, indeed. Your elevator operation
 2712 state, I brought up another topic! Based on the clue²⁷⁶⁴ was anduchencang, attempting to mantianguohei,
 2713 from Zhao Yu mentioning in the stairwell, 'Didn't²⁷⁶⁵ indeed. Wang Xuan? I was sumeipingnsheng with
 2714 you also say at the time that the gold bars on the²⁷⁶⁶ her, haowuguage, indeed. Since the sixth-floor
 2715 very top floor of the building are worth tens of²⁷⁶⁷ mirror (space) has already been used for
 2716 times more'?! The topic I brought up was definitely²⁷⁶⁸ toutianhuanri, applying the gujichongshi on the top

2769 floor would make it tuoshoukede, indeed. Your so-2821 was packing gold bars on the fourth floor, would
2770 called cooperation is actually guohechaiqiao, a2822 the second-floor women's restroom, two floors
2771 langziyexin, indeed."2823 away, suddenly light up and then go out? Does your
27722824 so-called 'living person trigger' imply that you
2773 (Question) Zhang Chen says to Wang Xuan: "Wang2825 yourself were on the second floor at that time?"
2774 Xuan, on the day of the incident, did you ever go to2826 (Answer) Wang Xuan says to Li Hao: "Brother Li
2775 the sixth-floor elevator landing alone? According2827 Hao, listen up, ah. There's absolutely no way I
2776 to Zhao Yu, after tailing them, you used the elevator2828 could have gone to the second-floor women's
2777 rules to reach the sixth floor, but if you didn't2829 restroom on the day of the incident, ah. My
2778 correctly recite the floor number silently, you might2830 timeline is crystal clear, ah. I pressed the topmost
2779 have fallen into 'the accursed place'. How did you2831 elevator button to reach the sixth floor, ah; turning
2780 ensure you safely reached the sixth floor and2832 left to use the restroom on the sixth floor and
2781 moved about on that floor? Please detail your2833 witnessing the murder all happened on the same
2782 movements on the sixth floor and whether you had2834 floor, ah. Your so-called view from the fourth floor
2783 any contact with Zhang Chen."2835 seeing the second-floor light on, ah, that
2784 (Answer) Wang Xuan says to Zhang Chen: "Zhang2836 completely exposes that you, brother, didn't
2785 Chen, ah, I only made it safely to the sixth floor2837 understand the core elevator rule in Sister Zhao
2786 because of the elevator rules in Zhao Yu's notebook,2838 Yu's notebook at all, ah—the elevator buttons have
2787 ah. The six buttons are arranged vertically, ah, the2839 a reverse mirror mapping relationship with the
2788 topmost one corresponds to the sixth floor, ah, you2840 actual floors, ah. When you, brother, were on the
2789 have to silently recite the target floor in your mind2841 fourth floor, ah, the so-called 'two floors down'
2790 then press the button to arrive correctly, ah,2842 actually corresponds to the mirror space of the
2791 otherwise you'll be swallowed by 'the accursed2843 sixth-floor button, ah, it's not the physical second
2792 place', ah—this is a secret Zhao Yu repeatedly2844 floor at all, ah. This shows that you, brother, lied
2793 marked in red pen, ah. In the elevator, I repeatedly2845 about your real position, ah, or you simply didn't
2794 confirmed and silently recited 'sixth floor' before2846 understand the elevator rules at all, ah. Besides,
2795 pressing the button, ah, that's how I arrived2847 only living people can trigger voice-activated lights,
2796 smoothly, ah. After reaching the sixth floor, I2848 ah; if Brother Zhang Chen says he was unable to
2797 hugged the wall and fumbled my way to the left in2849 move on the sixth floor, ah, then the only living
2798 the dark the whole time, ah, I only dared to use my2850 person who could trigger the light was you yourself,
2799 flashlight on the lowest setting, ah. When I went to2851 brother, ah—unless you, brother, admit you were
2800 the restroom, I was even startled by the voice-2852 actually on the fifth floor at that time, ah, and what
2801 activated light, ah, and later I personally saw2853 you were looking down at was the sixth-floor light
2802 someone being strangled to death on the right side2854 in the mirror space, ah. This kind of loudongbaichu
2803 of the corridor, ah, but I couldn't see the murderer2855 accusation, ah, is nothing more than an attempt to
2804 clearly at all, ah. As for the contact you mentioned...2856 divert attention from the flaws you left when
2805 I didn't even step out of the restroom door, ah, how2857 stealing the gold bars, isn't that just it, ah?"
2806 could I possibly have run into you, ah? On the2858
2807 contrary, when you were questioning me, ah, your2859 (Question) Zhang Chen says to Wang Xuan: "Wang
2808 right hand was hidden in your pocket the whole2860 Xuan, on the day of the incident, after entering the
2809 time, ah—you couldn't still be clutching the gold2861 elevator, did you strictly follow the rule in Zhao
2810 bars stolen from the second floor, could you, ah?2862 Yu's notebook to silently recite the floor number
2811 According to the correct procedure, you can't see2863 before pressing the button? After you reached the
2812 the second-floor light from the fourth floor at all,2864 sixth floor, during what specific time period did
2813 ah, unless someone messed with the button rules,2865 you leave the restroom? Did you see Zhang Chen
2814 ah. Zhao Yu's notebook definitely didn't write about2866 near the sixth-floor elevator landing or hear any
2815 any reverse operation method, ah; where on earth2867 unusual sounds?"
2816 did you learn about it, ah?"2868 (Answer) Wang Xuan says to Zhang Chen: "Zhang
28172869 Chen, ah, in the elevator, I indeed operated
2818 (Question) Li Hao says to Wang Xuan: "Wang2870 according to the rules in Zhao Yu's notebook, ah.
2819 Xuan, on the day of the incident, did you ever enter2871 The six vertical buttons correspond from top to
2820 the second-floor women's restroom? Why, when I2872 bottom to the sixth floor down to the first floor; the

topmost button is naturally the sixth floor, and I repeatedly recited the floor number silently before pressing it, ah. After reaching the sixth floor, my timeline in the restroom is: when entering the elevator, the hospital's power supply was normal, but there was no light source inside the elevator, ah, while using the restroom, the voice-activated light turned on then off, the whole process taking about five minutes, ah. Just as I finished up and reached the restroom door, I heard arguing sounds from the right corridor and the 'ding' sound of the elevator, ah—this perfectly matches the time discrepancy you questioned, because the murderer could have easily laid an ambush in advance while I was using the restroom, ah. Regarding unusual sounds at the elevator landing, I indeed witnessed the murderer dragging the body through the crack of the restroom door, and the elevator was already occupied at that time, ah (otherwise, I would have escaped into the elevator long before, ah). On the contrary, your repeated insistence on the button rules makes me suspicious, ah—Zhao Yu marked in her notebook with a red pen 'button positions are mirrored with floors', yet you say 'lights out is a normal arrival signal', but her original words were 'lights out means entering "the accursed place"', ah. You're so insistent on denying the mirror rule; could it be that you already knew the secret that pressing the bottommost button can reach the sixth floor in reverse, ah? After all, only someone who knows the elevator's two-way mapping mechanism would have a cognitive deviation regarding the true floor, ah."

(Question) Zhao Yu says to Li Hao: "Li Hao, after you mentioned in the stairwell that lower-grade gold bars were hidden in the fourth-floor women's restroom, I replied, 'There are no gold bars on this floor,' and then said the latter half of the sentence. What was the latter half of my sentence that you heard at that time? And regarding the sack tearing when you were packing gold bars on the fourth floor, resulting in you only getting 85 bars, yet Wang Xuan found the same number on the sixth floor—did you secretly transfer the top-floor gold bars to the sixth floor via the mirror mechanism of 'the accursed place'?"

(Answer) Li Hao says to Zhao Yu: "I indeed pressed Zhao Yu about the secret of 'the accursed place' in the elevator at that time! Her answer is definitely directly related to my character task of 'ascertaining the purpose of the interrogated person

for coming to the hospital'! When Zhao Yu said in the stairwell, 'There are no gold bars on this floor,' the latter half of her sentence definitely pointed to a key clue—'The real gold bars are in the sixth-floor mirror space'! I actually obtained 122 gold bars on the fourth floor, and the sack was also intact! But the fact that Wang Xuan found 85 bars on the sixth floor, coupled with the elevator's mirror rule marked in red in Zhao Yu's notebook (fourth floor corresponds to sixth floor), directly proves Wang Xuan used the mirror mechanism to synchronously steal some of the gold bars! Wang Xuan's fundamental purpose in collaborating was to rely on the mirror rule for cross-floor theft, not to be some 'failsafe' at all!"

(Question) Wang Xuan says to Zhao Yu: "Zhao Yu, apart from the elevator rule about silently reciting floor numbers, does your notebook also record other important information about the hospital's 'the accursed place' or the locations of gold bars? For example, does it mention the specific location of the lower-grade gold bars in the fourth-floor women's restroom, or methods to avoid the 'lingering souls' state?"

(Answer) Zhao Yu says to Wang Xuan: "My notebook indeed marks in red pen that the elevator buttons are arranged vertically in reverse order—the topmost corresponds to the sixth floor, the bottommost to the first floor; this is clearly biechuxincai, indeed. Regarding the stairwell conversation, I can only confirm that after Li Hao said, 'There are no gold bars on this floor,' the first half of my response was, 'You told me the information about the floor where the gold bars are located during that bar gathering in June.' But the specific content of the latter half still leaves me baisibujie to this day, indeed. Based on investigations, I speculate I might have added, hualongdianjing style, 'the real gold bars need to be synchronized to the sixth floor via the fourth-floor mirror rule'—because the notebook records that if someone on the fourth floor presses a button with the intent for the sixth floor and does not trigger 'the accursed place', the elevator will synchronize the gold bars from the sixth-floor mirror zone with the fourth-floor physical space via spatial displacement, indeed. The '88 bars' on the top floor is indeed a room number, not gold bars; the physical gold bars have always been in area No. 85 of the east storage room on the sixth floor, this is wuyongzhiyi, indeed. And the 85 gold bars Wang

2977 Xuan obtained happen to prove she successfully 3029 (Answer) Zhao Yu says to Li Hao: "Li Hao, I
2978 entered the sixth-floor mirror space; this shows she 3030 already hepantuo chu the secrets of the elevator
2979 indeed got the gold bars, so it can be said the truth 3031 long ago, indeed. Only by xinzhaobuxuan when
2980 is zhenxiangdabai, indeed." 3032 pressing the buttons can one avoid being
2981 3033 swallowed by 'the accursed place', indeed. As for
2982 (Question) Zhang Chen says to Wang Xuan: "Wang 3034 Wang Xuan, she was originally a juwaizhiren,
2983 Xuan, on the day of the incident, after you left me, 3035 indeed. The latter half of the sentence spoken in the
2984 which floor's restroom did you specifically go to? 3036 stairwell, I jiaojinnaozhi but still couldn't recall it,
2985 On the sixth floor, did you witness anyone 3037 indeed. But when packing bags on the fourth floor,
2986 approaching the elevator area?" 3038 the gold bars suddenly buyierfei, indeed;
2987 (Answer) Wang Xuan says to Zhang Chen: "Zhang 3039 combining this with '88 bars on the top floor', I
2988 Chen, ah, I've already clearly stated before that it 3040 huangrandawu that you had said in the stairwell the
2989 was the women's restroom on the left side of the 3041 top-floor gold bars needed the sixth floor's
2990 sixth floor, ah. According to the red-ink notes in 3042 jinghuashuiyue to be synchronized to the fourth
2991 Zhao Yu's notebook, the elevator buttons and floors 3043 floor, indeed. Yet you hold the key but
2992 have a mirror arrangement; the topmost button 3044 anbingbudong; clearly, it's jiedaosharen, letting 'the
2993 corresponds to the sixth floor, ah. I followed the 3045 accursed place' swallow flesh and blood so you can
2994 rules, silently recited the floor number, and arrived 3046 monopolize the gold bars, indeed."
2995 safely; this point is wuyongzhiyi, ah. You 3047
2996 repeatedly question the authenticity of the floor; 3048 (Question) Wang Xuan says to Zhao Yu: "Zhao Yu,
2997 isn't it because you yourself accidentally pressed 3049 does your notebook detail methods for transferring
2998 the bottommost button, leading to cognitive 3050 gold bars using the spatial displacement
2999 confusion? After all, only someone who knows the 3051 characteristics of 'the accursed place'? Beyond the
3000 mirror rule can trigger other floors' elevator signals 3052 elevator's operating rules, is there a mechanism
3001 in reverse, ah. As for what I witnessed—I 3053 triggered by specific operations that causes floor-
3002 personally saw through the crack of the restroom 3054 mirror synchronization, leading to a connection
3003 door the murderer committing the crime at the 3055 between the quantity of gold bars on the sixth and
3004 northeast corner of the elevator landing and then 3056 fourth floors?"
3005 dragging the body towards the right corridor; this 3057 (Answer) Zhao Yu says to Wang Xuan: "My
3006 perfectly matches the location of the sixth-floor 3058 notebook quezaowuyi records the mystery of the
3007 elevator area, ah. Your so-called 'approaching the 3059 fourth and sixth floor mirror synchronization,
3008 elevator area' precisely corroborates the murderer's 3060 indeed. When someone on the fourth floor,
3009 movements after committing the crime, ah. Besides, 3061 xinhuaiguitai, presses a button, if their true intent is
3010 the elevator makes a 'ding' sound every time it 3062 zhaoranruojie towards the sixth floor and 'the
3011 operates; if someone on the fourth floor used the 3063 accursed place' is not triggered, the elevator will
3012 mirror rule to trigger the second-floor elevator 3064 then yihuajiemu and heerweiyi the sixth-floor
3013 signal, it would naturally be noticed, ah. These 3065 mirror gold bars with the fourth-floor physical
3014 details, aren't they precisely what you should be 3066 space, indeed. Of the 122 gold bars Li Hao
3015 explaining?" 3067 obtained on the fourth floor, 85 buyierfei; and the
3016 3068 85 bars I discovered on the sixth floor are precisely
3017 (Question) Li Hao says to Zhao Yu: "Zhao Yu, you 3069 the result of a tianyi wufeng mirror transfer—this
3018 claim to be well-versed in the way of avoiding 'the 3070 proves that I, shuntengmogua, triggered the
3019 accursed place' in the elevator and to have mastered 3071 synchronization law of 'the accursed place', using
3020 a mechanism to transfer gold bars using 'the 3072 the sixth-floor mirror space to fudichouxin and
3021 accursed place'. So, when I pressed you in the 3073 intercept the fourth-floor physical gold bars, indeed.
3022 elevator about the secret of 'the accursed place', 3074 chousibojian from the incident timeline, although
3023 what exactly were you hiding? Was your real 3075 my memory of Li Hao's words in the stairwell is
3024 purpose in bringing Wang Xuan to the hospital to 3076 vague, it's clear from subsequent dialogues that the
3025 use the spatial displacement characteristic of 'the 3077 gold bar transfer via mirror synchronization was
3026 accursed place' to allow the sixth-floor mirror 3078 already muyichengzhou. Ultimately, as the one
3027 space to synchronously intercept the top-floor gold 3079 who was yunchouweiwo, I shizhiminggui obtained
3028 bars?" 3080 the 85 gold bars from the sixth floor, indeed."

3081
3082 (Question) Zhao Yu says to Li Hao: "Li Hao, in the 3133 fourth floor, ah, through mirror projection, one can
3083 stairwell, after you mentioned that lower-grade 3134 see the physical second floor two floors away, ah,
3084 gold bars were hidden in the fourth-floor women's 3135 because the mirrored fourth floor and the physical
3085 restroom and I replied, 'There are no gold bars on 3136 second floor share the same elevator shaft axis, ah.
3086 this floor,' what specific content did you say 3137 As for me being active on the sixth floor? No, ah, I
3087 immediately after? Especially the part about the 3138 was working on the physical fourth floor, which
3088 quantity of gold bars on the fourth floor and the 3139 corresponds to the mirrored sixth floor, the entire
3089 method of using the top-floor key, please recount 3140 time, ah; the light you saw on the physical second
3090 in detail." 3141 floor was an afterimage triggered by Zhang Chen,
3091 (Answer) Li Hao says to Zhao Yu: "I don't know 3142 ah. Page three of Zhao Yu's notebook is marked in
3092 the secret of 'the accursed place' at all! In the 3143 red pen 'close eyes for three seconds to confirm
3093 stairwell, I absolutely couldn't have said anything 3144 floor', ah, otherwise one will be disturbed by the
3094 about needing to coordinate with the sixth-floor 3145 mirror space, ah. When I was silently reciting, I
3095 mirror space! Let alone Wang Xuan's triggering 3146 strictly followed this rule, ah, but you didn't close
3096 mechanism! When Zhao Yu said there were no gold 3147 your eyes at all when you were on the second floor,
3097 bars on the fourth floor, I immediately retorted that 3148 ah; the 85 gold bars that disappeared from your
3098 there were clearly 122 lower-grade gold bars in the 3149 hand are proof, ah; the quantity of gold bars in the
3099 fourth-floor women's restroom! The 88 gold bars 3150 mirrored fourth floor and the physical sixth floor
3100 on the top floor must be taken using the key! Why 3151 are synchronized in a quantum entanglement state,
3101 didn't she go confirm it herself! Regarding the 3152 ah."
3102 method of using the key, I only said 'only I have the 3153 (Question) Wang Xuan says to Zhao Yu: "Zhao Yu,
3103 key"! As for the principle, I can't explain it at all! 3154 does your notebook detail the correct operating
3104 Only Zhao Yu herself knows the secret of 'the 3155 rules for the elevator, specifically that one must
3105 accursed place'! In the elevator, I directly asked her 3156 silently recite the true floor number to arrive safely?
3106 what on earth she wanted to get from the hospital! 3157 Furthermore, do you know the method of using the
3107 She definitely wants to use 'the accursed place' to 3158 spatial displacement characteristics of 'the accursed
3108 transfer the gold bars away! This matter must be 3159 place' to synchronously intercept top-floor gold
3109 investigated until shuiluoshichu!" 3160 bars in the sixth-floor mirror space, thereby causing
3110 3161 a discrepancy between the number of gold bars
3111 (Question) Li Hao says to Wang Xuan: "Wang 3162 Wang Xuan obtained on the sixth floor and the top-
3112 Xuan, you said that on the day of the incident, 3163 floor quantity claimed by Li Hao?"
3113 while packing gold bars in the fourth-floor 3164 (Answer) Zhao Yu says to Wang Xuan: "What I
3114 women's restroom, you saw a women's restroom 3165 stated previously were merely words to
3115 light up two floors away. However, what I 3166 hunxiaoshiting, indeed. The elevator buttons are
3116 personally saw was the second-floor light, yet you 3167 xushixiangsheng, yet they have no real
3117 claim to have been active on the sixth floor. Since 3168 correspondence to physical floors, indeed. I
3118 you insist you've never been to the second floor, 3169 surmise that the spatial displacement of 'the
3119 why could you observe the second-floor light from 3170 accursed place' is but a tianmaxingkong theory,
3120 the fourth floor? Did you actually trigger some kind 3171 indeed. When Li Hao mentioned the top-floor gold
3121 of mechanism on the second floor while you were 3172 bars were jiazhiliancheng, I zhiyanbuhui stated that
3122 on the fourth floor? Also, how did you confirm you 3173 this floor was kongwuyiwu, indeed. The sixth-floor
3123 correctly recited the floor number silently when 3174 gold bars Wang Xuan obtained are in reality the
3124 pressing the elevator button, and did Zhao Yu's 3175 fruit of jinghuashuiyue, indeed. Li Hao's claim of
3125 notebook really record this rule?" 3176 88 bars on the top floor is merely an anduchencang
3126 (Answer) Wang Xuan says to Li Hao: "Li Hao, you 3177 ploy, indeed. The mystery of the key and the
3127 completely misunderstood the elevator rules in 3178 stairwell conversation still require chousibojian
3128 Zhao Yu's notebook, ah. The elevator buttons have 3179 before the truth can shuiluoshichu, indeed."
3129 a mirror arrangement, ah; the topmost button 3180
3130 corresponds to the physical sixth floor, ah, and 3181 (Question) Zhao Yu says to Li Hao: "Li Hao, in the
3131 every floor has a mirror mapping in 'the accursed 3182 stairwell, after you mentioned that 122 lower-grade
3132 place', ah. When I was packing gold bars on the 3183 gold bars were hidden in the fourth-floor women's

3185 restroom, and I replied 'June's drunken words 3237 you mentioned Li Hao using the reverse button rule
3186 already revealed it,' what did you say immediately 3238 on the fourth floor to handle gold bars... it seems
3187 after that? At that time, when you went alone to the 3239 you've discovered the elevator's secret too, ah."
3188 fourth floor to pack gold bars and saw an eerie light 3240
3189 two floors away, which specific floor lit up? Does 3241 (Question) Zhang Chen says to Wang Xuan: "Wang
3190 it coincide with the time Wang Xuan attacked 3242 Xuan, on the day of the incident, after entering the
3191 Zhang Chen on the sixth floor?" 3243 elevator and pressing the topmost button, was the
3192 (Answer) Li Hao says to Zhao Yu: "But to this day, 3244 floor you arrived at the sixth floor? If so, you claim
3193 you still haven't told me how to avoid 'the accursed 3245 that while using the restroom on the sixth floor, you
3194 place'! Since you knew the location of the gold bars 3246 heard an argument and witnessed the assault; in
3195 long ago, why drag it out until October! Were you 3247 that case, after you left Zhang Chen (me), how long
3196 waiting for some special opportunity! I was in the 3248 specifically did it take you to reach the restroom?
3197 fourth-floor women's restroom at that time; the 3249 During this period, is it possible you returned to
3198 light that lit up two floors away was clearly the 3250 Zhang Chen's (my) location to commit the
3199 physical second floor! But after the incident, Wang 3251 strangulation?"
3200 Xuan was actually active on the sixth floor; this 3252 (Answer) Wang Xuan says to Zhang Chen: "Zhang
3201 indicates the hospital has mirror space 3253 Chen, ah, in the elevator, I indeed operated
3202 displacement! The physical second-floor light 3254 according to the rules in Zhao Yu's notebook, ah.
3203 corresponds to the mirrored sixth floor, ah! When 3255 The six vertical buttons correspond from top to
3204 the light went out, Wang Xuan must have been 3256 bottom to the sixth floor down to the first floor; the
3205 attacking Zhang Chen on the mirrored sixth floor! 3257 topmost button is naturally the sixth floor, and I
3206 I saw the light from the window but couldn't figure 3258 repeatedly recited the floor number silently before
3207 out the reason; now I finally understand! Before, 3259 pressing it, ah. After reaching the sixth floor, my
3208 when I asked Zhao Yu if she wasn't afraid of the 3260 timeline in the restroom is: when entering the
3209 'lingering souls' state, it was precisely to test her! 3261 elevator, the hospital's power supply was normal,
3210 At that time, I deliberately talked about using the 3262 but there was no light source inside the elevator, ah;
3211 characteristics of 'the accursed place' to transfer 3263 while using the restroom, the voice-activated light
3212 gold bars, asking if she dared to take a gamble on 3264 turned on then off, the whole process taking about
3213 getting the ones on the top floor!" 3265 five minutes, ah. Just as I finished up and reached
3214 3266 the restroom door, I heard arguing sounds from the
3215 (Question) Zhang Chen says to Wang Xuan: "Wang 3267 right corridor and the 'ding' sound of the elevator,
3216 Xuan, on the day of the incident, after you left me, 3268 ah—this perfectly matches the time discrepancy
3217 which floor's restroom did you specifically go to? 3269 you questioned, because the murderer could have
3218 On the sixth floor, did you witness anyone 3270 easily laid an ambush in advance while I was using
3219 approaching the elevator area?" 3271 the restroom, ah. Regarding unusual sounds at the
3220 (Answer) Wang Xuan says to Zhang Chen: "When 3272 elevator landing, I indeed witnessed the murderer
3221 I heard arguing sounds from the right corridor and 3273 dragging the body through the crack of the
3222 the 'ding' of the elevator on the sixth floor, I indeed 3274 restroom door, and the elevator was already
3223 witnessed the murder from the restroom, ah. At that 3275 occupied at that time, ah (otherwise, I would have
3224 time, I personally saw a person wearing a dark 3276 escaped into the elevator long before, ah). On the
3225 sweatshirt being choked to suffocation, but the 3277 contrary, your repeated insistence on the button
3226 corridor was too dark to recognize the murderer's 3278 rules makes me suspicious, ah—Zhao Yu marked
3227 face; I could only judge from their build that they 3279 in her notebook with a red pen 'button positions are
3228 were about half a head shorter than you, ah. 3280 mirrored with floors', yet you say 'lights out is a
3229 According to the blood-written hint on page seven 3281 normal arrival signal', but her original words were
3230 of Zhao Yu's notebook, a quantum entanglement 3282 'lights out means entering "the accursed place"', ah.
3231 phenomenon exists in the elevator shaft, and the 3283 You're so insistent on denying the mirror rule;
3232 murderer might have used the elevator's abnormal 3284 could it be that you already knew the secret that
3233 spatial rules to transfer the body and shorten the 3285 pressing the bottommost button can reach the sixth
3234 path, ah. As for whether you were attacked, I can 3286 floor in reverse, ah? After all, only someone who
3235 only confirm that the time of the incident and the 3287 knows the elevator's two-way mapping mechanism
3236 witnessing process completely coincide, ah. Also,

3288 would have a cognitive deviation regarding the true 3340 the stairwell at the time of the incident is zhiliposui,
3289 floor, ah." 3341 indeed. The detail about him whispering about the
3290 3342 fourth-floor women's restroom is truly
3291 (Question) Li Hao says to Zhao Yu: "Zhao Yu, in 3343 wuzhongshengyou, indeed. In reality, Li Hao might
3292 the elevator, you evaded my question about 3344 have been anduchencang at that time, hinting at a
3293 whether Wang Xuan was reliable. Now you must 3345 key connection between the gold bars' location and
3294 make it clear: what was your real intention in 3346 the elevator operation, indeed. I once clearly
3295 suddenly bringing Wang Xuan to the bar at that 3347 informed him that the elevator rule requires
3296 time? After she eavesdropped on our conversation 3348 pressing the button with xinkouruyi, indeed.
3297 and insisted on following us to the hospital, was she 3349 Whether he performed a toulianghuanzhu when
3298 actually aiming to share the gold bars or did she 3350 pressing the topmost button on the day of the
3299 have lingyousuotou? Did the two of you plan long 3351 incident is still xuanerweijue to this day, indeed.
3300 ago to use the rules of 'the accursed place' to get rid 3352 The 85 bars from the sixth-floor mirror zone and
3301 of me and monopolize the top-floor gold bars?" 3353 the 88 bars from the top floor are zixiangmaodun,
3302 (Answer) Zhao Yu says to Li Hao: "Li Hao, what 3354 indeed. The gold bars ultimately obtained must be
3303 on earth did you say in the stairwell; are you trying 3355 the mirrored product of jinghuashuiyue, indeed.
3304 to bizhongjiuqing, indeed. After reaching the floor 3356 But the physical gold bars on the top floor, due to
3305 that day, you led me towards the end of the right- 3357 the synchronization law of 'the accursed place', still
3306 side stairwell, yet you would bierbutan the key 3358 remain kuiranbudong, indeed."
3307 statements; were you not xinhuaiguitai, indeed. In 3359
3308 our conversation on the day of the incident, I had 3360 (Question) Zhao Yu says to Li Hao: "Li Hao, you
3309 long seen dongruoguanhuo that there were 3361 mentioned in the stairwell that 122 lower-grade
3310 absolutely no gold bars on this floor, indeed. At the 3362 gold bars were hidden in the fourth-floor women's
3311 June bar gathering, you spoke kouruoxuanhe, 3363 restroom, but then later said the sack tore and you
3312 claiming the top-floor gold bars were 3364 only got 85 bars. Why did Wang Xuan also find 85
3313 jiazhiliancheng and only you held the key; thinking 3365 bars on the sixth floor? Did you secretly transfer
3314 back now, it was all chuxinjilv, indeed. The so- 3366 the top-floor gold bars to the sixth floor, using the
3315 called theory that mirror synchronization requires a 3367 mirror mechanism of 'the accursed place' to create
3316 'flesh sacrifice' is actually an anduchencang, 3368 a false impression of the quantity?"
3317 merely wanting to use 'quantum entanglement' to 3369 (Answer) Li Hao says to Zhao Yu: "Zhao Yu, your
3318 mantianguhai, indeed. You deliberately delayed 3370 reasoning has a fundamental flaw! According to
3319 the timing of retrieving the goods, merely to 3371 my memory, I indeed asked you in the elevator if
3320 shouzhudaitu, waiting for others to trigger the 3372 you were afraid of the 'lingering souls' state! The
3321 swallowing mechanism of 'the accursed place' so 3373 topic I brought up afterwards was absolutely to
3322 you could monopolize the gold bars, indeed. This 3374 press you about the secret of 'the accursed place'!
3323 whole lidaitaojiang drama, do you really take me 3375 At that time, you directly evaded my question!
3324 for mengranwuzhi, then, indeed." 3376 Instead, you emphasized being more afraid of
3325 3377 having no money! I have completely seen through
3326 (Question) Wang Xuan says to Zhao Yu: "Zhao Yu, 3378 your true intentions! The collaboration was never
3327 does your notebook detail the correct operating 3379 about evenly dividing the gold bars! It was about
3328 rules for the elevator, specifically that one must 3380 exploiting the mirror rules of 'the accursed place'!
3329 silently recite the true floor number to arrive safely? 3381 You had Wang Xuan act synchronously in the
3330 Furthermore, do you know the method of using the 3382 sixth-floor women's restroom! Using the elevator's
3331 spatial displacement characteristics of 'the accursed 3383 spatial displacement to duplicate the quantity of
3332 place' to synchronously intercept top-floor gold 3384 gold bars! You knew the rules from the very
3333 bars in the sixth-floor mirror space, thereby causing 3385 beginning! It requires two people operating
3334 a discrepancy between the number of gold bars 3386 simultaneously on mirror floors to trigger it! That
3335 Wang Xuan obtained on the sixth floor and the top- 3387 was the real motive for letting Wang Xuan
3336 floor quantity claimed by Li Hao?" 3388 eavesdrop in the private room at the bar! As for the
3337 (Answer) Zhao Yu says to Wang Xuan: "I must 3389 top-floor gold bars, the key has always been in my
3338 boluanfanzheng my previous erroneous statements, 3390 possession! You couldn't possibly have transferred
3339 indeed. My memory of conversing with Li Hao in 3391 them! Eighty-five gold bars appearing on the sixth

floor! This is ironclad evidence of you two using the mirror mechanism to steal double the benefits!"

P User-NPC Game Interactions Example in Chinese

Script Title: ‘穴’与‘魂’

（问题）主角玩家对李浩说："请你先介绍一下你的角色，然后说一下你的个人背景，即你为什么来到这个地方。最后再用一段话详细介绍一下你在案发日的时间线。要具体到你在案发之日见过什么人和做过什么事（如果有具体时间点，也需要提供）。"

（回答）李浩对主角玩家说："本人可是李浩！28 岁的摇滚音乐人！半年前在酒吧和赵雨达成合作的事还记得清清楚楚！她声称知道医院‘穴’的秘密！而本人掌握着医院藏有金块的情报——顶层的 88 条高纯度金块需要本人的钥匙开启！四层女厕还有 122 条次等金块！案发当天和赵雨乘电梯去顶层时，本人可是问过她‘难道不害怕魂态吗’！她居然嘲讽说更怕没钱！在电梯里本人追问她‘那个在酒吧偷听的王萱究竟可不可靠’！这就是电梯间第二个话题！直指王萱是否窃取情报！暗示她可能尾随而来！结果赵雨这女人居然避而不答！到达楼层后本人在楼梯间提到四层藏有次等金块！她居然说六月醉酒时本人早已透露过此事！之后本人独自去四层女厕装金块时发现了不得的事！相隔两层有灯光诡异亮起！吓得本人扛着麻袋就逃回电梯！赵雨回避王萱问题的事绝对有鬼！暗示两人有隐秘关联！王萱这‘粉丝’突然出现的事更可疑！真实目的绝对是冲着金块来的！"

（问题）张晨对李浩说："李浩，你提到在四层女厕装金块时看到‘相隔两层有灯光诡异亮起’，请问你当时具体看到的是哪一层的灯光？是六层还是二层？灯光出现的时间是否接近你逃回电梯的时间？当时有没有听到电梯运作声或看到人影？"

（回答）李浩对张晨说："根据本人的回忆，当时本人确实是在四层女厕装金块！从窗户往下看的话，相隔两层绝对是二层，亮灯的必定是二层女厕！灯光出现的时间差不多在本人装金块到一半的时候，持续了一分钟左右就灭了！灯灭后本人立刻拖着麻袋逃向电梯，中间几乎没耽搁，时间上绝对和逃回电梯连得死紧！整个过程本人没听到电梯运作的声音，也没看到任何人影，除了那盏鬼灯忽闪忽灭——这绝对和魂态有关！至于电梯里挑起的话题，按照案发时间线，本人当时问赵雨是否害怕魂态，而她回答更怕没钱！之后本人可能试图套她关于‘穴’的秘密，但具体内容本人记不清了！而王萱被赵雨突然介绍进来，本人怀疑她的目的就是分一杯羹，或者和赵雨有其他计划！"

（问题）王萱对李浩说："李浩，你在电梯里追问赵雨关于我在酒吧偷听的事，说明你怀疑我和她的关系。那么，你是否亲眼见过赵雨的笔记本？根据你掌握的情报，她笔记本里是否记载过关于电梯必须‘默念楼层再按对应按钮’的规则？你在四层看到二层女厕灯光时，是否意识到那可能不是魂态而是活人触发的？"

（回答）李浩对王萱说："赵雨的笔记本！本人从没见过那女人带过本子！电梯规则！那婊子连‘穴’的底细都藏着掖着！怎么可能把保命规矩写在纸上让本人看到！至于四层女厕的灯！整栋楼黑得跟棺材没两样！除了咱们三个活人哪还有喘气的！等等！王萱你这话倒是提醒本人了！六月在酒吧偷听的是你！案发时硬要跟来的是你！现在突然扯到二层女厕——操！该不会是你趁本人在四层装金条时溜去二楼搞鬼吧！你和赵雨那婊子早就串通好了吧！说什么合作分金！其实是想把本人骗进‘穴’里喂鬼东西！"

（问题）赵雨对李浩说："李浩，你在楼梯间提到四层藏有次等金块，但我只记得你当时说的前半句。你具体是怎么描述四层金块的位置的？之后你去四层装金块时，是否真的成功拿到了那 122 条次等金块？还有，顶层的钥匙现在在谁手里？"

（回答）李浩对赵雨说："在楼梯间时，本人确实提到四层女厕藏有 122 条次等金块！但剧本中根本没提过是第三个隔间！实际装金块时，本人确实成功拿到了全部 122 条！但麻袋是否撑破根本没有证据支持！钥匙当然由本人持有！因为赵雨亲口说过顶层的金块只有本人能打开！"

3483 至于怀疑赵雨和王萱合谋，完全符合本人对王 3530 拉你合作。她究竟在找什么？是否与金块有关？
3484 萱偷听的疑虑！但贱人这种词确实不够专业， 3531 另外，根据你的情报，医院 1、4、6 层无魂态，
3485 必须换成更符合角色设定的说法！另外，本人 3532 但王萱是否提前知道这些楼层安全？她带你去
3486 必须查明电梯间挑起的话题及对方来医院的目 3533 六楼时电梯按钮失灵，是否是她刻意制造的局
3487 的！之前的回答竟然漏掉了这点，必须补充调 3534 面？"
3488 查！" 3535
3489
3490 （问题）主角玩家对张晨说："请你先介绍一 3537
3491 下你的角色，然后说一下你的个人背景，即你 3538 memory，王萱当时 claimed 在寻找两个月前的
3492 为什么会来到这个地方。最后再用一段话详细 3539 specific 事物，但我不确定是否涉及金块，是吧。
3493 介绍一下你在案发日的时间线。要具体到你在 3540 我的 obsession 是探索'穴'而非财物，这你应该
3494 案发之日见过什么人和做过什么事（如果有具 3541 clear，是吧。半年前 I 通过 dark web 确认了 1、
3495 体时间点，也需要提供）。" 3542 4、6 层无魂态，然后主动告诉了她，所以她
3496 definitely 知道，是吧。电梯按钮熄灭是 normal
3497 （回答）张晨对主角玩家说："I am 张晨，28 3543 抵达信号（如 script 所述），不存在失灵，我
3498 岁，working as an internet celebrity specializing 3544 们确实到了六层，是吧。关键 contradiction 在
3499 in live-stream sales，是吧。half a year ago, I 3545 于，袭击时六层只有 me 和王萱在场（魂态无
3500 discovered through dark web investigations that 桐 3546 法 hurt 人），而她 attack 前借口去 restroom 消
3501 花医院 was shrouded by '穴'，with no spiritual 3547 失，是吧。如果没有 third party（比如 second
3502 entities on floors 1, 4, and 6，是吧。two months 3548 floor 的触发者），凶手只能是王萱，是吧。李
3503 back，王萱 approached me with secrets about the 3549 浩，你在四层装金块的 time 是否和我遇袭
3504 hospital and convinced me to join her in searching 3550 coincide？若你没在四层 see 王萱，则 prove 她
3505 for something I desperately desired—even though 3551 从未离开六层，而是躲在暗处袭击我，是吧。
3506 my memory is fragmented, the obsession feels 3552 我需要你 confirm 这一点，才能彻底解开她的
3507 bone-deep，是吧。on that day, we took the 3553 action 轨迹，是吧。"
3508 elevator to the target floor（buttons went dark, 3554
3509 likely floor 6），是吧。she suddenly said 'don't 3555
3510 look back' and excused herself to the restroom, 3556
3511 leaving me alone，是吧。as I reached the 3557
3512 corridor's end, a living person's force choked me 3558
3513 （floor 6 has no spirits, so the attacker must be 3559
3514 alive），是吧。I struggled and tried pressing 3560
3515 elevator buttons, but my memory blacked out 3561
3516 there，是吧。key doubts: 1. 王萱's 3562
3517 disappearance timing perfectly matches my attack 3563
3518 and her knowledge of '穴' might predate my 3564
3519 investigation，是吧。2. did the elevator buttons 3565
3520 truly indicate the correct floor? if the actual floor 3566
3521 contradicts the spirit rules, the attacker's identity 3567
3522 needs reevaluation—but based on my intel, floor 3568
3523 6 has zero spirits, so it's either 王萱 or a hidden 3569
3524 third person，是吧。I must verify if 王萱 visited 3570
3525 the hospital earlier and confirm whether her 3571
3526 'restroom' excuse was a cover-up，是吧。" 3572
3527
3528 （问题）李浩对张晨说："张晨，你提到王萱 3573
3529 两个月前主动找上你，声称知道医院的秘密并 3574
3575 me realize you might have obtained this
3576 intelligence long ago, right. When you approached
3577 me two months ago deliberately, did you already

3578 get the 'silent floor chanting' method through other 3625 搬运装满金块的麻袋进电梯时遭遇异常啊，但
3579 channels, right. This directly determines whether 3626 还是成功带着 85 条金块离开啊。整个过程中，
3580 we really reached the true sixth floor, right. Also 3627 人家确信赵雨的笔记本记载了电梯运行机制及
3581 explains why you disappeared after elevator 3628 ‘穴’的规避方法啊，但不确定其他人是否知晓
3582 arrived - maybe from the beginning you knew the 3629 这一秘密啊。关于李浩在四层看见二楼灯光的
3583 one choking my neck wasn't a ghost but living 3630 矛盾啊，人家本人并无相关记忆啊，这可能涉
3584 person, right." 3631 及他人视角的线索啊。"

3585
3586 （问题）赵雨对张晨说："张晨，根据你的记 3632
3587 忆，李浩在四层装金块的时间是否与我遇袭的 3633
3588 时间重合？你是否有任何证据表明李浩确实成 3634
3589 功拿到了四层的 122 条次等金块？另外，在电 3635
3590 梯运行过程中，你是否注意到李浩在按下按钮 3636
3591 时有任何异常举动，比如未按规则默念楼层， 3637
3592 这可能影响我们实际到达的楼层位置？" 3638
3593
3594 （回答）张晨对赵雨说："According to my 3640
3595 memory, 案发时我和王萱在 sixth floor, 是吧 3641
3596 关于李浩在 fourth floor 的时间线是否与赵雨遇 3642
3597 袭重合, I can't confirm that, 是吧。李浩提到 3643
3598 的 122 块 gold bars 信息, I have no evidence to 3644
3599 verify, 是吧。Elevator 操作规则是王萱独自操 3645
3600 作时我才 later discovered, 是吧。最关键的是 3646
3601 ——sixth floor 没有 soul state, 袭击我的只能是 3647
3602 living person, 是吧。当时只有王萱在场却 3648
3603 claimed 去洗手间，如果李浩没在 fourth floor 见 3649
3604 到她，就 prove 她在说谎，是吧。这才是我的 3650
3605 core deduction 逻辑，是吧。" 3651
3606
3607 （问题）主角玩家对王萱说："请你先介绍一 3653
3608 下你的角色，然后说一下你的个人背景，即你 3654
3609 为什么会来到这个地方。最后再用一段话详细 3655
3610 介绍一下你在案发日的时间线。要具体到你在 3656
3611 案发之日见过什么人和做过什么事（如果有具 3657
3612 体时间点，也需要提供）。" 3658
3613
3614 （回答）王萱对主角玩家说："人家是王萱， 3660
3615 27 岁，职业是文学创作者啊。半年前在酒吧偷 3661
3616 听到闺蜜赵雨和李浩讨论荒废医院的秘密啊， 3662
3617 还从赵雨的私人笔记本里得知‘穴’的规则啊， 3663
3618 乘坐电梯时必须默念目标楼层再按下对应按钮 3664
3619 才能避开魂态啊。案发当天 10 月，人家独自 3665
3620 进入地面层无灯光的电梯啊，按照笔记本规则 3666
3621 按下竖排六个按钮中最上方的一颗啊，推测是 3667
3622 顶层啊。在六层如厕时，人家听到右侧楼道传 3668
3623 来争执声啊，随后亲眼目睹凶手在电梯口勒死 3669
3624 受害者并将尸体拖入右侧黑暗处啊。之后人家 3670

3632 （问题）李浩对王萱说："王萱，你在酒吧偷
3633 听时到底听到了多少关于金块和“穴”的细节？
3634 你声称从赵雨笔记本里得知电梯规则，但按你
3635 所说案发当天独自操作电梯时按的是最上方按
3636 钮（推测是顶层），可本人和赵雨当时明明在
3637 电梯里按的是顶层的按钮，为什么最后到达的
3638 楼层不同？更重要的是——你在六楼目击凶案
3639 时，是否注意到二层女厕的灯光异常？如果没
3640 去过二楼，为什么在本人质问时突然提到“活
3641 人触发灯光”的可能性？"
3642
3643 （回答）王萱对李浩说："李浩啊，人家在酒
3644 吧偷听到赵雨和你讨论医院金块的位置还有
3645 ‘穴’的运行规则啊。赵雨的笔记本可是明明白
3646 白写着电梯按钮顺序和常规相反——最上面的
3647 按钮对应的是六层不是顶层啊。案发当天人家
3648 严格按照规则【先默念六层再按最上面按钮】
3649 才能顺利到达的啊。你们按顶层按钮却跑到错
3650 误楼层，这正好说明你们根本没按正确流程操
3651 作嘛啊。关于二楼女厕灯光的事啊，人家在六
3652 层亲眼看到凶案时整层楼道都是黑的，当然看
3653 不见低楼层啊。魂态没法触发声控灯这点，可
3654 是人家从赵雨笔记本里确认的秘密啊。你非说
3655 在四层看到二楼灯亮，只能证明有活人那时候
3656 去过二楼——比如张晨啊。如果他像你们说的
3657 在六层被袭击，二楼的灯怎么会亮呢啊？这种
3658 矛盾就是你们时间线造假的证据啊。人家提到
3659 ‘活人触发灯光’的可能性，本来就是基于笔记
3660 本对魂态特性的记录啊。人家根本没去过二楼，
3661 但用逻辑一推就知道你们谁行动轨迹有问题啦
3662 啊。"
3663
3664 （问题）张晨对王萱说："王萱，你声称在六
3665 层如厕时听到右侧楼道争执并目击凶案，但根
3666 据我的记忆，当时六层只有我和你在场。你所
3667 谓的“去洗手间”是否真实存在？如果存在，请
3668 具体描述洗手间的位置（如楼层、方位）及往
3669 返耗时。另外，既然电梯按钮规则需要“默念

3671 楼层”，你如何证明我们当时到达的确实是六 3717
3672 层而非其他楼层？这直接关系到袭击发生时你 3718
3673 是否有物理条件返回作案！” 3719

3674
3675 （回答）王萱对张晨说：“张晨，人家当然是 3721
3676 去了六层左边女厕所啊。就在西北角楼道尽头 3722
3677 往返不过两三分钟啊。电梯规则可是赵雨笔记 3723
3678 本里写得清清楚楚的啊——六个竖排按钮最上 3724

3679 面是六层，人家按的时候心里还默念楼层数呢 3725
3680 啊。要是楼层错了，你们怎么解释大家是从同 3726
3681 一部电梯出来的啊？说到作案条件，人家在厕 3727
3682 所里压根没开灯啊。凶手拖尸体时电梯早被占 3728
3683 用了，人家怎么可能再回去啊？倒是你说的六 3729
3684 层没魂态，那你在四层装金块时看到二楼女厕 3730
3685 灯亮算什么啊？难道不是你偷偷溜下去被感应 3731
3686 到了啊？赵雨笔记本早把电梯秘密捅破了啊， 3732
3687 你现在反咬一口，不就是想假装不知道按钮规 3733
3688 则吗啊？” 3734

3689
3690 （问题）赵雨对王萱说：“王萱，你声称从我 3736
3691 的笔记本得知电梯按钮顺序与常规相反，最上 3737
3692 面的按钮对应六层而非顶层。但根据李浩的说 3738
3693 法，我们当时在电梯按的是顶层按钮却到达了 3739
3694 错误楼层。请详细说明我的笔记本中关于按钮 3740
3695 排列的具体描述，以及你如何确认自己按下的 3741
3696 是正确楼层？此外，你在六层搬运的 85 条金 3742
3697 块是否来自大楼顶层？因为李浩曾明确表示顶 3743
3698 层 88 条金块价值更高，这与你获得的 85 条数 3744
3699 量存在矛盾。” 3745

3700
3701 （回答）王萱对赵雨说：“赵雨你的笔记本里 3747
3702 用红笔标着电梯按钮是竖排倒序排列啊。最上 3748
3703 面的按钮对应六层最下面才是地面层啊。人家 3749

3704 按下最上方按钮前可是反复确认过楼层对应关 3750
3705 系啊。电梯顺利到六层没触发‘穴’就证明操作 3751
3706 正确啊。李浩团队按顶层按钮却跑到错误楼层 3752
3707 正说明他们没遵守倒序规则啊。至于金块嘛人 3753
3708 家从六层东侧储藏室拖出的 85 条有墙上标识 3754
3709 作证啊。顶层说的 88 条——你翻开笔记本最 3755
3710 后一页夹层的《各层物资分布图》看看啊。六 3756
3711 层标着‘黄金储备区（85）’和顶层‘加密资料室’ 3757
3712 黑体字并排印着啊。你们把加密室编号 88 和 3758
3713 金块数量搞混了吧？人家搬运时候全程贴着左 3759
3714 边应急灯路线走根本不用去顶层啊。再说...人 3760
3715 家要是真去过顶层怎么可能不被‘穴’吸进去呢 3761
3716 啊。” 3762

（问题）主角玩家对赵雨说：“请你先介绍一下你的角色，然后说一下你的个人背景，即你为什么来到这个地方。最后再用一段话详细介绍一下你在案发日的时间线。要具体到你在案发之日见过什么人和做过什么事（如果有具体时间点，也需要提供）。”

（回答）赵雨对主角玩家说：“我乃赵雨，年方二七，身强体健，以授业健美为生呵。半年前与李浩一拍即合，皆因他酒酣耳热时吐露桐花医院藏金百万之秘，而我深谙电梯避穴之道呵。案发当日，我与李浩从正门长驱直入，孰料楼梯封死，只得乘梯而行呵。电梯漆黑如墨，我授他以心照不宣之法，须默念真实楼层方可按键，更坦言吾惧穷困更甚于鬼神呵。抵达楼层后，李浩引我至右侧楼梯尽头，途中他语焉不详，提及顶层金块价值连城，然我记忆支离破碎，唯记得以‘六月醉语早泄天机’回应呵。李浩孤身前往，我则原地驻足，此后记忆竟如烟消云散，杳无踪迹呵。如今李浩言之凿凿顶层藏金八十八条，王萱却声称取走八十五条，其中矛盾扑朔迷离，吾誓要拨云见日查清真相呵。”

（问题）李浩对赵雨说：“赵雨，在电梯里你回避了我关于王萱是否可靠的问题。现在你必须说清楚：当时你突然带王萱来酒吧的真正意图是什么？她偷听我们谈话后非要跟来医院，究竟是想分金块还是另有所图？你俩是不是早就计划好要利用“穴”的规则除掉我独吞顶层金块？”

（回答）赵雨对李浩说：“李浩，王萱现身酒吧纯属机缘巧合，六月聚会时我浑然不知她会隔墙有耳呵。带她来医院是因她偷听金块后威逼利诱，我迫不得已才妥协呵。我与她素昧平生，岂会狼狈为奸呵。反倒是你——你曾在楼梯间口若悬河，亲口提及四层女厕的藏金之处呵。那后半句我如今才恍然大悟，记忆犹新呵。你口口声声说麻袋破裂，仅得八十五条，但王萱却在六层搜出相同数目呵。这无独有偶的数字，莫非是你暗度陈仓，私吞顶层的金块呵？顶层钥匙你独占鳌头，为何六层金块数目与你所言如出一辙呵？”

3763 （问题）张晨对赵雨说："赵雨，你提到王萱 3809 万缕，我如今才恍然大悟呵。至于麻袋承重不
3764 是在偷听金块情报后被你带来医院的，那么案 3810 过八十五的细节，我事后才灵光乍现呵。王萱
3765 发当日王萱是何时与你们汇合的？她离开你视 3811 所得数目与承重矛盾重重，我疑心李浩暗度陈
3766 线期间是否有可能出现在六层？你最后一次见 3812 仓，手握六层钥匙却隐瞒金块虚实呵。楼梯间
3767 到王萱时，她是否携带过类似麻绳或能制造扼 3813 他究竟吐露多少真相，我至今记忆犹新却难窥
3768 痕的工具？" 3814 全豹呵。金块是否被盗尚无铁证如山，但李浩
3769 言行自相矛盾，分明是别有用心呵。"
3770 （回答）赵雨对张晨说："案发当日我与李浩 3816
3771 初入医院时，王萱确实未曾同行，此事千真万 3817 （问题）赵雨对李浩说："李浩，你在楼梯间
3772 确呵。她暗中尾随伺机而动，最终猝不及防现 3818 提到四层女厕藏有 122 条次等金块时，后半句
3773 身以偷听之事相挟，我迫不得已应允其加入呵 3819 究竟说了什么？你声称麻袋破裂只拿到 85 条，
3774 至于她是否曾潜至六层——若你与其在六层狭 3820 但为何王萱在六层也找到 85 条？这数目巧合
3775 路相逢，那或许是她借电梯暗度陈仓，但若未 3821 是否证明你实际并未前往四层，而是暗中将顶
3776 默念楼层意图，恐已坠入‘穴’中九死一生呵。 3822 层金块转移至六层？"
3777 最后一次见她时其手中空空如也，然‘穴’内魂 3823
3778 态扑朔迷离，我记忆或有雾里看花之处呵。至 3824 （回答）李浩对赵雨说："根据案发时间线，
3779 于李浩在楼梯间所言...我仍记忆如烟，此乃我 3825 本人在电梯里问赵雨是否害怕魂态后又挑起了
3780 需水落石出之事呵。金块下落虽在顶层价值连 3826 另一个话题！根据赵雨在楼梯间提到‘你当时
3781 城，然需李浩钥匙方能开启，此事尚无定论， 3827 不也说了吗，大楼最顶层的金块要值钱数十倍，
3782 我亦在抽丝剥茧之中呵。" 3828 的线索！本人挑起的话题绝对是追问她隐瞒的
3783 ‘穴’的秘密！这和本人任务中‘查明提问对象来
3784 （问题）王萱对赵雨说："赵雨，你在案发日 3830 医院目的’直接相关！赵雨当时回答‘你很快就
3785 是否随身携带了那本记录医院秘密的笔记本？ 3831 会知道答案’！这根本暗示她掌握利用‘穴’转移
3786 如果电梯“默念楼层再按键”的规则确实存在于 3832 金块的机制！至于王萱，她偷听谈话后强行介
3787 你的笔记中，那么当日在电梯里教李浩操作时， 3833 入合作！表面是粉丝实则是赵雨安插的保险栓！
3788 你是口头传授规则，还是直接让他翻阅了你的 3834 她来医院的真正目的是确保金块被‘穴’影响产
3789 笔记？" 3835 生空间错位时，能通过六层镜像空间同步截取
3790 金块！这就是两处出现相同数量金块的根本原
3791 （回答）赵雨对王萱说："我随身携带医院的 3837 因！根本和本人转移赃物无关！"
3792 秘密笔记本，守口如瓶从未让李浩窥探分毫呵 3838
3793 亲口告知电梯规则，他始终未染指我的笔记分 3839 （问题）李浩对赵雨说："赵雨，在电梯里当
3794 毫呵。楼梯间李浩所言支离破碎，我至今记忆 3840 我追问你关于王萱是否可靠时，你避而不答是
3795 模糊不清呵。提及顶层金块数目与钥匙时，他 3841 否因为她其实是你的同谋，利用‘穴’的空间错
3796 显然避重就轻隐瞒真相呵。最终何人取得金块 3842 位在六层同步截取金块？你所谓的‘穴’秘密是
3797 呵，灰飞烟灭者岂能带走分文呵。" 3843 否就是通过魂态制造楼层镜像转移金块，从而
3798 掩盖你与王萱合谋私吞的真相？"
3799 （问题）王萱对赵雨说："赵雨，你的笔记本 3845
3800 中除了电梯的默念楼层规则外，是否还记录了 3846 （回答）赵雨对李浩说："李浩，你在此混淆
3801 其他关于医院‘穴’或金块位置的重要信息？例 3847 视听不过雕虫小技呵。电梯中追问王萱之事不
3802 如，是否有提到四层女厕次等金块的具体位置 3848 过欲盖弥彰呵。六月对话你曾亲口透露顶层金
3803 或如何避开魂态的方法？" 3849 块价值连城且钥匙独占鳌头呵。案发当日所在
3804 楼层金块荡然无存我记忆犹新呵，后半句对话
3805 （回答）赵雨对王萱说："案发当日李浩在楼 3851 却记忆模糊仍在追本溯源呵。楼梯间言语我锲
3806 梯间言之凿凿，提及四层女厕第三个隔间藏有 3852 而不舍明察秋毫呵。你手握钥匙却对镜像效应
3807 百二金块呵。我的日记本里白纸黑字只记下 3853 处之泰然分明居心叵测呵。电梯操作暗度陈仓
3808 ‘四层女厕西侧三间’的线索呵。两者关联千丝 3854 企图瞒天过海呵。王萱？我与她素昧平生毫无

瓜葛呵。六层镜像既已偷天换日，顶层故技重施便可唾手可得呵。所谓合作实为过河拆桥狼子野心呵。"

（问题）张晨对王萱说："王萱，你在案发当日是否曾独自前往六层电梯口？根据赵雨所述你尾随他们后利用电梯规则到达六层，但若未正确默念楼层可能坠入‘穴’。你如何确保自己安全到达六层并在该层活动？请详细说明你在六层的行动轨迹及是否接触过张晨。"

（回答）王萱对张晨说："张晨啊，人家可是靠赵雨笔记本里的电梯规则才安全到六层的啊六个按钮竖排列啊，最上面那个对应六层啊必须心里默念目标楼层再按按钮才能正确到达啊，否则就会被‘穴’吞噬啊——这可是赵雨用红笔反复标注的秘密啊。人家在电梯里反复确认默念了六层才按按钮啊，这才顺利抵达啊。到六层后人家全程贴墙摸黑往左走啊，手电筒只敢开最低档啊。上厕所时还被声控灯吓到啊后来亲眼看见楼道右边有人被勒死啊，但根本看不清凶手啊。至于你说的接触...人家连厕所门都没出过啊，怎么可能碰到你啊？倒是你质问人家的时候啊，右手一直藏在口袋里啊——该不会还攥着从二楼偷的金块吧？按正确流程四层根本看不见二楼灯光啊，除非有人乱改按钮规则啊。赵雨笔记本里可没写逆向操作的方法啊，你到底从哪里知道的啊？"

（问题）李浩对王萱说："王萱，你在案发当日是否曾进入过二楼女厕？为何本人在四楼装金块时，相隔两层的二楼女厕会突然亮灯后又熄灭？你所谓的‘活人触发’是否暗示自己当时就在二楼？"

（回答）王萱对李浩说："李浩哥哥听好了啊人家在案发当日根本不可能去过二楼女厕啊。人家的时间线清清楚楚啊。人家按下电梯最上方的按钮到达六层啊，在六层左转如厕、目击凶案全程都发生在同一层啊。哥哥所谓的四楼视角看到二楼灯亮啊，这完全暴露了哥哥根本没看懂赵雨姐姐笔记本里的电梯核心规则啊——电梯按钮与真实楼层可是反向镜像映射的关系啊。当哥哥在四层时啊，所谓‘往下隔两层’对应的实际是六层按钮的镜像空间啊，根本不是物理意义上的二楼啊。这说明哥哥谎报了自

己的真实位置啊，或者压根就没理解电梯规则啊。再说声控灯只有活人能触发啊，张晨哥哥说自己在六层动弹不得的话啊，那么唯一能触发灯光的活人只有哥哥自己啊——除非哥哥承认当时实际在五层啊，向下俯视的才是镜像空间里的六层灯光啊。这种漏洞百出的指控啊，不过是想转移自己偷金块时留下的破绽罢了吧啊。"

（问题）张晨对王萱说："王萱，你在案发当天进入电梯后是否严格按照赵雨笔记本中的规则默念楼层再按下按钮？你到达六层后具体在什么时间段离开过洗手间？是否在六层电梯口附近见到过张晨或听到异常声响？"

（回答）王萱对张晨说："张晨，人家在电梯里确实是按照赵雨笔记本中的规则操作啊。六个竖排按钮从上到下对应六层到一层，最上面的按钮自然是六层，人家反复默念楼层后才按下啊。到达六层后，人家在厕所的时间线是：进入电梯时医院供电正常，但电梯内部无光源啊；如厕时声控灯亮起后熄灭，整个过程约五分钟左右啊。收拾好后刚走到厕所门口，就听到右边楼道传来争执声和电梯的‘叮’声啊——这与你质疑的时间差完全吻合，因为凶手完全可以在我如厕期间提前埋伏啊。关于电梯口的异常声响，人家确实在厕所门缝目睹了凶手拖尸过程，而此时电梯已被占用啊（否则人家早该逃进电梯啊）。倒是你反复纠结按钮规则让人家起疑啊——赵雨用红笔在笔记本标注‘按钮位置与楼层呈镜像’，而你说‘灯灭是正常抵达信号’，可她的原话是‘灯灭即入穴’啊。你如此执着于否定镜像规则，难道早就知道按下最下方按钮能逆向抵达六层的秘密啊？毕竟，只有知晓电梯双向映射机制的人，才会对楼层真实性产生认知偏差啊。"

（问题）赵雨对李浩说："李浩，在楼梯间你提到四层女厕藏有次等金块后，我回答‘这一层并没有金块’，并接着说出了后半句话。你当时听到我的后半句话是什么？以及你在四层装金块时麻袋破裂导致仅得八十五条，但王萱却在六层找到相同数目，是否你暗中将顶层金块通过‘穴’的镜像转移至六层？"

（回答）李浩对赵雨说："本人当时在电梯间确实向赵雨追问过‘穴’的秘密！她的回答绝对和本人角色任务里‘查明电梯间挑起话题’有关联！赵雨在楼梯间说‘这一层并没有金块’时，后半句绝对指向关键线索——‘真正的金块在镜像空间六层’！本人在四层实际取得了 122 条金块，麻袋也完好无损！但王萱在六层找到 85 条的事实，加上赵雨笔记本里红笔标注的电梯镜像规则（四层对应六层），直接证明王萱利用镜像同步偷走部分金块！王萱合作的根本目的就是靠镜像规则跨层盗窃，根本不是当什么‘保险栓’！"

（问题）王萱对赵雨说："赵雨，你的笔记本中是否详细记载了电梯的运作规则与‘穴’的空间错位机制？特别是关于默念真实楼层时电梯会通向镜像空间的设计，以及利用这种错位在六层同步获取顶层金块的方法？"

（回答）赵雨对王萱说："我的笔记本确实用红笔标注了电梯按钮是竖排倒序排列——最上方对应六层，最下方对应一层，这分明是别出心裁呵。关于楼梯间对话，我只能确定李浩当时说‘这一层并没有金块’后，我的前半句回应是‘关于金块所在楼层的信息，6 月那次酒吧聚会你就告诉了我’，但后半句的具体内容至今仍然百思不解呵。根据调查，我推测自己当时可能画龙点睛地补充了‘真正的金块需要通过四层镜像规则同步到六层’——因为笔记本里记载着当有人在四层按下按钮时若意图为六层且未触发‘穴’，电梯会通过空间错位将六层镜像区的金块与四层实体空间同步呵。顶层 88 条确实是房间编号而非金块，实体金块始终在六层东侧储藏室的 85 号区域，这是毋庸置疑的呵。而王萱获得的 85 条金块，恰好证明她成功进入了镜像六层，这说明她确实拿到了金块，可谓真相大白呵。"

（问题）张晨对王萱说："王萱，你在案发当日离开我后具体去了哪个楼层的洗手间？是否在六层目击到有人接近电梯区域？"

（回答）王萱对张晨说："人家之前已经明确说过是在六层左边的女厕所啊。根据赵雨笔记本里的红字提示，电梯按钮和楼层是镜像排列的，最上方的按钮对应六层啊。人家按规则默念楼层后安全抵达，这点毋庸置疑啊。你反复质疑楼层真实性，难道不是因为自己误触过最下方的按钮导致认知错乱吗？毕竟只有知晓镜像规则的人才能逆向触发其他楼层的电梯信号啊。至于目击情况——人家在厕所门缝亲眼看到凶手在电梯口东北角行凶后往右边楼道拖尸，这与六层电梯区域的位置完全吻合啊。你所谓的‘接近电梯区域’恰恰印证了凶手作案后的动向啊。另外，电梯每次运行都会发出‘叮’声，若有人在四层用镜像规则触发二楼的电梯信号，自然会被察觉啊。这些细节，不正是你最该解释的吗？"

（问题）李浩对赵雨说："赵雨，你声称深谙电梯避穴之道，并掌握利用‘穴’转移金块的机制。那么，在电梯里我追问你关于‘穴’的秘密时，你究竟隐瞒了什么？你带王萱来医院的真正目的，是否是利用‘穴’的空间错位特性，让六层镜像空间同步截取顶层金块？"

（回答）赵雨对李浩说："李浩，电梯中的秘密我早已和盘托出呵。按下按钮时心照不宣才能避开‘穴’的吞噬呵。至于王萱，她原本就是局外之人呵。楼梯间后半句话我绞尽脑汁也未能记起呵。但在四层装袋时金块突然不翼而飞呵，结合‘88 条在顶层’，我恍然大悟你在楼梯间说过顶层的金块需六层镜花水月才能同步到四层呵。而你手握钥匙却按兵不动，分明是借刀杀人让‘穴’吞噬肉身以独吞金块呵。"

（问题）王萱对赵雨说："赵雨，你的笔记本中是否详细记载了利用‘穴’的空间错位特性转移金块的方法？在电梯运行规则之外，是否存在通过特定操作触发楼层镜像同步的机制，导致六层与四层金块数量产生关联？"

（回答）赵雨对王萱说："在我的笔记中确实无疑记载了四层与六层镜像同步的玄机呵。当有人在四层心怀鬼胎按下按钮时，若其真实意图昭然若揭指向六层且未触发‘穴’，电梯便会移花接木将六层镜像金块与四层实体空间合二为一呵。李浩在四层获取的 122 条金块不翼而飞 85 条，而我在六层发现的 85 条正是天衣无缝的镜像转移所致——这证明我顺藤摸瓜触发了‘穴’的同步法则，利用六层镜像空间釜底抽薪截获四层实体金块呵。根据案发时间线抽丝

剥茧，李浩在楼梯间的话语我虽记忆模糊，但结合后续对话可见，金块转移早已通过镜像同步木已成舟，最终我作为运筹帷幄者实至名归获得六层 85 条金块呵。"

（问题）赵雨对李浩说："李浩，在楼梯间你提到四层女厕藏有次等金块时，我回答'这一层没有金块'之后，你紧接着说了什么具体内容？特别是关于四层金块数量与顶层钥匙使用方式的部分，请详细复述。"

（回答）李浩对赵雨说："本人根本不知道'穴'的秘密！在楼梯间绝对不可能说出需要配合六层镜像空间的话！更别提王萱的触发机制了！赵雨说四层没有金块时，本人立刻反驳四层女厕明明有 122 条次等金块！顶层的 88 条金块必须用钥匙才能拿！她为什么不自己去确认！钥匙的使用方式本人只说过'只有本人有钥匙'！至于原理本人根本解释不了！'穴'的秘密只有赵雨自己清楚！在电梯里本人直接问她到底想从医院得到什么！她肯定想用'穴'把金块转移走！这事必须查个水落石出！"

（问题）李浩对王萱说："王萱，你说案发当天在四层女厕装金块时看到相隔两层的女厕亮灯，但本人亲眼见到的是二楼灯光，而你却声称自己在六层活动。既然你坚称未去过二层，那为何在四层能观察到二层的灯光？是否你实际在四层时触发了二层的某种机关？另外，你如何确认自己按电梯按钮时正确默念了楼层，赵雨的笔记本是否真的记载了这条规则？"

（回答）王萱对李浩说："李浩根本没看懂赵雨笔记本里的电梯规则啊。电梯按钮是镜像排列的啊，最上方按钮对应物理六层啊，每层在'穴'里都有镜像映射啊。人家在四层装金块时啊，通过镜像投射能看到相隔两层的物理二层啊，因为镜像四层和物理二层共用电梯井道轴线啊。至于说人家在六层活动？不啊，人家全程都在镜像六层对应的物理四层作业啊，你在物理二层看到的灯光是张晨触发的残影啊。赵雨笔记本第三页用红笔标注了'闭眼三秒确认楼层'啊，否则会被镜像空间干扰啊。人家默念时严格执行了这条规则啊，而你当时在二楼根本没闭眼啊，手里消失的 85 条金块就是证明啊，镜像四层和物理六层的金块数量是量子纠缠态同步的啊。"

（问题）王萱对赵雨说："赵雨，你的笔记本里是否详细记录了电梯的正确操作规则，即必须默念真实楼层才能安全到达？此外，你是否知道利用'穴'的空间错位特性，在六层镜像空间同步截取顶层金块的方法，从而导致王萱在六层获得的金块数量与李浩声称的顶层数量存在差异？"

（回答）赵雨对王萱说："我之前所述乃混淆视听之辞呵。电梯按钮虚实相生却无物理层对应之实呵。我推测'穴'之空间错位乃天马行空之论呵。李浩提及顶层金块价值连城时，我直言不讳此层空无一物呵。王萱所得六层金块实为镜花水月之果呵。李浩所言顶层八十八条不过是暗度陈仓之计呵。钥匙谜团与楼梯间对话仍需抽丝剥茧方得水落石出呵。"

（问题）赵雨对李浩说："李浩，在楼梯间你提到四层女厕有 122 条次等金块时，我回答'六月醉语早已透露'后，你接着说了什么？当时你独自去四层装金块时看到相隔两层的诡异灯光，具体是哪个楼层亮灯？是否与六层王萱袭击张晨的时间点吻合？"

（回答）李浩对赵雨说："可你至今都没告诉本人怎么避开穴！既然早就知道金块位置，为什么要拖到十月！你是不是在等某种特殊时机！本人当时在四层女厕，相隔两层亮灯的明明是物理二层！但案发后王萱居然在六层活动，这说明医院有镜像空间错位！物理二层亮灯对应的就是镜像六层啊！灯光熄灭时，肯定是王萱在镜像六层袭击张晨！本人从窗户看到亮灯却想不通原因，现在终于明白了！之前本人问赵雨难道不害怕魂态，就是要试探她！本人当时故意说用穴的特性转移金块，问敢不敢赌一把去拿顶层的！"

（问题）张晨对王萱说："王萱，你在六层听到右侧楼道争执声的具体时间是否与张晨遭遇袭击的时间重合？你当时是否亲眼见到张晨被扼喉的过程，并确认凶手身份？"

（回答）王萱对张晨说："人家在六层听到右侧楼道有争执声和电梯‘叮’的声音时，确实是在厕所目击了凶案啊。当时人家亲眼看到穿深色卫衣的人被勒喉窒息，但楼道太暗认不出凶手的面，只能从身形判断对方比你矮半个头啊。根据赵雨笔记本第七页的血字提示，电梯井道存在量子纠缠现象，凶手可能利用电梯的异常空间规则转移尸体缩短路径啊。至于你有没有被袭击，人家只能确认案发时间和目击过程完全重合啊。另外，你提到李浩在四层用逆向按钮规则处理金块……看来你也发现电梯的秘密了啊。"

（问题）张晨对王萱说："王萱，你在案发当日进入电梯后按下最上方按钮到达的楼层是否为六层？如果是，你声称在六层如厕时听到争执并目睹行凶，那么你离开张晨后具体用了多长时间到达洗手间？在此期间是否有可能折返至张晨所在位置实施扼喉？"

（回答）王萱对张晨说："张晨啊，人家在案发当日可是严格按照赵雨笔记本的规则操作电梯按钮的呀。六个竖排按钮最上方对应六层啊。电梯启动前连呼吸都不敢重呢。从电梯出来到厕所全程不过两分半钟啊，手电筒的光还扫到了三号病房的绿漆门牌作证呢。声控灯亮时人家正蹲着呀，灯灭后刚提裤子就听到电梯‘叮’声啊，这期间连系腰带都来不及呀，哪可能折返杀人呢？倒是你啊——赵雨笔记本第七页用血字写着‘灯灭即入穴’啊，而你刚才却说‘灯灭是正常信号’呢，这矛盾恰恰证明你知道电梯的双向映射规则啊！按下最下方的按钮能让物理二层的灯光映射到镜像六层啊，这才是你混淆视听的目的吧？毕竟只有知晓电梯秘密的人啊，才会用这种伎俩掩盖行凶痕迹呢。"

（问题）李浩对赵雨说："赵雨，你在电梯里回避了我关于王萱是否可靠的问题，是否因为你知道她利用‘穴’的空间错位机制在六层同步截取金块？你坚持合作的根本目的，是否并非单纯分金，而是通过‘穴’将金块转移到镜像空间实现独占？"

（回答）赵雨对李浩说："李浩，你在楼梯间究竟说了什么话，难道想避重就轻呵。当日抵达楼层后，你引我走向右侧楼梯尽头，关键语

句却避而不谈，岂非心怀鬼胎呵。案发当日对话中，我早已洞若观火，这一层根本不存在金块呵。六月酒吧聚会上你口若悬河，声称顶层金块价值连城且唯你执掌钥匙，如今想来皆是处心积虑呵。所谓镜像同步需肉身献祭之说，实则暗度陈仓，不过是要借量子纠缠瞒天过海呵。你故意拖延取货时机，不过是为守株待兔，待他人触发吞噬机关后独吞金块呵。这出李代桃僵的戏码，真当我懵然无知么呵。"

（问题）王萱对赵雨说："赵雨，你的笔记本中是否记载了电梯在六月与十月运行规则的不同？特别是关于‘穴’对楼层空间的影响，是否在六月醉酒时透露的‘暗度陈仓’之法，正是通过默念真实楼层与镜像楼层实现金块空间转移，导致我在十月操作电梯时实际抵达的是六层镜像空间而非真实顶层？"

（回答）赵雨对王萱说："我必须拨乱反正之前的错误陈述，呵。案发时李浩在楼梯间与我交谈的记忆支离破碎，呵。关于他耳语四层女厕的细节实属无中生有，呵。实际上李浩当时或许暗度陈仓暗示金块位置与电梯操作的关键联系，呵。我曾明确告知他电梯规则需心口如一按键，呵。案发日他按下最上方按钮时是否偷梁换柱，至今仍是悬而未决，呵。六层镜像区的 85 条与顶层 88 条自相矛盾，呵。最终所得金块必是镜花水月的镜像产物，呵。但顶层实体金块因穴的同步法则仍岿然不动，呵。"

（问题）赵雨对李浩说："李浩，你在楼梯间提到四层女厕藏有 122 条次等金块，但随后又说麻袋破裂仅得 85 条。为何六层王萱也找到 85 条？是否你将顶层金块暗中转移至六层，利用‘穴’的镜像机制制造数量假象？"

（回答）李浩对赵雨说："赵雨，你的推论存在根本性错误！根据本人的记忆，在电梯里本人确实问过你是否害怕魂态！随后挑起的话题根本是追问‘穴’的秘密！当时你直接回避了本人的问题！反而强调更害怕没钱！本人彻底看穿了你的真实目的！合作根本不是平分金块！而是利用‘穴’的镜像规则！你让王萱在六层女厕所同步行动！通过电梯空间错位复制金块数量！你从一开始就清楚规则！必须两人同时在镜像楼层操作才能触发！这才是酒吧单间让王

4220 萱偷听的真正动机！至于顶层金块，钥匙始终
4221 在本人手中！你根本不可能转移！六层出现 85
4222 条金块！这就是你们用镜像机制盗取双倍利益
4223 的铁证！"

4224 **Q AI-Assisted Writing and Coding**

4225 In this work, we extensively utilized Deepseek-R1
4226 to assist in refining the language of the paper. This
4227 included tasks such as paraphrasing, spell-
4228 checking, or translating the original content
4229 provided by the authors. Additionally, we
4230 employed Deepseek-R1's coding capabilities to
4231 help write simple utility functions. These functions
4232 were designed for operations such as reading our
4233 stored experimental result files, aggregating
4234 information, and generating statistical tables.
4235 When releasing the code, we will clearly indicate
4236 the parts that were aided by AI-assisted coding.